SinOne

SC92F8363B/8362B/8361B

High-speed 1T 8051 core Flash MCU, 1 Kbytes SRAM, 8 Kbytes Flash, 128 bytes independent EEPROM, 23-channel low-power dual-mode touch circuitry, 12-bit ADC, six 10-bit PWM channels, 3 timers, multipliers/dividers, UART. SSI, Check Sum verification module

1. General Description

The SC92F8363B/8362B/8361B (hereinafter referred to as SC92F836XB) is an enhanced high-speed 1T 8051 core industrial-grade Flash microcontroller with integrated touch button function. Its instruction set is fully compatible with the traditional 8051 product series.

The SC92F836XB has a built-in 23-channel low-power dual-mode capacitor. The touch circuit can be selected to operate in STOP Mode.

The SC92F836XB also integrates 8 Kbytes Flash ROM, 1 Kbytes SRAM, 128 bytes EEPROM, up to 26 GP I/Os, 13 I/Os with external interrupt capability, 3 16-bit timers, 11 12-bit high-precision ADCs, 6 independent 10-bit PWMs, an internal 1% high-precision high-frequency 12/6/2MHz oscillator and a 4% precision low-frequency 128K oscillator, an external crystal oscillator, one UART, and one UART/SPI/IIC 3-to-1 communication port (SSI). To improve reliability and simplify customer circuitry, the SC92F836XB also integrates high-reliability circuitry such as a 4-level selectable voltage LVR and a 2.4V reference ADC voltage.

The SC92F836XB boasts excellent anti-interference performance and superb touch button performance, making it ideal for touch button and main control applications in various scenarios, such as smart home appliances and smart homes, IoT, wireless communication, game consoles, and other industrial and consumer applications.

2. Main Functions

Operating voltage: 2.4V~5.5V;

Operating temperature: -40 ~ 85ÿ;

Package:

SC92F8363B (SOP28/TSSOP28)
SC92F8362B (SOP20/TSSOP20)
SC92F8361B (SOP16) Core: 1TB
8051 **Flash ROM: 8**

Kbytes Flash ROM (MOVC addressable disabled)
(0000H~00FFH) Can be rewritten 10,000 times

IAP: Codeable option to 0K, 0.5K, 1K, or 8K. **EEPROM: 128** bytes, no erase required, 100,000 write cycles, over 10 years of storage life.

SRAM: 256 bytes internal + 768 bytes external

System Clock (fSYS): ÿ Built-in high-frequency 24MHz oscillator (fHRC) ÿ When used as the system clock source, fSYS can be programmed to be set to 12/6/2MHz.

ÿ Frequency error: Exceeds ±1% across (3.0V~5.5V) and (-20~85ÿ) application environments. ÿ Built-in high-frequency crystal oscillator circuit.

ÿ Can be connected to an external 2~16MHz oscillator ÿ When used as a system clock source, fSYS can be programmed to use one of four frequency divisions: external crystal oscillator / 1 / 2 / 4 / 12. ÿ Operating voltage range corresponding to the IC system clock (fSYS)

ÿ >12MHz @2.9~5.5V ÿ ÿ12MHz @2.4~5.5V

Built-in **128kHz low-frequency LRC** oscillator: ÿ Can be used as a clock source for a Base Timer, and can wake up a Stop Timer ÿ Can be used as a clock source for a WDT (Wake-up Timer)

ÿ Frequency error: spanning (4.0V ~ 5.5V) and (-20 ~ 85ÿ) application loops
The frequency error does not exceed ±4%.

Low Voltage Reset (LVR): ÿ Four reset voltage levels are available: 4.3V, 3.7V, 2.9V, ... 2.3V

ÿ The default value is the value selected by the user when writing the Code Option.

Flash programming and emulation: ÿ 2-wire JTAG programming and emulation interface

Interrupts (INT): ÿ Timer0, Timer1, Timer2, INT0~2, ADC, PWM,
There are 12 interrupt sources in total: UART, SSI, Base Timer, and TK.
External interrupts have 3 interrupt vectors, for a total of 13 interrupt ports, all of which are available.
The interrupt priorities can be set to rise edge, fall edge, and double edge interrupts.

Digital peripherals:

ÿ Up to 26 bidirectional, independently controllable I/O ports, each with independent configuration capabilities. Pull-up resistors; P0 and P2 port source drive capability is controlled in four levels; all I/Os have large sink current drive capability (70mA); 11-bit WDT with selectable clock division ratio; three standard 80C51 timers: Timer0, Timer1, and Timer2; Timer2 can implement capture function; six 10-bit PWMs with shared period and individually adjustable duty cycle, capable of simultaneously outputting three sets of complementary PWM waveforms with dead time; five I/Os can be used as 1/2 BIAS LCD COM outputs; one independent UART communication port (switchable I/O port); one UART/SPI/IIC three-way selector communication port SSI (switchable I/O port); integrated 16 × 16-bit hardware multiplier and divider.

Analog Peripherals: ÿ 23-channel low-power dual-mode touch circuit, configurable to high-sensitivity or high-reliability modes: ÿ
High-sensitivity mode is suitable for touch applications requiring high sensitivity, such as air-button touch and proximity sensing. ÿ High-reliability mode has strong anti-interference capabilities and can withstand 10V... Dynamic CS Testing
ÿ Enables 23-channel touch button and derivative functions ÿ
Highly flexible development software library support, low development difficulty ÿ Automated debugging software support, intelligent development ÿ Low-power touch mode, overall chip power consumption can be as low as 11uA when a single touch button wakes up ÿ 11-channel 12-bit ±2LSB
ADC ÿ Built-in 2.4V reference voltage ÿ Two reference voltage options for the ADC: VDD and internal 2.4V ÿ One internal ADC can directly measure
VDD voltage ÿ ADC conversion completion interrupt can be set

Power saving modes: ÿ IDLE Mode, can be woken up by any interrupt ÿ
STOP Mode, can be woken up by INT0~2, BaseTimer, and TK

92 Series Product Naming Rules

Name	SC	92	F	8	4	6	3	X	M	28	IN
Serial Number	ÿ	ÿ	ÿ	ÿ	ÿ	ÿ	ÿ	ÿ	ÿ	ÿ	ÿ

Serial Number	meaning
ÿ	Sinone Chip (abbreviation)
ÿ	Product series name
ÿ	Product type (F: Flash MCU)
ÿ	Serial Number: 7: GP series, 8: TK series
ÿ	ROM Size: 1 is 2K, 2 is 4K, 3 is 8K, 4 is 16K, 5 is 32K…
ÿ	Subsequence numbering: 0~9, A~Z
ÿ	Number of pins: 0: 8 pins, 1: 16 pins, 2: 20 pins, 3: 28 pins, 5: 32 pins, 6: 44 pins, 7: 48 pins, 8: 64pinÿ9ÿ100pinÿ
ÿ	Version number: (Default, B, C, D)
ÿ	Package types: (D: DIP; M: SOP; X: TSSOP; F: QFP; P: LQFP; Q: QFN; K: SKDIP)
ÿ	Number of pins
ÿ	Packaging methods: (U: tube; R: reel; T: tape)

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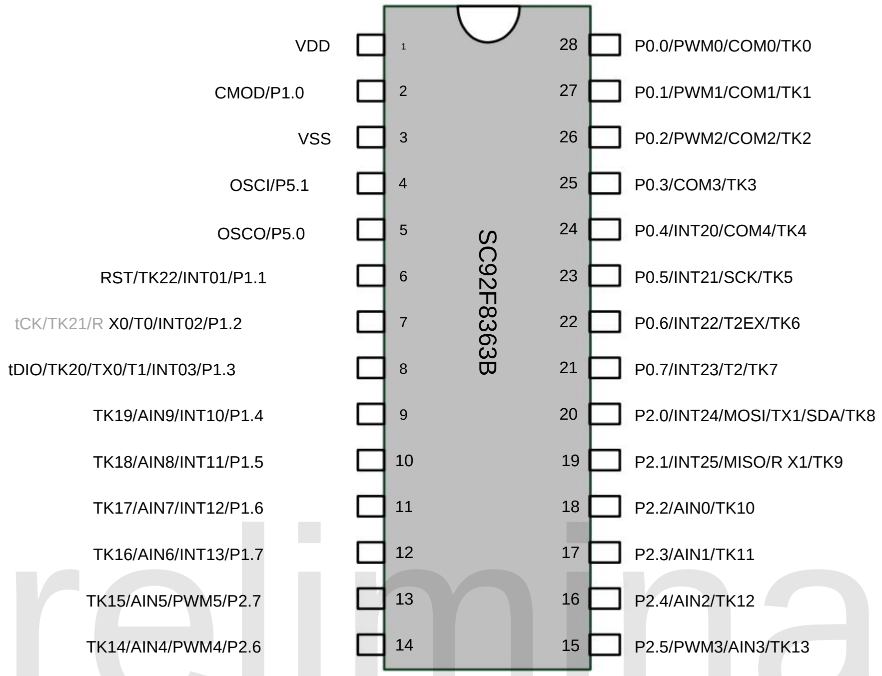
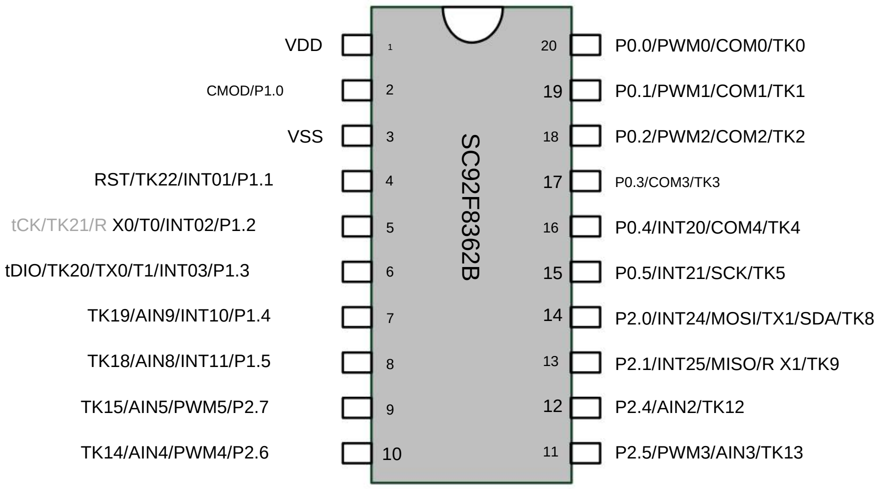
3- pin definition

3.1 Pin Configuration

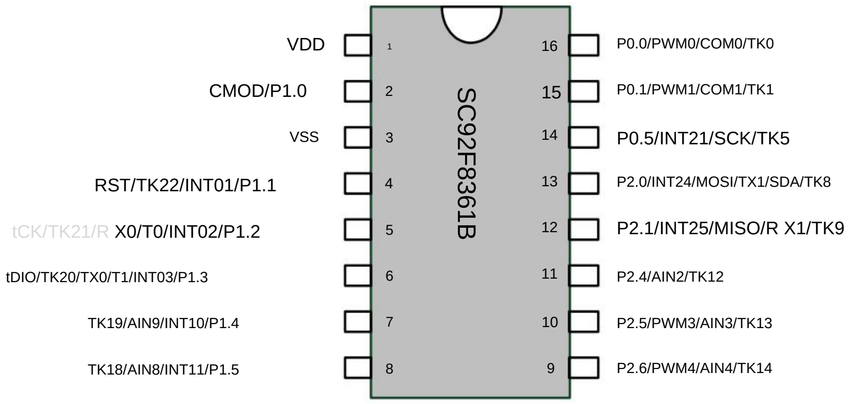
Special note: The TK20/TK21 and **TK** debugging communication ports of the SC92F836XB are multiplexed. If you need to use the **TK** debugging function, please try to avoid using **TK20/TK21!**



SC92F8363B Pin Configuration Diagram



SC92F8362B Pin Configuration Diagram



SC92F8361B Pin Configuration Diagram

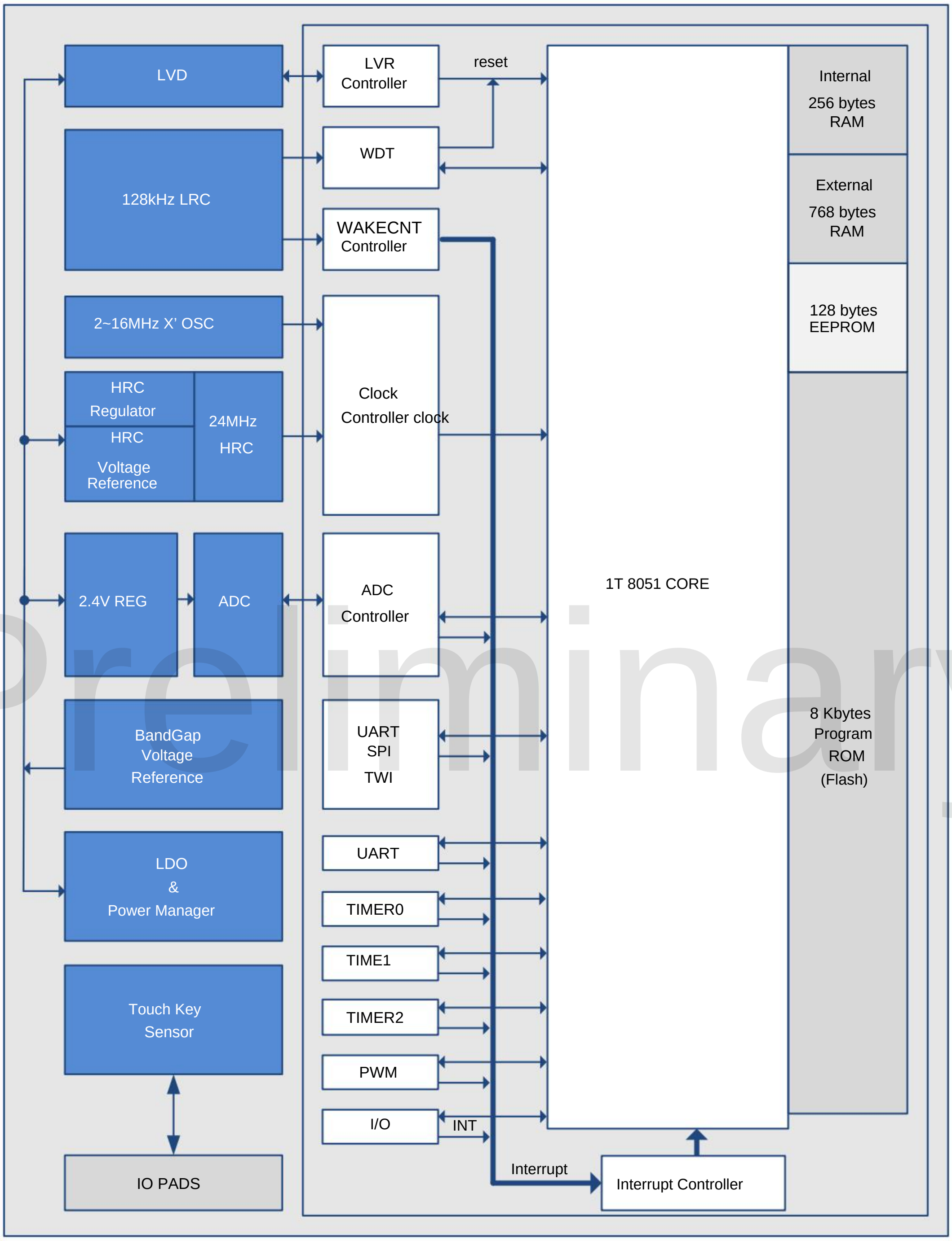
3.2 Pin Definitions

Pin number			Pin name	Pin type	Function Description
28PIN 20PIN 16PIN					
1	1	1	**VDD**	Power	
2	2	2	**P1.0/CMOD**	I/O	P1.0: GPIO P1.0 CMOD: Touch Key External Capacitor
3	3	3	**VSS**	Power grounding	
4	-	-	**P5.1/OSCI**	I/O	P5.1: GPIO P5.1 OSCI: External crystal oscillator input pin
5	-	-	**P5.0/OSCO**	I/O	P5.0: GPIO P5.0 OSCO: External crystal oscillator output pin
6	4	4	**P1.1/INT01/TK22/RST**	I/O	P1.1: GPIO P1.1 INT01: Input 1 of External Interrupt 0 TK22: Channel 22 of TK RST: External reset pin
7	5	5	**P1.2/INT02/T0/RX0/TK21** **/tCK**	I/O	P1.2: GPIO P1.2 INT02: Input 2 of External Interrupt 0 T0: Counter 0 External Input RX0: UART0 Receive TK21: Channel 21 of TK, used for adjusting **TK.** When testing this feature, please try to avoid using this **TK** channel! tCK: Programming and emulation port clock line
8	6	6	**P1.3/INT03/T1/TX0/TK20** **/tDIO**	I/O	P1.3: GPIO P1.3 INT03: Input 3 of External Interrupt 0 T1: Counter 1 External Input TX0: UART0 transmission TK20: Channel 20 of TK. If you need to use **TK** adjustment... When testing this feature, please try to avoid using this **TK** channel! tDIO: Programming and emulation port data cable
9	7	7	**P1.4/INT10/AIN9/TK19**	I/O	P1.4: GPIO P1.4 INT10: Input 0 of External Interrupt 1 AIN9: ADC Input Channel TK19: TK's Channel 19
10	8	8	**P1.5/INT11/AIN8/TK18**	I/O	P1.5: GPIO P1.5 INT11: Input 1 of External Interrupt 1 AIN8: ADC Input Channel 8 TK18: TK's Channel 18
11	-	-	**P1.6/INT12/AIN7/TK17**	I/O	P1.6: GPIO P1.6

					INT12: Input 2 of External Interrupt 1; AIN7: ADC Input Channel 7 TK17: TK's Channel 17
12	-	-	**P1.7/INT13/AIN6/TK16**	I/O	P1.7: GPIO P1.7 INT13: Input 3 of External Interrupt 1 AIN6: ADC Input Channel 6 TK16: TK's Channel 16
13	**9**	-	**P2.7/PWM5/AIN5/TK15**	I/O	P2.7: GPIO; P2.7 PWM5: PWM5 output port AIN5: ADC Input Channel 5 TK15: TK's Channel 15
14	**10**	**9**	**P2.6/PWM4/AIN4/TK14**	I/O	P2.6: GPIO; P2.6 PWM4: PWM4 output port AIN4: ADC Input Channel 4 TK14: TK's Channel 14
15	**11**	**10**	**P2.5/PWM3/AIN3/TK13**	I/O	P2.5: GPIO; P2.5 PWM3: PWM3 output port AIN3: ADC Input Channel 3 TK13: TK's Channel 13
16	**12**	**11**	**P2.4/AIN2/TK12**	I/O	P2.4: GPIO P2.4 AIN2: ADC Input Channel 2 TK12: Channel 12 of TK
17	-	-	**P2.3/AIN1/TK11**	I/O	P2.3: GPIO P2.3 AIN1: ADC Input Channel 1 TK11: TK's Channel 11
18	-	-	**P2.2/AIN0/TK10**	I/O	P2.2: GPIO P2.2 AIN0: ADC Input Channel 0 TK10: Channel 10 of TK
19	**13**	**12**	**P2.1/INT25/MISO/RX1/TK9**	I/O	P2.1: GPIO P2.1 INT25: Input 5 of External Interrupt 2 MISO: SPI Master Input Slave Output RX1: UART1 Receiver TK9: TK's Channel 9
20	**14**	**13**	**P2.0/INT24/MOSI/TX1/SYES/TK8**	I/O	P2.0: GPIO P2.0 INT24: Input 4 of External Interrupt 2 MOSI: SPI Master Output Slave Input TX1: UART1 Send SDA: TWI's SDA TK8: Channel 8 of TK
21	-	-	**P0.7/INT23/T2/TK7**	I/O	P0.7: GPIO P0.7 INT23: Input 3 of External Interrupt 2 T2: External Input of Counter 2 TK7: TK's Channel 7
22	-	-	**P0.6/INT22/T2EX/TK6**	I/O	P0.6: GPIO P0.6 INT22: Input 2 of External Interrupt 2 T2EX: External Capture Signal Input of Timer 2 TK6: TK's Channel 6
23	**15**	**14**	**P0.5/INT21/SCK/TK5**	I/O	P0.5: GPIO P0.5 INT21: Input 1 of External Interrupt 2 SCK: SCK of SPI and TWI TK5: TK's Channel 5
24	**16**	-	**P0.4/INT20/COM4/TK4**	I/O	P0.4: GPIO P0.4 INT20: Input 0 of External Interrupt 2 COM4: LCD driver common terminal COM4 TK4: TK's Channel 4
25	**17**	-	**P0.3/COM3/TK3**	I/O	P0.3: GPIO P0.3 COM3: LCD driver common terminal COM3 TK3: TK's Channel 3

26	**18**	-	**P0.2/PWM2/COM2/TK2**	I/O	P0.2: GPIO; P0.4: PWM2: PWM2 output port COM2: LCD driver common terminal COM2 TK2: TK's Channel 2
27	**19**	**15**	**P0.1/PWM1/COM1/TK1**	I/O	P0.1: GPIO P0.1 PWM1: PWM1 output port COM1: LCD driver common terminal COM1 TK1: Channel 1 of TK
28	**20**	**16**	**P0.0/PWM0/COM0/TK0**	I/O	P0.0: GPIO P0.0 PWM0: PWM0 output port COM0: LCD driver common terminal COM0 TK0: Channel 0 of TK

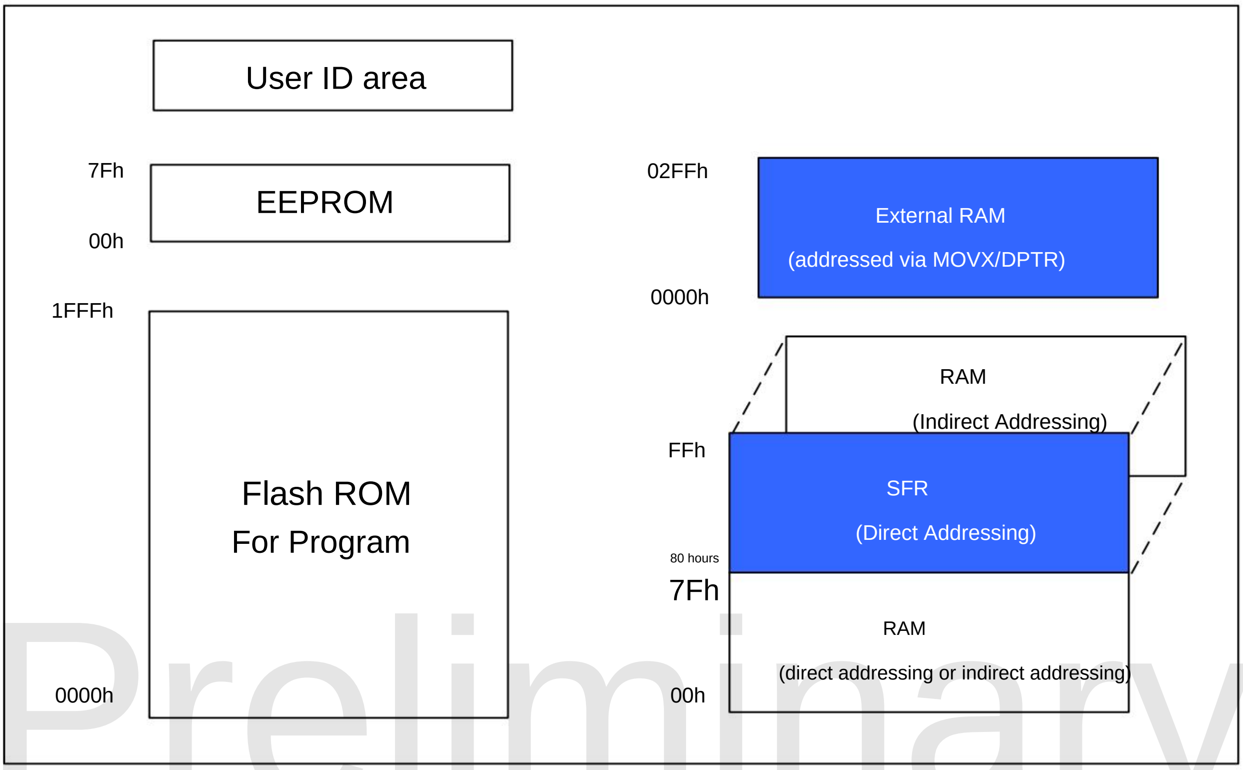
4 Internal block diagram



SC92F836XB BLOCK DIAGRAM

5. FLASH ROM and SRAM Structure

The Flash ROM and SRAM structures of the SC92F836XB are as follows:



Flash ROM and SRAM structure diagram

5.1 FLASH ROM

The SC92F836XB has an 8 Kbyte Flash ROM, with ROM addresses ranging from 0000H to 1FFFH. This 8 Kbyte Flash ROM can be rewritten 10,000 times and can be programmed and erased using a dedicated ICP programmer (SOC PRO52/DPT52/SC LINK) provided by SinOne. The address is...

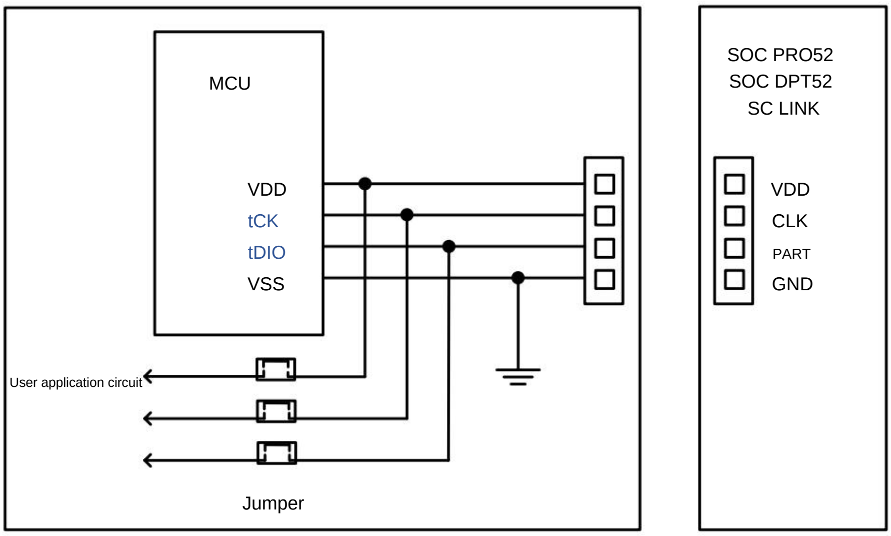
The MOVC instruction cannot address the 256-byte range of addresses 0000H to 00FFH.

EEPROM is a separate 8-kilobyte ROM segment, located at addresses 00H to 7FH, which can be read and written byte-by-byte within the program. For specific operating methods, please refer to section 20, EEPROM and IAP Operations.

User ID area: The user ID is written at the factory. Users can only read it. For specific operation methods, refer to the 20 EEPROM and IAP operation.

The SC92F836XB's 8 Kbyte Flash ROM provides blanking, programmable, verifiable, and erase functions, but does not provide read functionality. This Flash ROM and EEPROM typically do not require an erase operation before writing; new data can be overwritten directly. The Flash ROM is programmed via tDIO,

tCK, VDD, and VSS, with the specific connections as follows: SC92F836XB



ICP mode Flash Writer programming connection diagram

5.2 CUSTOMER OPTION area (User programming settings)

The SC92F836XB has a separate Flash area for storing the customer's power-on initial settings; this area is called the Code Option area.

When programming the IC, the user writes this part of the code into the IC. During the reset and initialization of the IC, the user will load this setting into the SFR as the initial setting.

Option-related SFR operation instructions:

Read and write operations on Option-related SFRs are controlled by two registers: OPINX and OPREG. The specific location of each Option SFR is determined by OPINX.

Confirmed, as shown in the table below:

Symbolic address		illustrate	7	6	5	4	3	2	1	0
OP_HRCR	83H@FFH	System Clock Change Register	OP_HRCR[7:0]							
OP_CTM0	C1H@FFH	Customer Option Register 0	ENWDT	ENXTL	SCLKS[1:0]		DISRST	DISLVR	LVRS[1:0]	
OP_CTM1	C2H@FFH	Customer Option Register 1	VREFS	XTLHF	·	·	IAPS[1:0]		·	·

OP_HRCR (83H@FFH) System Clock Change Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	OP_HRCR[7:0]							
read/write	Reading/Writing							
power-on initial value n		n	n	n	n	n	n	n

Position	The sign	illustrate
number 7~0	**OP_HRCR[7:0]** is used for internal high-frequency RC frequency tuning.	The center value of 10,000,000b corresponds to the HRC center frequency; a larger value indicates a faster frequency, and a smaller value indicates a faster frequency. The frequency has slowed down.

OP_CTM0 (C1H@FFH) Customer Option Register **0 (Read/Write)**

Bit number	7	6	5	4	3	2	1	0
symbol	ENWDT	ENXTL	SCLKS[1:0]		DISRST	DISLVR	LVRS[1:0]	
	Read/Write	Read/Write	Read/Write		Read/Write	Read/Write		
Power-on initial value n							Reading/Writing	
		n	n		n	n	n	

Position number	Bit symbol	illustrate
7	**INDIVIDUAL**	WDT switch 0: WDT invalid 1: WDT is valid (but the IC stops counting during IAP execution).
6	**ENXTL**	External high-frequency crystal oscillator selection switch 0: External high-frequency crystal oscillator is off; P5.0 and P5.1 are active. 1: External high-frequency crystal oscillator is on, P5.0 and P5.1 are ineffective.
5~4	**SCLKS[1:0]**	System clock frequency selection: 00: Reserved; 01: The system clock frequency is the high-frequency oscillator frequency divided by 2; 10: The system clock frequency is the high-frequency oscillator frequency divided by 4; 11: The system clock frequency is the high-frequency oscillator frequency divided by 12.
3	**DISRST**	IO/RST Reset Switching Control 0: P1.1 When the reset pin is used 1: P1.1 When using normal I/O pins
2	**DISLVR**	LVR Enable Settings 0: LVR is in normal use 1: LVR invalid
1~0	**LVRS [1:0]**	LVR Voltage Selection Control 11: 4.3 V Reset 10: 3.7V Reset 01: 2.9V Reset 00: 2.3V Reset

OP_CTM1 (C2H@FFH) Customer Option Register **1 (Read/Write)**

Bit number	7	6	5	4	3	2	1	0
symbol VREFS XTLHF Read/Write Read/Write Read/			-	-	IAPS[1:0] Read/		-	-
Power-on initial value n		Write	-	-	Write Read/Write		-	-
		n	x	x	n	n	x	x

Position number	Bit symbol	illustrate
7	**VREFS**	Reference voltage selection (initial value is loaded from **Code Option ,** user can modify the setting). 0: Set the ADC's VREF to VDD 1: Set the ADC's VREF to the internally accurate 2.4V.
6	**XTLHF**	External crystal oscillator control mode register 0: External crystal oscillator frequency <12MHz 1: External crystal oscillator frequency ÿ 12MHz
3~2	**IAPS[1:0]**	IAP Spatial Range Selection 00: IAP operations are prohibited in the Code area; only the EEPROM area can be used for data storage. 01: The last 0.5K Code area allows IAP operations (1E00H~1FFFH) 10: The last 1K Code area allows IAP operations (1C00H~1FFFH) 11: All Code areas are allowed for IAP operations (0000H~1FFFH)
5~4,1~0	-	reserve

5.2.1 OPTION- related SFR operation instructions

The read and write operations of Option-related SFRs are controlled by two registers: OPINX and OPREG. The specific location of each Option SFR is determined by OPINX.

It is confirmed that the write values for each Option SFR are determined by OPREG:

Symbol address description				Power-on initial values
OPINX	FEH Option pointer		OPINX[7:0]	00000000b
SINCERELY	FFH Option Register		SINCERE[7:0]	nnnnnnnnb

When operating on Option-related SFRs, the OPINX register stores the address of the relevant OPTION register, and the OPREG register stores the corresponding value.

For example, to configure OP_HRCR to 0x01, the specific steps are as follows:

C language routines:

```
OPINX = 0x83;      // Write the address of OP_HRCR to the OPINX register // Write
REAL = 0x01;       0x01 (the value to be written to the OP_HRCR register) to the OPREG register
```

Assembly routines:

```
MOV OPINX,#83H       Write the address of OP_HRCR to the OPINX register; Write 0x01
MOV STRAIGHT,#01H    (the value to be written to the OP_HRCR register) to the OPREG register.
```

Note: Do not write values outside the **Customer Option** SFR **address** to the **OPINX** register ! Doing so will cause system malfunctions!

5.3 SRAM

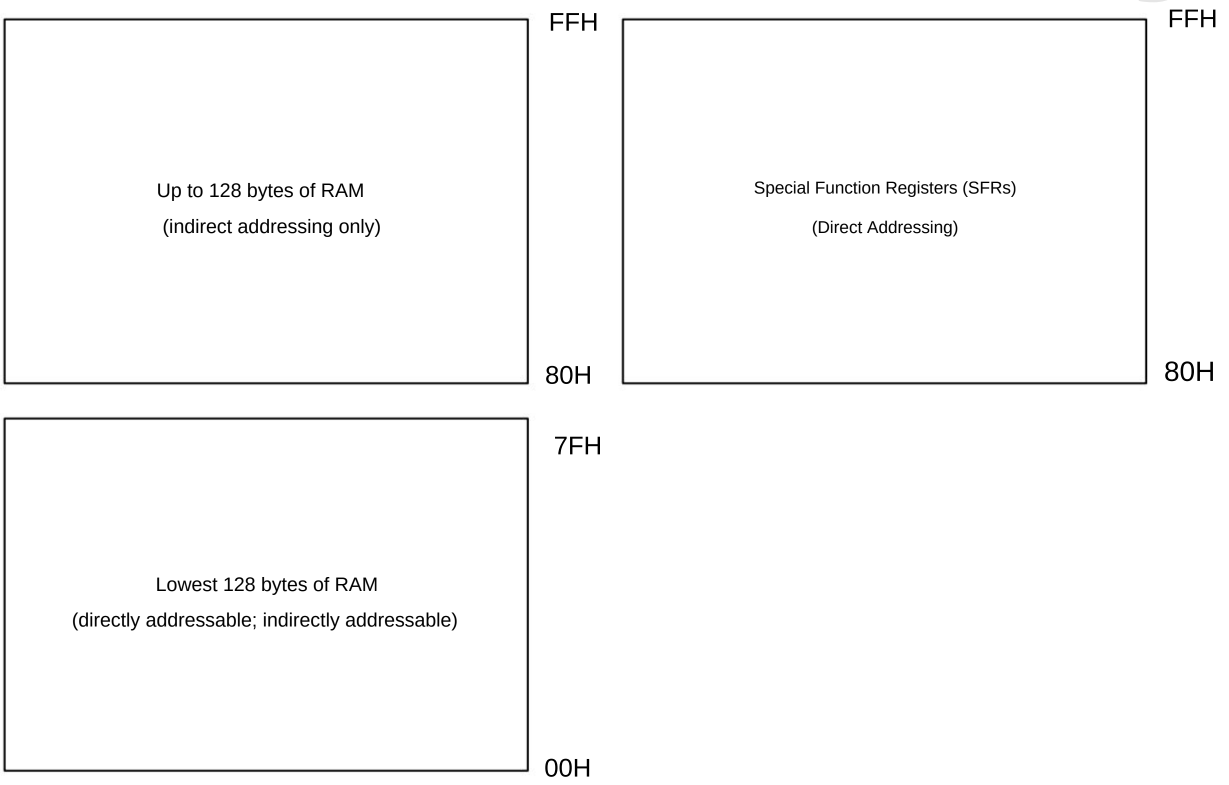
The SC92F836XB microcontroller integrates 1 Kbyte of SRAM, divided into 256 bytes of internal RAM and 768 bytes of external RAM. The internal RAM's address range is 00H~FFH, with the higher 128 bytes (addresses 80H~FFH) only allowing indirect addressing, and the lower 128 bytes (addresses 80H~FFH) only allowing indirect addressing. (00H~7FH) can be addressed directly or indirectly.

The address of the Special Function Register (SFR) is also 80H~FFH. However, the difference between the SFR and the internal high 128 bytes of SRAM is that the SFR register is a direct... It can be directly addressed, while the internal high 128 bytes of SRAM can only be indirectly addressed.

The external RAM has addresses from 0000H to 02FFH, but it needs to be addressed using the MOVX instruction.

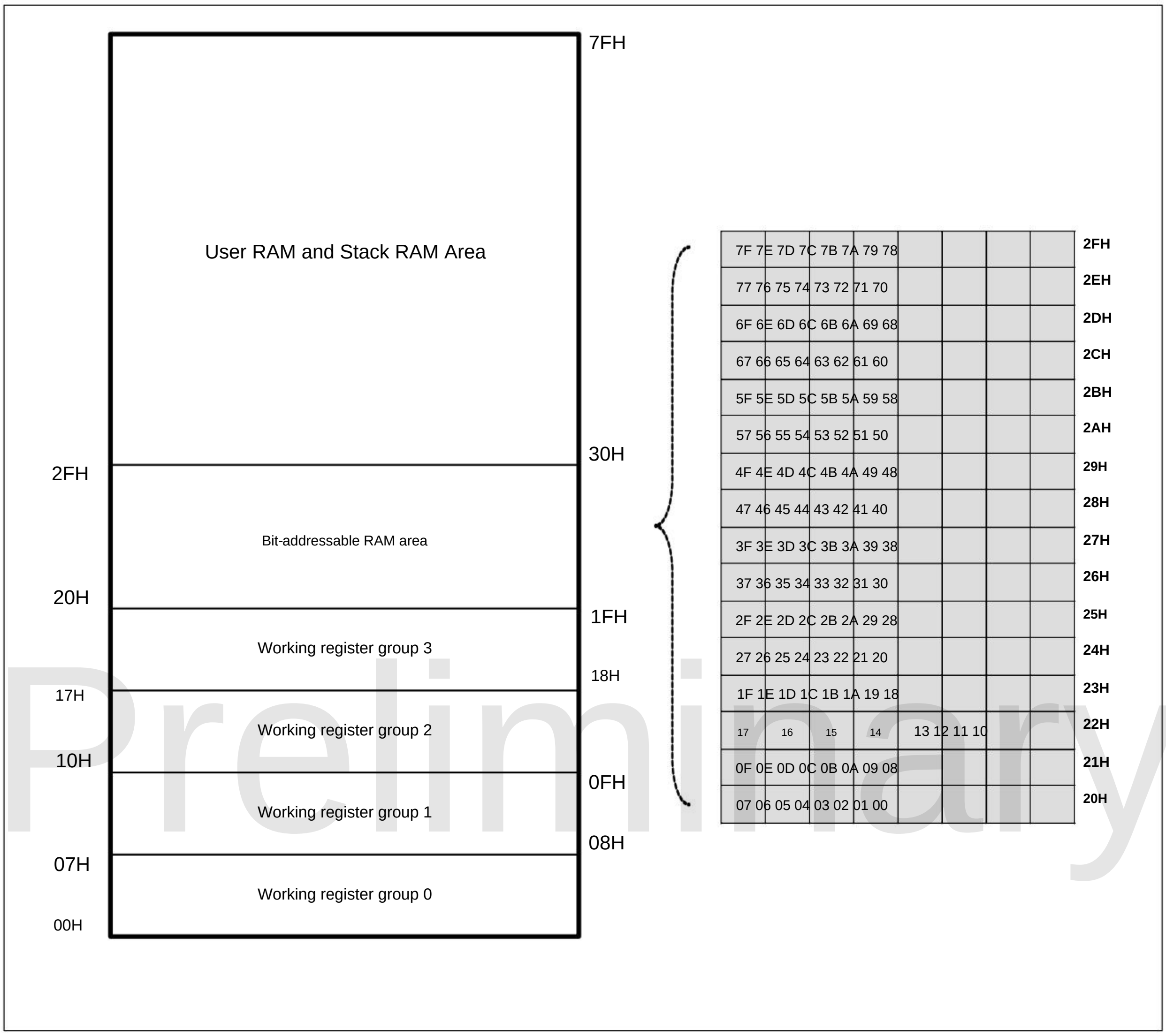
5.3.1 Internal 256 BYTES SRAM

The internal lower 128 bytes of SRAM can be divided into three parts: ÿ Working register group 0~3, address 00H~1FH. The combination of RS0 and RS1 in the program status word register PSW determines the currently used working register. Using working register group 0~3 can speed up the operation; ÿ Bit-addressable area 20H~2FH: This area can be used by the user as general RAM or bit-addressable RAM. When bit-addressable, the address of the bit is 00H~7FH (this address is bit-based, unlike the byte-based addressing of general SRAM), which can be distinguished by instructions in the program; ÿ User RAM and stack area: After the SC92F836XB is reset, the 8-bit stack pointer points to the stack area. Users usually set the initial value when initializing the program, and it is recommended to set it in the cell range of E0H~FFH.



Internal 256-byte RAM structure diagram

The internal lower 128 bytes of RAM are structured as follows:



SRAM Structure Diagram

5.3.2 External **768 BYTES SRAM**

The external 768 bytes of RAM can be accessed via MOVX @DPTR, A; alternatively, MOVX A, @Ri or MOVX @Ri, A can be used in conjunction with it.

The EXADH register is used to access the external 768-byte RAM: the EXADH register stores the high-order address of the external SRAM, and the Ri register stores the external SRAM address.

The lower 8 bits of the SRAM address.

EXADH (F7H) External **SRAM** access address high bit (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	-	-	-	EXADH [1:0]	
power-on initial value x		x	x	x	x	x	0	0

Position	Bit symbol	illustrate
number 1~0	**EXADH [1:0]** High bits of the external SRAM access address	
7~2	-	reserve

6. Special Function Registers (SFRs)

6.1 SFR video

The SC92F836XB series has some special function registers, which we call SFRs. These SFR registers are located at addresses 80H~FFH, and some are...

Bit-addressing is used for some registers, but not all registers can be bit-addressed. Registers that can be bit-addressed have addresses ending in "0" or "8". These registers are used when a single bit needs to be changed.

It is very convenient when dealing with single-digit values. All SFR special function registers must be addressed using direct addressing.

The names and addresses of the special function registers of the SC92F836XB are shown in the table below:

	0/8	1/9	2/A	3/B	4/C	5/D	6/E	7/F
F8h	·	·	·	·	CHKSUML	CHKSUMH	OPINX	SINCERELY
F0h	B	IAPKEY	IAPADL	IPADH	IAPADE	IAPDAT	IAPCT	EXADH
E8h	·	EXA0	EXA1	EXA2	EXA3	EXBL	EXBH	OPERCON
Yes	ACC	·	·	·	·	·	·	·
D8h	P5	P5CON	P5PH	·	PWMDTYB	PWMDTY3	PWMDTY4	PWMDTY5
D0h	PSW	PWMCFG	PWMCON	PWMPRD	PWMDTYA	PWMDTY0	PWMDTY1	PWMDTY2
C8h	T2CON	T2MOD	RCAP2L	RCAP2H	TL2	TH2	BTMCON	WDTCON
C0h	·	·	·	·	·	·	INT2F	INT2R
B8h	IP	IP1	INT0F	INT0R	INT1F	INT1R	·	·
B0h	·	·	·	·	·	·	·	·
A8h	IE	IE1	ADCCFG2	ADCCFG0	ADCCFG1	ADCCON	ADCVL	ADCVH
A0h	P2	P2CON	P2PH	·	·	·	·	·
98h	SCON	SBUF	P0CON	P0PH	P0VO	SSCON0	SSCON1	SSDAT
90h	P1	P1CON	P1PH	·	·	SSCON2	·	IOHCON
88h	TCON	TMOD	TL0	TL1	TH0	TH1	TMCON	OTCON
80 hours	P0	SP	DPL	VAT	·	·	·	PCON
	Bit addressable	Non-bit addressable						

illustrate:

1. An empty portion in the SFR register indicates that there is no RAM for this register, and it is not recommended for users to use it.
2. F1H~FFH in the SFRs are special function registers used for system configuration. User use of these registers may cause system malfunctions. Users should ensure proper initialization.

   During system operation, these registers cannot be cleared or subjected to other operations.

6.2 **SFR** Explanation

The specific explanation of the Special Function Registers (SFRs) is as follows:

Symbol	address	description	7	6	5	4	3	2	1	0	initial power-on value
P0	80H	P0 port data register	P07	P06	P05	P04	P03	P02	P01	P00	00000000b
SP	81H	Stack pointer	SP[7:0]								00000111b
DPL	82H	DPTR data pointer low bit	DPL[7:0]								00000000b
VAT	83H	DPTR data pointer high bit	VAT[7:0]								00000000b
PCON	87H	Power Management Control Register	SMOD	-	-	-	-	-	STOP	IDL	0xxxxx00b
TCON	88H	Timer Control Register	TF1	TR1	TF0	TR0	IE1	-	IE0	-	00000x0xb
TMOD	89H	Timer Operating Mode Register	-	C/T1	M11	M01	-	C/T0	M10	M00	x000x000b
TL0	8AH	Timer 0 Low 8 Bits	TL0[7:0]								00000000b
TL1	8BH	Timer 1 Low 8 Bits	TL1[7:0]								00000000b
TH0	8CH	Timer 0 High 8 Bits	TH0[7:0]								00000000b
TH1	8DH	Timer 1 High 8 Bits	TH1[7:0]								00000000b
TMCON	8EH	Timer Frequency Control Register	-	-	-	-	-	T2FD	T1FD	T0FD	xxxxx000b
OTCON	8FH	Output Control Register	SSMOD[1:0]		-	-	VID[1:0]		-	-	00xx00xxb
P1	90H	P1 port data register	P17	P16	P15	P14	P13	P12	P11	P10	00000000b
P1CON	91H	P1 Port Input/Output Control Register	P1C7	P1C6	P1C5	P1C4	P1C3	P1C2	P1C1	P1C0	00000000b
P1PH	92H	P1 port pull-up resistor control register	P1H7	P1H6	P1H5	P1H4	P1H3	P1H2	P1H1	P1H0	00000000b
SSCON2	95H	SSI Control Register 2	SSCON2[7:0]								00000000b
IOHCON	97H	IOH Set Register	P2H[1:0]		P2L[1:0]		P0H[1:0]		P0L[1:0]		00000000b
SCON	98H	Serial Port Control Register	SM0	SM1	SM2	REN	TB8	RB8	OF	RI	00000000b
SBUF	99H	Serial Port Data Buffer Register	SBUF[7:0]								00000000b
P0CON	9AH	P0 Port Input/Output Control Register	P0C7	P0C6	P0C5	P0C4	P0C3	P0C2	P0C1	P0C0	00000000b
P0PH	9BH	P0 port pull-up resistor control register	P0H7	P0H6	POH5	P0H4	P0H3	P0H2	P0H1	P0H0	00000000b
P0VO	9CH	P0 port LCD voltage output register	-	-	-	P04VO	P03VO	P02VO	P01VO	P00VO	xxx00000b
SSCON0	9DH	SSI Control Register 0	SSCON0[7:0]								00000000b
SSCON1	9EH	SSI Control Register 1	SSCON1[7:0]								00000000b
SSDAT	9FH	SSI Data Register	SSD[7:0]								00000000b
P2	A0H	P2 port data register	P27	P26	P25	P24	P23	P22	P21	P20	00000000b
P2CON	A1H	P2 Port Input/Output Control Register	P2C7	P2C6	P2C5	P2C4	P2C3	P2C2	P2C1	P2C0	00000000b
P2PH	A2H	P2 port pull-up resistor control register	P2H7	P2H6	P2H5	P2H4	P2H3	P2H2	P2H1	P2H0	00000000b
IE	A8H	Interrupt Enable Register	YES	EADC	ET2	EUART	ET1	EINT1	ET0	EINT0	00000000b
IE1	A9H	Interrupt Enable Register 1	-	-	-	ETC	ONE2	EBTM	EPWM	THEY	xxx00000b
ADCCFG2	AAH	ADC Setting Register 2	-	-	-	-	LOWSP	ADCCK[2:0]			xxxx0000b
ADCCFG0	ABH	ADC setting register 0	EAIN7	EAIN6	EAIN5	EAIN4	EAIN3	EAIN2	EAIN1	EAIN0	00000000b
ADCCFG1	ACH	ADC Setting Register 1	-	-	-	-	-	-	EAIN9	EAIN8	xxxxxx00b
ADCCON	ADH	ADC Control Register	ADCEN	ADCS	EOC/ ADCIF	ADCIS[4:0]					00000000b
ADCVL	AEH	ADC Result Register	ADCV[3:0]				-	-	-	-	0000xxxxb
ADCVH	AFH	ADC Result Register	ADCV[11:4]								00000000b
IP	B8H	Interrupt Priority Control Register	-	IPADC	IPT2	PORT	IPT1	POINT1	IPT0	IPINT0	x0000000b
IP1	B9H	Interrupt Priority Control Register 1	-	-	-	IPTK	POINT2	IPBTM	IPPWM	IPSSI	xxx00000b
INT0F	BAH	INT0 Falling Edge Interrupt Control Register	-	-	-	-	INT0F3	INT0F2	INT0F1	-	xxxx000xb
INT0R	BBH	INT0 Rising Edge Interrupt Control Register	-	-	-	-	INT0R3	INT0R2	INT0R1	-	xxxx000xb
INT1F	BCH	INT1 Falling Edge Interrupt Control Register	-	-	-	-	INT1F3	INT1F2	INT1F1	INT1F0	xxxx0000b
INT1R	BDH	INT1 Rising Edge Interrupt Control Register	-	-	-	-	INT1R3	INT1R2	INT1R1	INT1R0	xxxx0000b
INT2F	C6H	INT2 Falling Edge Interrupt Control Register	-	-	INT2F5	INT2F4	INT2F3	INT2F2	INT2F1	INT2F0	xx000000b
INT2R	C7H	INT2 Rising Edge Interrupt Control Register	-	-	INT2R5	INT2R4	INT2R3	INT2R2	INT2R1	INT2R0	xx000000b
T2CON	C8H	Timer 2 Control Register	TF2	EXF2	RCLK	TCLK	EXEN2	TR2	C/T2	CP/RL2	00000000b
T2MOD	C9H	Timer 2 Operating Mode Register	-	-	-	-	-	-	T2OE	DCEN	xxxxxx00b
RCAP2L	CAH	Timer 2 Reload Low 8 Bits	RCAP2L[7:0]								00000000b
RCAP2H	CBH	Timer 2 Overload High 8 Bits	RCAP2H[7:0]								00000000b
TL2	CCH	Timer 2 Low 8 Bits	TL2[7:0]								00000000b
TH2	CDH	Timer 2 High 8 Bits	TH2[7:0]								00000000b
BTMCON	CEH	Low-Frequency Timer Control Register	ENBTM	BTMIF	-	-	BTMFS[3:0]				00xx0000b
WDTCON	CFH	WDT control register	-	-	-	CLRWDT	-	WDTCKS[2:0]			xxx0x000b
PSW	D0H	Program Status Word Register	EN	AC	F0	RS1	RS0	OV	F1	P	00000000b

PWMCFG	D1H	PWM Setting Register	PWMCKS[1:0]		INV5	INV4	INV3	INV2	INV1	INV0	00000000b
PWMCON	D2H	PWM Control Register	ENPWM	PWMIF	ENPWM5	ENPWM4	ENPWM3	ENPWM2	ENPWM1	ENPWM0	00000000b
PWMPRD	D3H	PWM Period Setting Register	PWMPRD[9:2]								00000000b
PWMDTYA	D4H	PWM Duty Cycle Setting Register A	PWMPRD[1:0]		PDT2[1:0]		PDT1[1:0]		PDT0[1:0]		00000000b
PWMDTY0	D5H	PWM0 Duty Cycle Setting Register	PDT0[9:2]								00000000b
PWMDTY1	D6H	PWM1 Duty Cycle Setting Register	PDT1[9:2]								00000000b
PWMDTY2	D7H	PWM2 Duty Cycle Setting Register	PDT2[9:2]								00000000b
P5	D8H	P5 port data register	-	-	-	-	-	-	P51	P50	xxxxxx00b
P5CON	D9H	P5 Port Input/Output Control Register	-	-	-	-	-	-	P5C1	P5C0	xxxxxx00b
P5PH	DAH	P5 port pull-up resistor control register	-	-	-	-	-	-	P5H1	P5H0	xxxxxx00b
PWMDTYB	DCH	PWM duty cycle setting register B PWMMOD		-	PDT5[1:0]		PDT4[1:0]		PDT3[1:0]		0x000000b
PWMDTY3	DDH	PWM3 Duty Cycle Setting Register / PWM dead time configuration register	PDT3[9:2]								00000000b
PWMDTY4	DEH	PWM4 Duty Cycle Setting Register	PDT4[9:2]								00000000b
PWMDTY5	DFH	PWM5 Duty Cycle Setting Register	PDT5[9:2]								00000000b
ACC	E0H	Accumulator	ACC[7:0]								00000000b
EXA0	E9H	Extended Accumulator 0	EXA[7:0]								00000000b
EXA1	EAH	Extended Accumulator 1	EXA[15:8]								00000000b
EXA2	EBH	Extended Accumulator 2	EXA[23:16]								00000000b
EXA3	ECH	Extended Accumulator 3	EXA[31:24]								00000000b
EXBL	EDH	Extended B Register L	EXB [7:0]								00000000b
EXBH	EEH	Extended B Register H	EXB [15:8]								00000000b
OPERCON	EFH	Operation Control Register	OPERS	MD	-	-	-	-	-	CHKSUMS	00xxxxx0b
B	F0H	B register	B[7:0]								00000000b
IAPKEY	F1H	IAP Protection Register	IAPKEY[7:0]								00000000b
IAPADL	F2H	IAP Write to Low-order Address Register	IAPDR[7:0]								00000000b
IPADH	F3H	IAP Write to High-Number Register	-	-	-	IAPADR[12:8]					xxx00000b
IAPADE	F4H	IAP Write to Extended Address Register	IAPADER[7:0]								00000000b
IAPDAT	F5H	IAP Data Register	IAPDAT[7:0]								00000000b
IAPCT	F6H	IAP Control Register	-	-	-	-	PAYTIMES[1:0]		CMD[1:0]		xxxx0000b
EXADH	F7H	External SRAM Operation Address High Bit	-	-	-	-	-	-	EXADH [1:0]		xxxxxx00b
CHKSUML	FCH	Check Sum Result Register Low Byte	CHKSUML[7:0]								00000000b
CHKSUMH	FDH	Check Sum Result Register High Byte	CHKSUMH[7:0]								00000000b
OPINX	FEH	Option pointer	OPINX[7:0]								00000000b
SINCERELY	FFH	Option Register	SINCERE[7:0]								nnnnnnnnb

6.2.1 Introduction to Commonly Used Special Function Registers in the 8051 CPU Core

Program counter **PC**

The program counter (PC) is not part of the SFR registers. The PC is 16 bits long and is a register used to control the order of instruction execution. This occurs when the microcontroller is powered on or reset. After that, the PC value is 0000H, which means that the microcontroller program starts executing from address 0000H.

Accumulator **ACC (E0H)**

The accumulator (ACC) is one of the most frequently used registers in the 8051 core microcontroller, and its mnemonic is A in the instruction set. It is commonly used to store references. Operands and results of arithmetic or logical operations.

Register **B (F0H)**

Register B must be used in conjunction with accumulator A in multiplication and division operations. The multiplication instruction MUL A, B multiplies the 8 bits of data in accumulator A and register B. When signed numbers are multiplied, the least significant byte of the resulting 16-bit product is stored in A, and the most significant byte is stored in B. The division instruction DIV A, B divides A by B, and the integer... The quotient is stored in register A, and the remainder is stored in register B. Register B can also be used as a general-purpose temporary register.

Stack pointer **SP (81H)**

The stack pointer is an 8-bit special-purpose register that indicates the location of the top of the stack in general-purpose RAM. After a microcontroller reset, the initial value of SP is... 07H indicates that the stack will start increasing upwards from 08H. 08H~1FH are working register groups 1~3.

PSW (D0H) Program Status Word Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	EN	AC	F0	RS1	RS0	OV	F1	P
read/write Read/write Power-on		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	0	0	0	0	0

Position number	Bit symbol	illustrate
7	**EN**	Flag 1: When there is a carry in the highest bit of an addition operation, or a borrow in the highest bit of a subtraction operation. 0: No carry in the highest bit of addition, or no borrow in the highest bit of subtraction.
6	**AC**	Carry auxiliary flag (can be easily adjusted during BCD code addition and subtraction operations) 1: When there is a carry in bit 3 during addition, or a borrow in bit 3 during subtraction. 0: No borrowing or carrying
5	**F0**	User flag
4~3	RS1 and RS0 working register group selection bits:	
2	**OV**	Overflow flag
1	**F1**	F1 logo User-defined flags
0	**P**	Parity flag. This flag indicates the parity value of the number of 1s in the accumulator ACC. 1: The number of 1s in ACC is odd. 0: The number of 1s in ACC is even (including 0).

RS1	RS0 is currently	using working register groups 0-3.
0		Group 0 (00H~07H)
0	Group 0, 1	(08H~0FH)
1	0	Group 2 (10H~17H)
1	1	Group 3 (18H~1FH)

Data pointer **DPTR** (82H, 83H)

The data pointer DPTR is a 16-bit special purpose register, consisting of the lower 8 bits DPL (82H) and the higher 8 bits DPH (83H). DPTR is...

It is the only register in the traditional 8051 core microcontroller that can be directly operated on in 16-bit mode, and DPL and DPH can also be operated on by byte.

7. Power, Reset, and Clock

7.1 Power Supply Circuit

The SC92F836XB power supply core includes circuits such as BG, LDO, POR, and LVR, enabling reliable operation within a 2.4~5.5V range. Furthermore, The IC incorporates a calibrated, precise 2.4V voltage, which can be used as an internal reference voltage for the ADC. Users can find the specific settings in the 18 analog-to-digital converter ADC documentation . content.

7.2 Power-on reset process

After the SC92F836XB is powered on, the following process will occur before the client software executes:

ÿ Reset Phase

ÿ Information Retrieval Stage

ÿ Normal operation phase

7.2.1 Reset Phase

This means that the SC92F836XB will remain in a reset state until the voltage supplied to the SC92F836XB exceeds a certain voltage, at which point its internal operation will become active. The duration of the reset phase depends on the rise rate of the external power supply. The reset phase will only be completed after the external power supply reaches the built-in POR voltage.

7.2.2 Information Retrieval Stage

The SC92F836XB contains an internal warm-up counter. During the reset phase, this warm-up counter is kept cleared to 0 until the voltage exceeds the POR. After the voltage is applied, the internal RC oscillator starts oscillating, and the preheating counter begins counting. Once the internal preheating counter has counted to a certain number, it will start counting every certain number of seconds. Measuring an HRC clock cycle will read a byte of data from the IFB (including the Code Option) in the Flash ROM and store it in the internal system register. The reset signal will not end until the preheating is complete.

7.2.3 Normal Operation Phase

After the information loading phase ends, the SC92F836XB begins reading instruction codes from Flash, thus entering the normal operation phase. The LVR voltage value at this time... This is the setting value that the user writes into the Code Option.

7.3 Reset Method

The SC92F836XB has four reset methods: ÿ External RST reset ÿ Low voltage reset (LVR) ÿ Power-on reset (POR) ÿ Watchdog timer (WDT) reset.

7.3.1 External **RST** Reset

External RST reset is achieved by sending a reset pulse signal of a certain width to the SC92F836XB via an external RST signal.

Before programming, users can configure the P1.1 pin as RST (reset pin) by configuring the Customer Option item in the host computer programming software.

7.3.2 Low Voltage Reset **LVR**

The SC92F836XB has a built-in low-voltage reset circuit. The reset threshold voltage has four options: 4.3V, 3.7V, 2.9V, and 2.3V, with the default being... The value Default is the Option value written by the user.

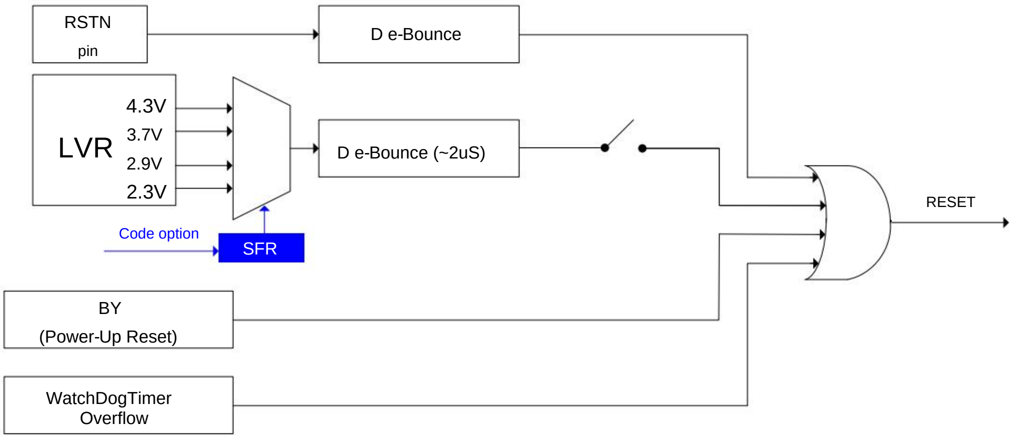
OP_CTM0 (C1H@FFH) Customer Option Register **0 (Read/Write)**

Bit number	7	6	5	4	3	2	1	0
symbol ENWDT ENXTL	Read/Write	Read/Write Read/Write	SCLKS[1:0] Read/		DISRST DISLVR		LVRS[1:0]	
Power-on initial value n			Write		Reading/Writing	Reading/Writing	Reading/Writing	
		n	n		n	n	n	

Position number	Bit symbol	illustrate
2	**DISLVR**	LVR Enable Settings 0: LVR is in normal use 1: LVR invalid
1~0	**LVRS [1:0]**	LVR Voltage Selection Control 11: 4.3 V Reset 10: 3.7V Reset

		01: 2.9V Reset 00: 2.3V Reset

The reset circuit diagram of SC92F836XB is as follows:



SC92F836XB Reset Circuit Diagram

7.3.3 Power-on reset **(POR)**

The SC92F836XB has an internal power-on reset circuit. When the power supply voltage VDD reaches the POR reset voltage, the system automatically resets.

7.3.4 Watchdog Timer Reset **(WDT)**

The SC92F836XB has a WDT (Wave Detector) whose clock source is an internal 128kHz oscillator. The user can select this via the programmer's Code Option.

Enable watchdog reset function.

OP_CTM0 (C1H@FFH) Customer Option Register **0 (Read/Write)**

Bit number	7 Symbol	6	5 4		3	2	1	0
ENWDT	ENXTL Read/Write	Read/Write	SCLKS[1:0] Read/Write		DISRST DISLVR		LVRS[1:0]	
Power-on initial value n		Reading/Writing			Reading/Writing	Reading/Writing	Reading/Writing	
		n	n		n	n	n	

Position number	Bit symbol	illustrate
7	**INDIVIDUAL**	WDT switch (this bit is loaded by the system from the value set by the user's Code Option) 1: WDT begins work 0: WDT Off

WDTCON (CFH) Watchdog Control Register (Read/Write)

Bit number	7 symbol read/	6	5	4	3	2	1	0
write	-	-	-	CLRWDT	-	WDTCKS[2:0]		
power-	-	-	-	Reading/Writing	-	Reading/Writing		
on initial value x		x	x	0	x	0	0	0

Position number	Bit symbol	illustrate
4	**CLRWDT**	WDT clears the "0" bit (writing 1 is valid). 1: The WDT counter starts counting from 0. This bit is automatically set to 0 by the system hardware.
2~0	**WDTCKS [2:0]** Watchdog Clock Selection	WDTCKS[2:0] WDT Overflow Time 000 500ms

		001	250ms
		010	125ms
		011	62.5ms
		100	31.5ms
		101	15.75ms
		110	7.88ms
		111	3.94ms
7~5,3	-	reserve	

7.3.5 Reset to initial state

When the SC92F836XB is in reset state, most registers return to their initial state. The watchdog timer (WDT) is disabled, and the PORT is... The register is FFh. The program counter PC is initialized to 0000h, and the stack pointer SP is initialized to 07h. A "warm boot" reset (such as WDT, LVR) is used. (etc.) will not affect SRAM; SRAM values will always remain as they were before the reset. Loss of SRAM content will occur when the power supply voltage drops below a certain level, causing the RAM to be unable to retain the data. until.

The initial power-on reset values for the SFR registers are shown in the table below:

SFR Name Initial Value		**SFR** name	initial value
ACC 00000000b B 00000000b PSW 00000000b		P1PH	00000000b
SP 00000111b DPL 00000000b DPH		P2	00000000b
00000000b PCON 0xxxxx00b ADCCFG0		P2CON	00000000b
00000000b ADCCFG1 xxxxxx00b ADCCFG2		P2PH	00000000b
xxxx0000b ADCCON 00000000b ADCVH		P5	xxxxxx00b
00000000b ADCVL 0000xxxxb BTMCON		P5CON	xxxxxx00b
00xx0000b IAPADE 00000000b IAPADH		P5PH	xxxxxx00b
xxx00000b IAPADL 00000000b IAPCTL xxxx0000b		PWMCFG	00000000b
IAPDAT 00000000b IAPKEY 00000000b IE		PWMCON	00000000b
00000000b IE1 xxx00000b INT0R xxxx000xb INT1R		PWMDTYA	xx000000b
xxxx0000b INT2R xx000000b INT0F xxxx000xb		PWMDTY0	00000000b
INT1F xxxx0000b INT2F xx000000b IP x0000000b		PWMDTY1	00000000b
IP1 xxx00000b OPINX 00000000b OPREG		PWMDTY2	00000000b
nnnnnnnnb EXADH xxxxxx00b OTCON 00xx00xxb		PWMPRD	00000000b
00000000b IOHCON 00000000b P0 00000000b		PWMDTYB	0x000000b
P0CON 00000000b P0PH P0VO xxx00000b P1		PWMDTY3	00000000b
00000000b		PUBLIC HOUSE4	00000000b
		PWMDTY5	00000000b
		RCAP2H	00000000b
		RCAP2L	00000000b
		SBUF	00000000b
		SCON	00000000b
		SSDAT	00000000b
		SSCON0	00000000b
		SSCON1	00000000b
		SSCON2	00000000b
		TCON	00000x0xb
		TMCON	xxxxx000b
		TMOD	x000x000b
		TH0	00000000b
		TL0	00000000b
		TH1	00000000b
		TL1	00000000b
		T2CON	00000000b
		TH2	00000000b
		TL2	00000000b
		T2MOD	xxxxxx00b
		WDTCON	xxx0x000b
		CHKSUMH	00000000b
		CHKSUML	00000000b

P1CON	00000000b	-	-

7.4 High-Frequency System Clock Circuit

The SC92F836XB integrates a high-precision HRC oscillator with adjustable frequency and a crystal oscillator circuit, allowing users to choose one for operation.

This is the system clock. The HRC is precisely calibrated to 24MHz@5V/25ÿ at the factory. Users can set the system clock via the programmer's Code Option.

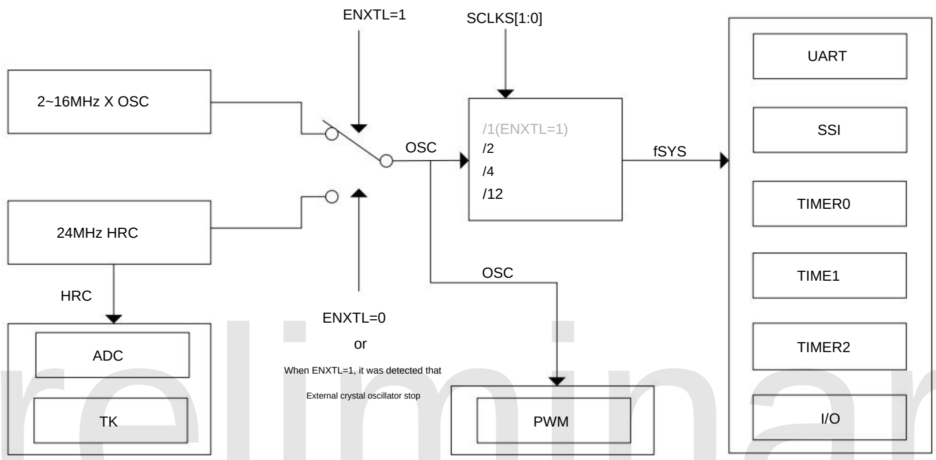
Set to 12/6/2MHz for use. The calibration process filters out the impact of manufacturing deviations on accuracy. This HRC is affected by the ambient temperature and operating conditions.

Voltage fluctuations can cause some drift. For voltage drift (3.0V~5.5V) and temperature drift (-20ÿ~85ÿ), the deviation is generally within ±1%.

To enhance system reliability, the SC92F836XB has a built-in system clock monitoring circuit. When the user selects a crystal oscillator as the system clock source...

Furthermore, when the crystal oscillator circuit stops oscillating, the system clock source will be automatically switched to the built-in HRC and will remain in this state until the next reset.

Note: The clock source for **the ADC** and dual-mode touch circuit is fixed at **fHRC =** 24MHz and will not change with the switching of internal and external system clocks.



SC92F836XB Internal Clock Relationship

OP_CTM0 (C1h@FFH) Customer Option Register **0 (Read/Write)**

Bit number	7	6	5	4	3	2	1	0
symbol ENWDT ENXTL Read/Write Read/Write Power-on			SCLKS[1:0] Read/Write		DISRST DISLVR Read/Write Read/Write		LVRS[1:0]	
initial value n		Reading/Writing					Reading/Writing	
		n	n		n	n	n	

Position number	Bit symbol	illustrate
6	ENXTL	External high-frequency crystal oscillator selection switch 0: External high-frequency crystal oscillator is off; P5.0 and P5.1 are active. 1: External high-frequency crystal oscillator is on, P5.0 and P5.1 are ineffective. Note: The clock source for **the ADC** and dual-mode touch circuit is fixed at **fHRC =** 24MHz and will not change with frequency. It changes due to the switching of internal and external system clocks.
5~4	SCLKS[1:0] System clock frequency	selection: 00: The system clock frequency is the high-frequency oscillator frequency divided by 1, and is only valid when ENXTL=1; 01: The system clock frequency is the high-frequency oscillator frequency divided by 2; 10: The system clock frequency is the high-frequency oscillator frequency divided by 4; 11: The system clock frequency is the high-frequency oscillator frequency divided by 12.

OP_CTM1 (C2h@FFH) Customer Option Register **1 (Read/Write)**

Bit number	7	6	5	4	3	2	1	0
symbol VREFS XTLHF Read/Write Read/Write Power-on initial			-	-	IAPS[1:0] Read/		-	-
value n		Reading/Writing	-	-	Write Read/Write		-	-
		n	x	x	n	n	x	x

Position number	Bit symbol	illustrate
6	**XTLHF**	External crystal oscillator control mode register 0: External crystal oscillator frequency <12MHz 1: External crystal oscillator frequency ÿ 12MHz

The SC92F836XB has a special feature: users can modify the SFR value to adjust the HRC frequency within a certain range. Users can configure...

This is achieved by setting the OP_HRCR register. For instructions on configuring this register, please refer to section 5.2.1. " Option-related SFR Operations".

OP_HRCR (83h@FFH) System Clock Change Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	OP_HRCR[7:0]							
read/	Reading/Writing							
write power-on initial value n		n	n	n	n	n	n	n

Position	Bit sign	illustrate
number 7~0	OP_HRCR[7:0] **HRC** frequency change register	Users can change the frequency fHRC of the high-frequency oscillator by modifying the value of this register , thereby changing... The system clock frequency fSYS of the variable IC : 1. The initial value of OP_HRCR[7:0] after power-on: OP_HRCR[s] is a fixed value, which is... Ensure fHRC is 24MHz; the OP_HRCR[s] may vary for each IC. 2. When the initial value is OP_HRCR[s], the IC's system clock frequency fSYS can be adjusted via Option. The item is set to an accurate 12/6/2MHz, and for every 1 change in OP_HRCR [7:0], the fSYS frequency is adjusted. The rate changed by approximately 0.23%. The relationship between OP_HRCR [7:0] and the output frequency of fSYS is as follows:

OP_HRCR [7:0] value	fSYS actual output frequency (taking 12MHz as an example)
OP_HRCR [s]-n	12000*(1-0.23%*n)kHz
...	....
OP_HRCR [s]-2	12000*(1-0.23%*2) = 11944.8kHz
OP_HRCR [s]-1	12000*(1-0.23%*1) = 11972.4kHz
OP_HRCR [s]	12000kHz
OP_HRCR [s]+1	12000*(1+0.23%*1) = 12027.6kHz
OP_HRCR [s]+2	12000*(1+0.23%*2) = 12055.2kHz
...	...
OP_HRCR [s]+n	12000*(1+0.23%*n)kHz

Notice:

1. After each power-on, the value of OP_HRCR[7:0] is closest to the high-frequency oscillator frequency fHRC.
The value is 24MHz; users can use EEPROM to correct the HRC value after each power-on to ensure...
The IC's system clock frequency, fSYS, operates at the frequency required by the user.
2. To ensure reliable IC operation, the IC's maximum operating frequency should not exceed 10% of 12MHz.
That is, 13.2MHz; 3.
Please confirm that the change of HRC frequency will not affect other functions.

7.5 Low-frequency oscillator and low-frequency clock timer

The SC92F836XB has a built-in 128kHz RC oscillation circuit, which serves as the clock source for the low-frequency clock timer (Base Timer) and the WDT.

Enabling the Base Timer or enabling the WDT can start the 128kHz low-frequency oscillator.

The low-frequency clock timer (Base Timer) can wake the CPU from STOP mode and generate an interrupt.

BTMCON (CEH) Low-Frequency Timer Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol ENBTM BTMIF Read/Write Read/Write Power-on			-	-	BTMFS[3:0]			
initial value 0		Reading/Writing	-	-	Reading/Writing			
		0	x	x	0	0	0	0

Position number	Bit symbol	illustrate
7	**ENBTM**	Low-frequency Base Timer Startup Control 0: Base Timer not started 1: Start the Base Timer
6	**BTMIF**	Base Timer Interruption Request Flag This flag is automatically cleared by the hardware after the CPU receives an interrupt from the Base Timer.
3~0	**BTMFS [3:0]** Low-frequency clock interrupt frequency selection	0000: An interrupt is generated every 15.625ms. 0001: An interrupt is generated every 31.25ms. 0010: An interrupt is generated every 62.5ms. 0011: An interrupt is generated every 125ms. 0100: An interrupt is generated every 0.25 seconds. 0101: An interrupt is generated every 0.5 seconds. 0110: An interrupt is generated every 1.0 second. 0111: An interrupt is generated every 2.0 seconds. 1000: An interrupt is generated every 4.0 seconds. 1001~1111: Reserved
5~4	-	reserve

7.6 **STOP** and **IDLE** Modes

The SC92F836XB provides a special function register PCON. Configuring bits 0 and 1 of this register can control the MCU to enter different operating modes.

model.

Writing a 1 to PCON.1 will stop the internal high-frequency system clock, entering STOP mode for power saving. In STOP mode, use... .

Users can wake up the SC92F836XB via external interrupts INT0~2, low-frequency clock interrupt, and TK interrupt, or wake it up via external reset.

Awake.

Writing 1 to PCON.0 stops the program from running and puts it into IDLE mode. However, external devices and the clock continue to run, and all functions that were active before entering IDLE mode are restored.

The CPU state is preserved. IDLE mode can be woken up by any interrupt.

PCON (87H) Power Management Control Register (Write-only, *Unreadable *****)

Bit number	7	6	5	4	3	2	1	0
symbol SMOD Read/Write Write-only		-	-	-	-	-	STOP	IDL
Power-on initial value 0		-	-	-	-	-	(Write only)	
		x	x	x	x	x	0	0

Position number	Bit symbol	illustrate
1	**STOP**	STOP mode control 0: Normal operating mode 1: Energy-saving mode: The high-frequency oscillator stops working, while the low-frequency oscillator and WDT can be set according to the parameters. Choose whether or not to work.
0	**IDL**	IDLE mode control

		0: Normal operating mode; 1: Power saving mode. The program stops running, but external devices and the clock continue to run. All CPU states are saved before entering IDLE mode.

Note: When configuring **the MCU** to enter **STOP** or **IDLE** mode, the statement that performs configuration operations on the PCON register must be followed by 8 **NOP instructions .** It cannot be followed directly by other instructions, otherwise subsequent instructions will not be able to be executed normally after waking up!

For example: Setting the MCU to enter STOP mode:

C language
example: #include"intrins.h"

```
PCON |= 0x02; _nop_();       //Write 1 to PCON's bit 1 STOP bit to configure the MCU to enter STOP mode.//At least 8
_nop_();                     _nop_() operations are required.
_nop_();
_nop_();
_nop_();
_nop_();
_nop_();

……
```

Assembly routines:

```
ORL PCON,#02H       Write 1 to PCON's bit 1 (STOP bit) to configure the MCU to enter STOP mode; at least 8
NOP                 NOPs are required.
NOP
NOP
NOP
NOP
NOP
NOP
NOP
……
```


8. Central Processing Unit **(CPU)** and Instruction Set

8.1 CPU

The CPU used in the SC92F836XB is a high-speed 1T standard 8051 core, whose instructions are fully compatible with traditional 8051 core microcontrollers.

8.2 Addressing Modes

The addressing modes of the SC92F836XB's 1T 8051 CPU instructions are: ÿ Immediate addressing ÿ Direct addressing ÿ Indirect addressing ÿ Register addressing ÿ Relative addressing

Address ÿ Indexed addressing ÿ Bit addressing

8.2.1 Immediate Addressing

Immediate addressing, also known as immediate data addressing, directly specifies the operands to be performed in the instruction operand list. Examples of instructions are as follows:

MOV A, #50H (This instruction loads the immediate value 50H into accumulator A)

8.2.2 Direct Addressing

In direct addressing mode, the instruction operand field provides the addresses of the operands to be performed. Direct addressing mode can only be used to represent special function registers. The system includes registers, internal data registers, and a bit-address space. Special function registers and the bit-address space can only be accessed using direct addressing. For example:

`ANL 50H, #91H` (means the number in memory location 50H is ANDed with the immediate value 91H, and the result is stored in memory location 50H. Here, 50H is the direct address space). (The address represents a cell in the internal data register RAM.)

8.2.3 Indirect Addressing

Indirect addressing is represented by adding an "@" symbol before R0 or R1. Assuming the data in R1 is 40H and the data in internal data memory location 40H is 55H, the instruction would be:

MOV A, @R1 (Transfers data 55H to accumulator A).

8.2.4 Register Addressing

Register addressing operates on the selected working registers R7-R0, accumulator A, general-purpose register B, address register, and carry C. Registers R7-R0 are represented by the lower 3 bits of the instruction code, while ACC, B, DPTR, and carry C are implicit in the instruction code. Therefore, register addressing also includes an implicit addressing mode. The selection of the working register area is determined by RS1 and RS0 in the program status word register PSW. Registers specified by instruction operands refer to registers in the current working area.

INC R0 means (R0) + 1 ÿ R0

8.2.5 Relative Addressing

Relative addressing adds the current value in the program counter (PC) to the number given in the second byte of the instruction; the result is used as the transfer address of the jump instruction. This transfer address is also called the destination address, the current value in the PC becomes the base address, and the number given in the second byte of the instruction becomes the offset. Because the destination address is relative to the base address in the PC, this addressing mode is called relative addressing. The offset is a signed number, and its range is +127 to -128. This addressing mode is mainly used for jump instructions.

JC ÿ+50H

This means that if the carry bit C is 0, the content in the program counter PC remains unchanged, i.e., no transfer occurs. If the carry bit C is 1, the destination address of the transfer instruction is the result of adding the current value in PC and the base address to the offset 50H.

8.2.6 Indexed Addressing

In indexed addressing mode, the instruction operand specifies an index register to store the index base address. During indexed addressing, the offset is added to the index base value, and the result is used as the address of the operand. The index register consists of the program counter PC and the address register DPTR.

MOVC A, @A+DPTR indicates that

accumulator A is an offset register, its contents are added to the contents of address register DPTR, and the result is used as the address of the operand. The number in that unit is then retrieved and loaded into accumulator A.

8.2.7 Bit Addressing

Bit addressing refers to the addressing mode when performing bit operations on internal data memory (RAM) and special function registers that can be manipulated. During bit operations, the carry bit C acts as a bit accumulator. The instruction operand directly provides the address of the bit, and then the bit is manipulated according to the nature of the opcode. The bit address encoding method is exactly the same as that of byte addressing in direct byte addressing; the main difference lies in the nature of the operation instruction, and special attention should be paid when using it.

MOV C, 20H (Loads the value of the bit manipulation register at address 20H into the carry bit C.)

9. Interrupt

The SC92F836XB microcontroller provides 12 interrupt sources: Timer0, Timer1, Timer2, INT0~2, ADC, PWM, UART, SSI, Base Timer, TK. These 12 interrupt sources are divided into 2 interrupt priorities, and can be individually set to high or low priority. Three external... Each interrupt source can be individually configured with a rising, falling, or rising/falling edge trigger condition, and each interrupt has its own independent priority setting bit. Interrupt flags, interrupt vectors, and enable bits; the total enable bit EA can enable or disable all interrupts.

9.1 Interrupt Sources and Vectors

The interrupt sources, interrupt vectors, and related control bits of the SC92F836XB are listed below:

Interrupt source	Interruption occurred time	Interrupt flag:	Interrupt enabled control	Interruption priority control	Interrupt vector	lookup priority	interrupt number ÿC51ÿ	Flag removal Way	Can it be awakened? STOP
INT0	External Interrupt 0 Conditions met	IE0	EINT0	IPINT0	0003H	1 (High)	0	H/W Auto	
Timer0	Timer0 Overflow out	TF0	ET0	IPT0	000BH	2	1	H/W Auto	cannot
INT1	External Interrupt 1 Conditions met	IE1	EINT1	POINT1	0013H	3	2	H/W Auto	
Timer1	Timer1 Overflow out	TF1	ET1	IPT1	001BH	4	3	H/W Auto	cannot
UART	receive or transmit Finish	SEE/FOLLOW	EUART	IPUART	0023H	5	4.	Must be used by users Clear	cannot
Timer2	Timer2 Overflow out	TF2	ET2	IPT2	002BH	6	5.	Must be used by users Clear	cannot
ADC	conversion Finish	ADCIF	EADC	IPADC	0033H	7	6.	Must be used by users Clear	cannot
SSI	Receive or Send Finish	SPIF/TWIF	THEY	IPSPI	003BH	8	7.	Must be used by users Clear	cannot
PWM	overflow	PWMIF	EPWM	IPPWM	0043H	9	8	H/W Auto	cannot
BTM	Base timer overflow	BTMIF	EBTM	IPBTM	004BH	10	9	H/W Auto	
INT2	External Interrupt 2 Conditions met	·	ONE2	POINT2	0053H	11	10	·	able
TK	Touch Key counter overflow	TKIF	ETC	IPTK	005B	12	11	H/W Auto	

With EA=1 and all interrupt enable controls set to 1, the interrupt occurrences are as follows:

Timer interrupts: When Timer0 and Timer1 overflow, an interrupt is generated and the interrupt flags TF0 and TF1 are set to "1". This interrupt occurs when the microcontroller executes the timer. When an interrupt occurs, interrupt flags TF0 and TF1 are automatically cleared to "0" by the hardware. When Timer2 overflows, an interrupt is generated and interrupt flag TF2 is set to "1". After a Timer2 interrupt occurs, the hardware does not automatically clear the TF2 bit; this bit must be cleared by the user's software.

ADC Interrupt: The ADC interrupt occurs when the ADC conversion is complete, and its interrupt flag is the ADC conversion end flag EOC/ADCIF. (ADCCON.5). When the user sets ADCS to start the conversion, EOC will be automatically cleared to "0" by the hardware; when the conversion is complete, EOC will be... The hardware automatically sets it to "1". After an ADC interrupt occurs, the user must clear it using software when entering the interrupt service routine.

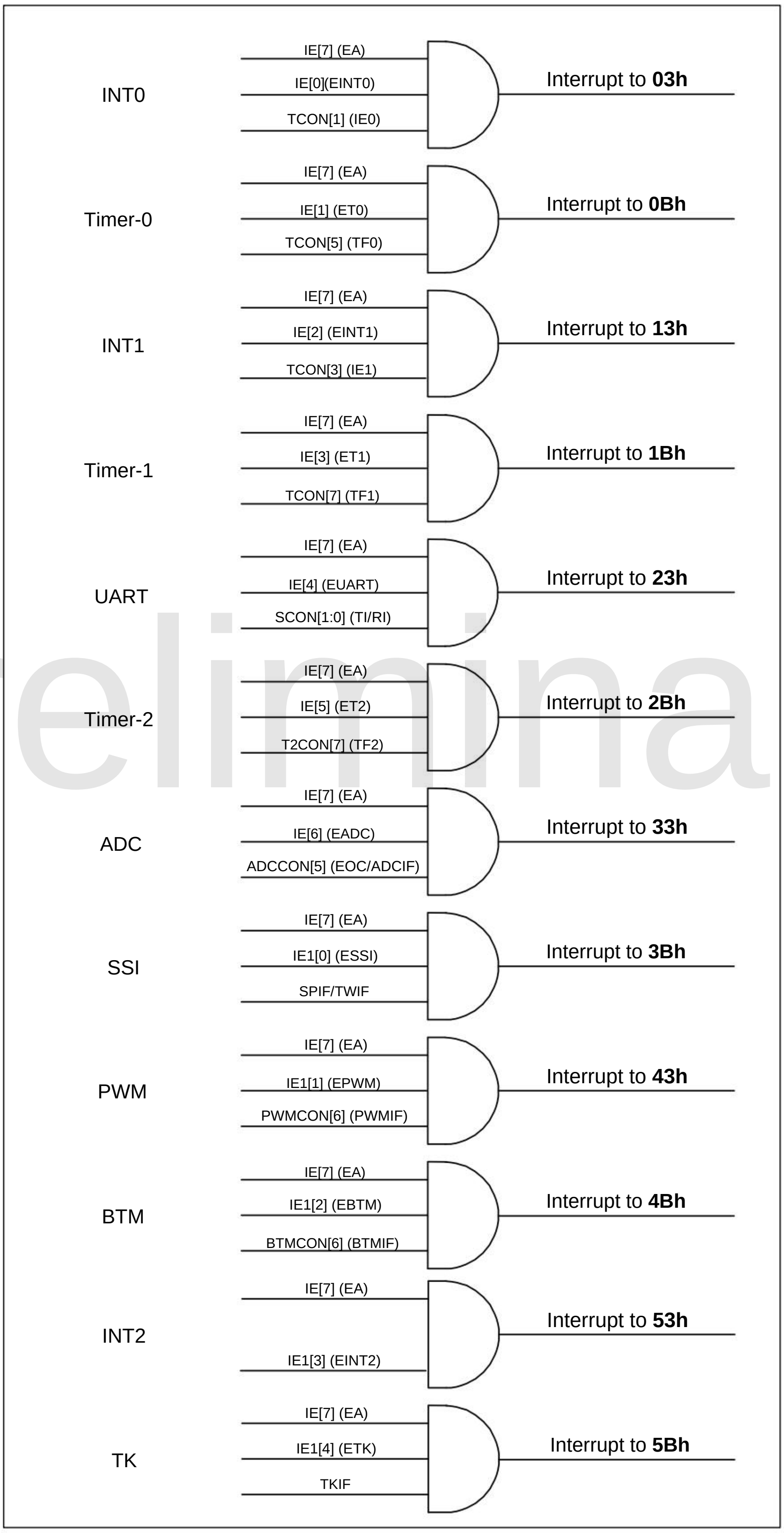
SSI Interrupt: When the SSI receives or transmits a frame of data, the SPIF/TWIF bit is automatically set to "1" by the hardware, generating an SSI interrupt. When the microcontroller executes... When this SSI interrupt occurs, the interrupt flags SPIF/TWIF must be cleared by the user's software.

PWM Interrupt: When the PWM counter overflows (that is, when the counter counts beyond PWMPRD), the PWMIF bit (PWM Interrupt Flag) is automatically set to "1" by the hardware, and a PWM interrupt is generated. When the microcontroller executes this PWM interrupt, the interrupt flag PWMIF is automatically cleared to "0" by the hardware.

External interrupts INT0~2: An external interrupt occurs when an interrupt condition is met at an external interrupt port. INT0 has three external interrupt sources, and INT1 has... There are four external interrupt sources, and INT2 has six external interrupt sources. Users can configure them as rising-edge, falling-edge, or double-edge interrupts as needed, which can be achieved by setting SFRs (INTxF and INTxR). Users can set the priority level of each interrupt through the IP register. External interrupts INT0~2 can also wake up the system. STOP for microcontrollers.

9.2 Interruption Structure Diagram

The interrupt structure of the SC92F836XB is shown in the figure below:



SC92F836XB Interrupt Structure and Vector

9.3 Interrupt Priority

The SC92F836XB microcontroller has two interrupt priorities, and these interrupt source requests can be programmed to be either high-priority interrupts or low-priority interrupts. This allows for nested interrupt service routines at two levels. A low-priority interrupt that is currently executing can be interrupted by a high-priority interrupt request, but cannot be interrupted by a high-priority interrupt request. Interrupted by another interrupt request of the same priority, execution continues until it reaches completion. Upon encountering a RETI return instruction, it returns to the main program and executes one more instruction before proceeding. Respond to the new interruption request.

In other words:

ÿ A low-priority interrupt can be interrupted by a high-priority interrupt request, but the reverse is not true;

ÿ No interrupt can be interrupted by an interrupt request of the same priority during the response process. Interrupt polling order: For the SC92F836XB microcontroller, if several interrupts of the same priority occur simultaneously, the interrupt response priority is the same.

In C51, interrupt query numbers are the same; that is, the one with the smaller query number will be responded to first, and the one with the larger query number will be responded to slowly.

9.4 Interrupt Handling Procedure

When an interrupt occurs and is responded to by the CPU, the main program execution is interrupted, and the following operations will be performed.

ÿ The currently executing instruction has been completed;

ÿ The PC value is pushed onto the stack to save the context;

ÿ The interrupt vector address is loaded into the program counter PC;

ÿ Execute the corresponding interrupt service routine;

ÿ The interrupt service routine ends and RETI is executed;

ÿ Pop the PC value from the stack and return to the program that was executed before the interrupt.

During this process, the system will not immediately execute other interrupts of the same priority, but will retain the interrupt requests that have occurred until the current interrupt handling ends. Then, it proceeds to execute a new interrupt request.

9.5 Interrupt-related SFR registers

IE (A8H) Interrupt Enable Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	YES	EADC	ET2	EUART	ET1	EINT1	ET0	EINT0
read/write Read/write Power-on		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	0	0	0	0	0

Position number	Bit symbol	illustrate
7	**YES**	Interrupt enable master control 0: Disable all interrupts 1: Enable all interrupts
6	**EADC**	ADC interrupt enable control 0: Disable ADC interrupt 1: Enable interrupt generation when ADC conversion is complete
5	**ET2**	Timer2 Interrupt Enable Control 0: Disable TIMER2 interrupt 1: Enable TIMER2 interruption
4	**EUART**	UART interrupt enable control 0: Disable UART interrupt 1: Enable UART interrupts
3	**ET1**	Timer1 Interrupt Enable Control 0: Disable TIMER1 interrupt 1: Enable TIMER1 interruption
2	**EINT1**	External Interrupt 1 Enable Control 0: Disable INT1 interrupt

		1: Enable INT1 interrupt
1	**ET0**	Timer0 Interrupt Enable Control 0: Disable TIMER0 interrupt 1: Enable TIMER0 interruption
0	**EINT0**	External interrupt 0 enable control 0: Disable INT0 interrupt 1: Enable INT0 interrupt

IP (B8H) Interrupt Priority Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	IPADC	IPT2	PORT	IPT1	POINT1	IPT0	IPINT0
read/	-	Read/	Read/	Read/	Read/	Read/	Read/	Reading/Writing
write power-on initial value x		Write 0	Write 0	Write 0	Write 0	Write 0	Write 0	0

Position number	Bit symbol	illustrate
6	**IPADC**	ADC interrupt priority selection 0: ADC interrupt priority is low 1: ADC interrupt priority is high
5	**IPT2**	Timer2 Interrupt Priority Selection 0: Timer2 interrupt priority is low. 1: Timer2 has high interrupt priority.
4	**PORT**	UART interrupt priority selection 0: UART interrupt priority is low 1: UART interrupt priority is high
3	**IPT1**	Timer1 Interrupt Priority Selection 0: Timer1 interrupt priority is low. 1: Timer1 has high interrupt priority.
2	**POINT1**	INT1 counter interrupt priority selection 0: INT1 interrupt priority is low. 1: INT1 interrupt has high priority.
1	**IPT0**	Timer0 Interrupt Priority Selection 0: Timer0 interrupt priority is low. 1: Timer0 has high interrupt priority.
0	**IPINT0**	INT0 counter interrupt priority selection 0: INT0 interrupt priority is low. 1: INT0 interrupt priority is high.
7	-	reserve

IE1 (A9H) Interrupt Control Register **1 (Read/Write)**

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	ETC	ONE2	EBTM EPWM ESSI		
read/	-	-	-	Reading and writing		Reading/Writing	Reading/Writing	Reading/Writing
write power-on initial value x		x	x	0	0	0	0	0

Position number	Bit symbol	illustrate
4	**ETC**	Touch Key Interrupt Enable Control 0: Disable Touch Key interruption 1: Enable Touch Key Interruption
3	**ONE2**	External Interrupt 2 Enable Control 0: Disable INT2 interrupt

		1: Enable INT2 interrupt
2	**EBTM**	Base Timer Interrupt Enable Control 0: Disable Base Timer interrupts 1: Allow Base Timer interruption
1	**EPWM**	PWM interrupt enable control 0: Disable PWM interrupt 1: Enable interrupt generation when PWM counter overflows (counts to PWMPRD)
0	**THEY**	Three-in-one serial port interrupt enable control 0: Disable serial port interrupt 1: Enable serial port interrupts
7~5	-	reserve

IP1 (B9H) Interrupt Priority Control Register **1 (Read/Write)**

Bit number **6** symbol reads/writes power-on initial value x	7		5	4	3	2	1	0
	-	-	-	IPTK	POINT2	IPBTM IPPWM		IPSSI
	-	-	-	Read/	Read/	Read/	Read/	Reading/Writing
		x	x	Write 0	Write 0	Write 0	Write 0	0

Position number	Bit symbol	illustrate
4	**IPTK**	Touch Key interrupt priority selection 0: Touch Key interrupt priority is low. 1: Touch Key interrupt priority is high.
3	**POINT2**	INT2 counter interrupt priority selection 0: INT2 interrupt priority is low. 1: INT2 interrupt has high priority.
2	**IPBTM**	Base Timer Interrupt Priority Selection 0: Base Timer interrupt priority is low. 1: Base Timer interrupt priority is high.
1	**IPPWM**	PWM interrupt enable selection 0: PWM interrupt priority is low. 1: PWM interrupt priority is high.
0	**IPSSI**	Three-in-one serial port interrupt priority selection 0: SSI interrupt priority is low 1: SSI interrupt priority is high.
7~5	-	reserve

TCON (88H) Timer Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	TF1	TR1	TF0	TR0	IE1	-	IE0	-
read/write Read/write Power-on		Read/	Read/	Read/	Read/	-	Read/	-
initial value 0		Write 0	Write 0	Write 0	Write 0	x	Write 0	x

Position number	Bit symbol	illustrate
3	**IE1**	INT1 Overflow interrupt request flag. When INT1 overflows and an interrupt occurs, the hardware sets IE1 to 0. If the value is "1", an interrupt is requested, and the hardware clears it to "0" when the CPU responds.
1	**IE0**	INT0 Overflow interrupt request flag. When an overflow occurs at INT0, the hardware sets INT0 to INT0. If the value is "1", an interrupt is requested, and the hardware clears it to "0" when the CPU responds.
2,0	-	reserve

INT0F (BAH) INT0 Falling Edge Interrupt Control Register (Read/Write)

Position	7	6	5	4	3	2	1	0
number symbol	-	-	-	-	INT0F3	INT0F2	INT0F1	-

Read/	-	-	-	-	Reading/Writing	Reading/Writing	Reading/Writing	-
write power-on initial value x		x	x	x	0	0	0	x

Position	Bit symbol	illustrate
number 3~1	**INT0Fn** (n=1~3)	INT0 Falling Edge Interrupt Control 0: INT0n Falling edge interrupt disabled 1: INT0n Falling edge interrupt enable
7~4,0	-	reserve

INT0R (BBH) INT0 rising edge interrupt control register (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	-	INT0R3 INT0R2 INT0R1			-
read/write	-	-	-	-	Reading/Writing	Reading/Writing	Reading/Writing	-
power-on initial value x		x	x	x	0	0	0	x

Position	Bit symbol	illustrate
number 3~1	**INT0Rn** (n=1~3)	INT0 Rising Edge Interrupt Control 0: INT0n Rising edge interrupt disabled 1: INT0n Rising edge interrupt enable
7~4,0	-	reserve

INT1F (BCH) INT1 Falling Edge Interrupt Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	-	INT1F3	INT1F2	INT1F1	INT1F0
read/write	-	-	-	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value x		x	x	x	0	0	0	0

Position	Bit symbol	illustrate
number 3~0	**INT1Fn** (n=0~3)	INT1 Falling Edge Interrupt Control 0: INT1n Falling edge interrupt disabled 1: INT1n Falling edge interrupt enable
7~4	-	reserve

INT1R (BDH) INT1 rising edge interrupt control register (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	-	INT1R3 INT1R2 INT1R1 INT1R0			
read/write	-	-	-	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value x		x	x	x	0	0	0	0

Position	Bit symbol	illustrate
number 3~0	**INT1Rn** (n=0~3)	INT1 Rising Edge Interrupt Control 0: INT1n rising edge interrupt disabled 1: INT1n Rising edge interrupt enable
7~4	-	reserve

INT2F (C6H) INT2 Falling Edge Interrupt Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	INT2F5 INT2F4		INT2F3	INT2F2	INT2F1	INT2F0
read/write	-	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value x		x	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 5~0	**INT2Fn** **(n=0~5)**	INT2 Falling Edge Interrupt Control 0: INT2n Falling edge interrupt disabled 1: INT2n Falling edge interrupt enable
7~6	-	reserve

INT2R (C7H) INT2 rising edge interrupt control register (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	INT2R5 INT2R4 INT2R3 INT2R2 INT2R1 INT2R0					
read/write	-	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value x		x	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 5~0	**INT2Rn** **(n=0~5)**	INT2 Rising Edge Interrupt Control 0: INT2n rising edge interrupt disabled 1: INT2n rising edge interrupt enable
7~6	-	reserve

10 Timers **TIMER0 and TIMER1**

The SC92F836XB microcontroller contains two 16-bit timers/counters, which operate in both counting and timing modes. Special functions...

The TMOD register contains a control bit C/Tx to select whether T0 and T1 are timers or counters. They are essentially both adder counters.

The only difference is the source of the count. Timers are powered by the system clock or its divided clock, while counters are powered by input pulses from external pins. Only...

The counting of T0 and T1 will only be turned on when TRx=1.

In counter mode, for each pulse on pins P1.2/T0 and P1.3/T1, the count values of T0 and T1 increase by 1 respectively.

In timer mode, the count source for T0 and T1 can be selected via the special function register TMCON as either fSYS/12 or fSYS (fSYS is the frequency divider).

(System clock).

Timer/counter T0 has 4 operating modes, and timer/counter T1 has 3 operating modes (mode 3 does not exist):

ÿMode 0: 13-bit timer/counter mode

ÿMode 1: 16-bit Timer/Counter Mode

ÿMode 2: 8-bit auto-reload mode

ÿ Mode 3: Two 8-bit timer/counter modes

In the above modes, modes 0, 1, and 2 of T0 and T1 are the same, while mode 3 is different.

10.1 Special Function Registers Related to **T0** and **T1**

Symbolic address		illustrate	7	6	5 4 3			2	1	0	**Reset** value
TCON	88H	Timer Control Register	TF1	TR1 TF0 TR0 IE1				-	IE0	-	00000x0xb
TMOD	89H	Timer Operating Mode Register	-	C/T1 M11 M01			-	C/T0 M10 M00			x000x000b
TL0	8AH	Timer 0 Low 8 Bits	TL0[7:0]								00000000b
TL1	8BH	Timer 1 Low 8 Bits	TL1[7:0]								00000000b
TH0	8CH	Timer 0 High 8 Bits	TH0[7:0]								00000000b
TH1	8DH	Timer 1 High 8 Bits	TH1[7:0]								00000000b
TMCON	8EH	Timer Frequency Control Register	-	-	-	-	- T2FD T1FD T0FD xxxxx000b				

The explanations of each register are as follows:

TCON (88H) Timer Control Register (Read/Write)

Bit code	7	6	5	4	3	2	1	0
symbol	TF1	TR1	TF0	TR0	IE1	-	IE0	-
read/write	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	-	Reading/Writing	-

Initial power-on value 0		0	0	0	0	x	0	x

Position number	Bit symbol	illustrate
7	**TF1**	T1 Overflow interrupt request flag. When T1 overflows and an interrupt occurs, the hardware sets TF1 to the specified value. "1" indicates an interrupt request; when the CPU responds, the hardware clears the value to "0".
6	**TR1**	The run control bit for timer T1. This bit is set to 1 and cleared to 0 by software. When TR1=1, it enables... T1 starts counting. T1 counting is disabled when TR1=0.
5	**TF0**	T0 Overflow Interrupt Request Flag. When T0 overflows and an interrupt occurs, the hardware sets TF0 to [value missing]. "1" indicates an interrupt request; when the CPU responds, the hardware clears the value to "0".
4	**TR0**	The run control bit for timer T0. This bit is set and cleared by software. When TR0=1, it enables... T0 starts counting. T0 counting is disabled when TR0=0.
2,0	-	reserve

TMOD (89H) Timer Operating Mode Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	C/T1	M11	M01	-	C/T0	M10	M00
read/	-	Reading/Writing	Reading/Writing	Reading/Writing	-	Reading/Writing	Reading/Writing	Reading/Writing
write power-on initial value x		0	0	0	x	0	0	0
	T1				T0			

Position number	Bit symbol	illustrate
6	**C/T1**	TMOD[6] controls timer 1 0: Timer T1 count is derived from fSYS frequency division. 1: Counter, T1 counts are derived from external pin T1/P1.3
5~4	**M11,M01**	Timer/Counter 1 Mode Selection 00: 13-bit timer/counter, TL1 high 3 bits invalid. 01: 16-bit timer/counter, TL1 and TH1 fully functional. 10: An 8-bit auto-reload timer that automatically reloads the value stored in TH1 into TL1 upon overflow. 11: Timer/Counter 1 is invalid (counting stopped)
2	**C/T0**	TMOD[2] controls timer 0 0: Timer T0 count is derived from fSYS frequency division. 1: Counter, T0 count is derived from external pin T0/P1.2
1~0	**M10,M00**	Timer/Counter 0 Mode Selection 00: 13-bit timer/counter, TL0 high 3 bits invalid. 01: 16-bit timer/counter, TL0 and TH0 fully functional. 10: An 8-bit auto-reload timer that automatically reloads the value stored in TH0 into TL0 upon overflow. 11: Timer 0 is currently used as a dual 8-bit timer/counter. TL0 is used as an 8-bit timer. The counter is controlled by the control bits of standard Timer 0; TH0 is only used as an 8-bit timer. The device is controlled by the control bit of Timer 1.
7,3	-	reserve

In the TMOD register, TMOD[0]~TMOD[2] sets the working mode of T0; TMOD[4]~TMOD[6] sets the working mode of T1.

The timer and counter (Tx) function is selected by the control bit C/Tx of the special function register TMOD. M0x and M1x are both used to select the function of Tx. Operating mode. TRx acts as a switch control for T0 and T1; T0 and T1 are only turned on when TRx=1.

TMCON (8EH) Timer Frequency Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	-	-	T2FD	T1FD	T0FD
read/	-	-	-	-	-	Reading/Writing	Reading/Writing	Reading/Writing
write power-on initial value x		x	x	x	x	0	0	0

Position number	Bit symbol	illustrate
1	**T1FD**	T1 Input Frequency Selection Control 0: T1 frequency originates from fSYS/12 1: T1 frequency originates from fSYS
0	**T0FD**	T0 Input Frequency Selection Control 0: T0 frequency originates from fSYS/12 1: T0 frequency originates from fSYS

IE (A8H) Interrupt Enable Register (Read/Write)

The symbol for position number **6**	**7**		**5**	**4**	**3**	**2**	**1**	**0**
	YES	EADC Read/	ET2	-	ET1	EINT1	ET0	EINT0
Write: Power-on initial value 0		Reading/Writing	Reading/Writing	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
		0	0	x	0	0	0	0

Position number	Bit symbol	illustrate
3	**ET1**	Timer1 Interrupt Enable Control 0: Disable TIMER1 interrupt 1: Enable TIMER1 interruption
1	**ET0**	Timer0 Interrupt Enable Control 0: Disable TIMER0 interrupt 1: Enable TIMER0 interruption

IP (B8H) Interrupt Priority Control Register (Read/Write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol	-	IPADC	IPT2	-	IPT1	POINT1	IPT0	IPINT0
read/write	-	Reading/Writing	Reading/Writing	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value x		0	0	x	0	0	0	0

Position number	Bit symbol	illustrate
3	**IPT1**	Timer1 Interrupt Priority 0: Sets the interrupt priority of Timer 1 to "low". 1: Set the interrupt priority of Timer 1 to "high".
1	**IPT0**	Timer0 Interrupt Priority 0: Sets the interrupt priority of Timer 0 to "low". 1: Set the interrupt priority of Timer 0 to "high".

10.2 T0 Working Mode

By setting M10 and M00 (TMOD[1], TMOD[0]) in the TMOD register, Timer/Counter 0 can achieve four different operations.

model.

Operating mode **0: 13 -bit counter/timer.**

The TH0 register stores the high 8 bits (TH0.7~TH0.0) of a 13-bit counter/timer, and TL0 stores the low 5 bits (TL0.4~TL0.0). The high three bits of TL0...

(TL0.7~TL0.5) are indeterminate values and should be ignored when reading. When the 13-bit timer/counter overflows, the system will set the timer overflow flag.

Set TF0 to 1. If Timer 0 interrupts are enabled, an interrupt will be generated.

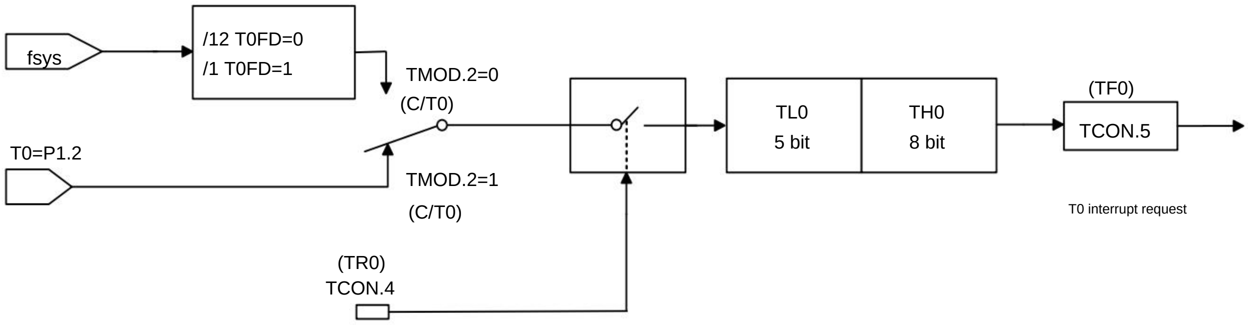
C/T0 selects the clock input source for the counter/timer. If C/T0=1, the level of the timer 0 input pin T0 (P1.2) changes from high to low.

This will increment the Timer 0 data register by 1. If C/T0=0, the system clock divider will be selected as the clock source for Timer 0.

When TR0 is set to 1, timer T0 is enabled. Setting TR0 to 1 does not forcibly reset the timer, meaning that if TR0 is set to 1, the timer register will start from the last time...

The count begins when TR0 is cleared to 0. Therefore, the timer register should be initialized before enabling the timer.

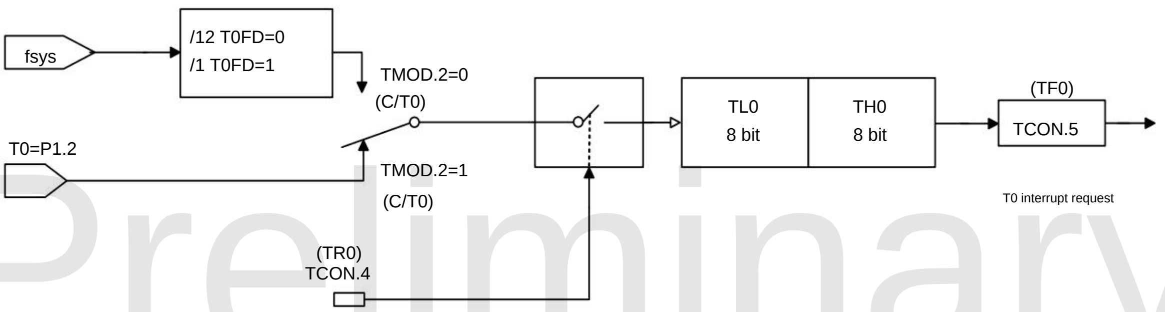
When used as a timer, T0FD can be configured to select the division ratio of the clock source.



Timer/counter operating mode 0: 13-bit timer/counter

Operating Mode **1: 16 -bit Counter/Timer.**

Except for using a 16-bit counter/timer (all 8 bits of TL0 data are valid), Mode 1 operates the same as Mode 0. Enable and Configure. The counter/timer method is the same.



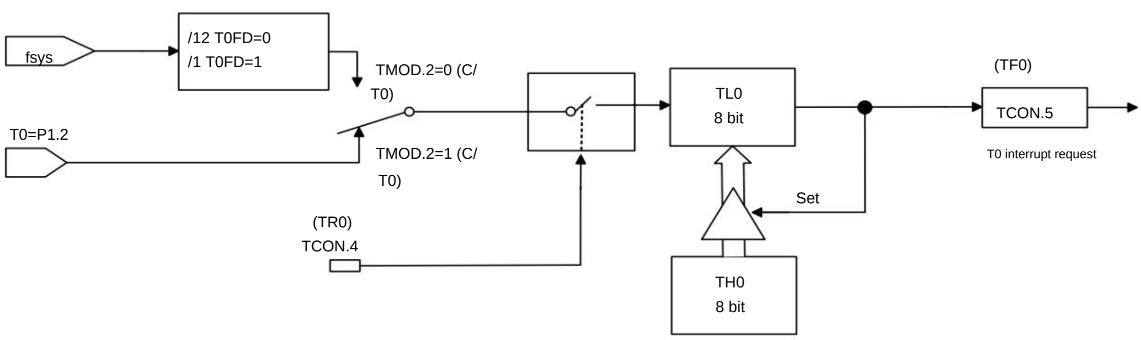
Timer/counter operating mode 0: 16-bit timer/counter

Operating Mode **2: 8** -bit Auto-Reload Counter/Timer. In

Operating Mode 2, Timer 0 is an 8-bit auto-reload counter/timer. TL0 stores the count value, and TH0 stores the reload value. When the counter in TL0 overflows to 0x00, the timer overflow flag TF0 is set to 1, and the value in register TH0 is reloaded into register TL0. If timer interrupts are enabled, an interrupt will be generated when TF0 is set to 1, but the reload value in TH0 will not change. TL0 must be initialized to the required value before the timer can start counting correctly.

Except for the automatic reload function, the enabling and configuration methods for the counter/timer in operating mode 2 are the same as in modes 0 and

1. When used as a timer, the configurable register TMCON.0 (T0FD) can be used to select the ratio by which the timer clock source is divided by the system clock fSYS.



Timer/counter operating mode 2: 8-bit timer/counter with automatic reload

Operating Mode **3:** Two **8-** bit counters/timers (Timer 0 only)

In operating mode 3, Timer 0 functions as two independent 8-bit counters/timers, controlled by TL0 and TH0 respectively. TL0 is controlled via Timer 0's control bits (in TCON) and status bits (in TMOD): TR0, C/T0, and TF0. Timer 0 can be accessed via TMOD.2 (C/T0) of T0.

Choose between timer mode and counter mode.

TH0 is controlled via the timer 1 control bit TCON, but TH0 is limited to timer mode and cannot be set to counter mode via TMOD.2 (C/T0). TH0 is enabled by the timer control bit TR1, which must be set to 1. When an overflow or interrupt occurs, TF1 is set to 1, and the interrupt is handled accordingly. When T0 is set to operating mode 3, the TH0 timer occupies the interrupt resources of T1 and the registers

in TCON, and the 16-bit counter of T1 will stop.

Stop counting, equivalent to "TR1=0". When using the TH0 timer, TR1 needs to be set to 1.

10.3 T1 Working Mode

By setting M11 and M01 (TMOD[5], TMOD[4]) in the TMOD register, Timer/Counter 1 can achieve three different working modes.

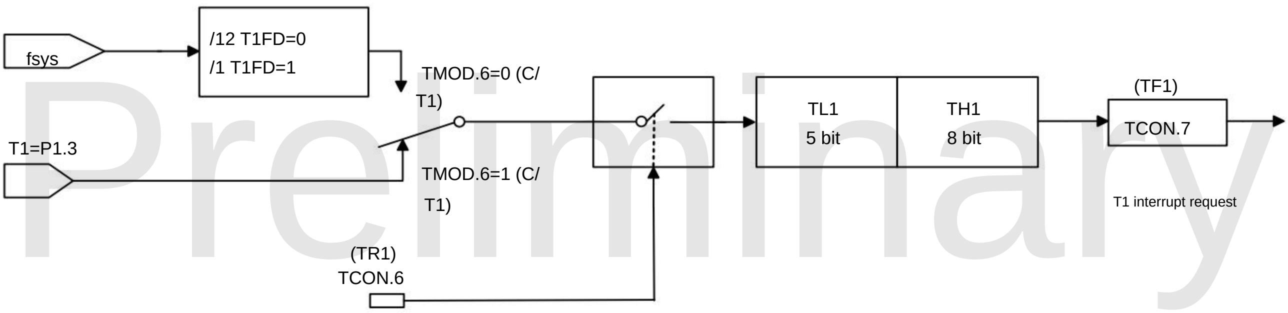
Operating mode **0: 13 -bit counter/timer.**

The TH1 register stores the high 8 bits (TH1.7~TH1.0) of the 13-bit counter/timer; TL1 stores the low 5 bits (TL1.4~TL1.0). The high three bits (TL1.7~TL1.5) of TL1 are indeterminate values and should be ignored when reading. When the 13-bit timer counter overflows, the system sets the timer overflow flag TF1 to 1. If Timer 1 interrupts are enabled, an interrupt will be generated. The C/T1 bit selects the clock source for the counter/timer.

If C/T1=1, a high-to-low transition of the Timer 1 input pin T1 (P1.3) will increment the Timer 1 data register by 1. If C/T1=0, the system clock divider is selected as the clock source for Timer 1.

Setting TR1 to 1 enables the timer. Setting TR1 to 1 does not force a timer reset; it means that if TR1 is set to 1, the timer register will start counting from the value when TR1 was last cleared to 0. Therefore, the initial value of the timer register should be set before enabling

the timer. When used as a timer, T1FD can be configured to select the clock source division ratio.

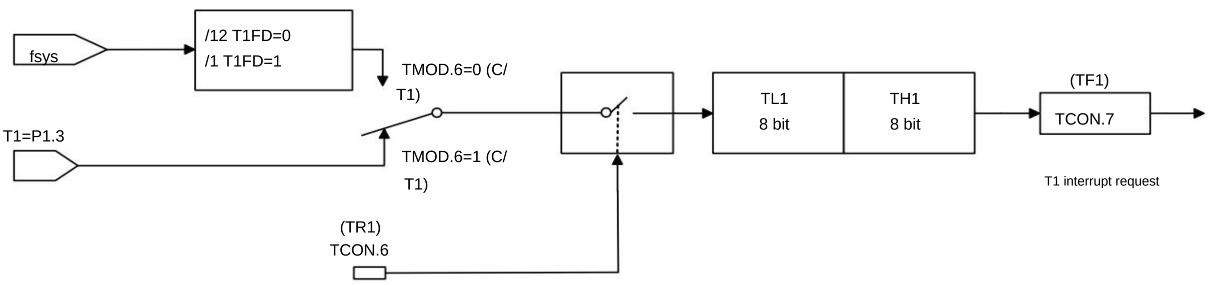


Timer/counter operating mode 0: 13-bit timer/counter; Operating

mode **1: 16 -bit counter/timer.** Except for

using a 16-bit counter/timer (all 8 bits of TL1 data are valid), mode 1 operates identically to mode 0. Enable and configure.

The counter/timer method is the same.



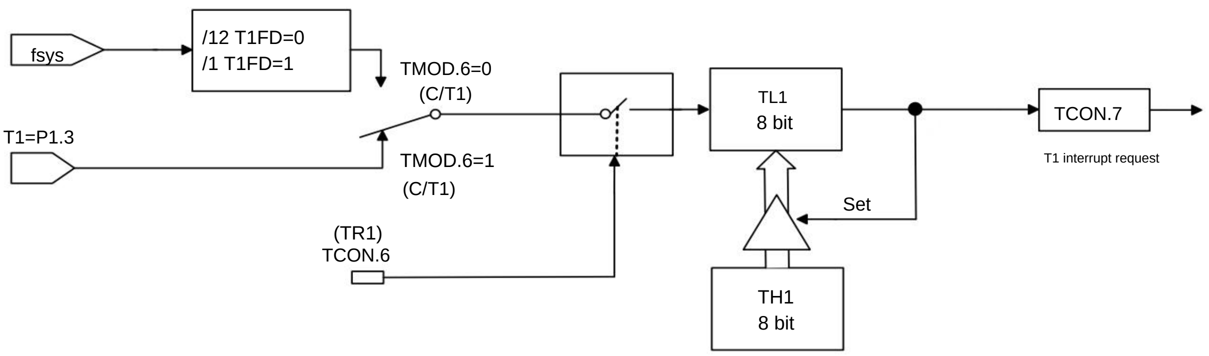
Timer/counter operating mode 0: 16-bit timer/counter

Operating Mode **2: 8** -bit Auto-Reload Counter/Counter. In

Operating Mode 2, Timer 1 is an 8-bit auto-reload counter/timer. TL1 stores the count value, and TH1 stores the reload value. When the counter in TL1 overflows to 0x00, the timer overflow flag TF1 is set to 1, and the value in register TH1 is reloaded into register TL1. If timer interrupts are enabled, an interrupt will be generated when TF1 is set to 1, but the reload value in TH1 will not change. TL1 must be initialized to the required value before the timer can start counting correctly.

Except for the automatic reload function, the enable and configuration methods for the counter/timer in working mode 2 are the same as those in modes 0 and 1.

When used as a timer, the TMCON.4(T1FD) register can be configured to select the ratio by which the timer clock source is divided by the system clock fSYS.



Timer/counter operating mode 2: 8-bit timer/counter with automatic reload

11. Timer 2

The SC92F836XB microcontroller's internal Timer2 has two operating modes: counting mode and timing mode. The special function register T2CON contains...

The control bit C/T2 selects whether T2 is a timer or a counter. They are essentially both adder-counters, just with different counting sources. (Timer)

The source of the clock signal is the system clock or its divided clock, but the source of the counter signal is the input pulse from an external pin. TR2 is the timer/counter mode of T2.

The counting is controlled by an on/off switch; T2 will only be turned on for counting when TR2=1.

In counter mode, for each pulse on pin T2, the count value of T2 increases by 1.

In timer mode, the count source for T2 can be selected as fSYS/12 or fSYS via the special function register TMCON.

Timer/counter T2 has 4 operating modes:

ÿMode 0: 16-bit capture mode

ÿMode 1: 16-bit Auto Reload Timer Mode

ÿMode 2: Baud Rate Generator Mode

ÿ Mode 3: Programmable clock output mode

11.1 T2 Related Special Function Registers

Symbolic address		illustrate	7	6	5	4	3	2	1	0	Reset value
T2CON	C8H	Timer 2 Control Register	TF2	EXF2	RCLK	TCLK	EXEN2	TR2	C/T2	CP/RL2	00000000b
T2MOD	C9H	Timer 2 Operating Mode Register	-	-	-	-	-	-	T2OE	DCEN	xxxxxx00b
RCAP2L	CAH	Timer 2 Reload/Capture Low 8 Bits	RCAP2L[7:0]								00000000b
RCAP2H	CBH	Timer 2 Reload/Capture High 8 Bits	RCAP2H[7:0]								00000000b
TL2	CCH	Timer 2 Low 8 Bits	TL2[7:0]								00000000b
TH2	CDH	Timer 2 High 8 Bits	TH2[7:0]								00000000b
TMCON	8EH	Timer Frequency Control Register	-	-	-	-	-	T2FD	T1FD	T0FD	xxxxx000b

The explanations of each register are as follows:

T2CON (C8H) Timer **2 Control Register (Read/Write)**

Bit number	7	6	5	4	3	2	1	0
symbol	TF2	EXF2	RCLK	TCLK	EXEN2	TR2	C/T2	CP/RL2
read/write	Read/write	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
Power-on initial value	0	0	0	0	0	0	0	0

Position number	Bit symbol	illustrate
7	**TF2**	Timer 2 Overflow Flag 0: No overflow (must be cleared to 0 by software) 1: Overflow (set by hardware if RCLK = 0 and TCLK = 0)

6	**EXF2**	T2EX pin external event input (falling edge) flag detected 0: No external event input (must be cleared to 0 by the software) 1: External input detected (set to 1 by hardware if EXEN2 = 1)
5	**RCLK**	UART Receive Clock Control Bit 0: Timer 1 generates the receive baud rate. 1: Timer 2 generates the receive baud rate.
4	**TCLK**	UART transmit clock control bits 0: Timer 1 generates the transmit baud rate. 1: Timer 2 generates the transmit baud rate.
3	**EXEN2**	The external event input (falling edge) on the T2EX pin is used as a reload/capture trigger to enable/disable the circuit. system: 0: Ignore events on the T2EX pin 1: When Timer 2 is not used as the UART clock (T2EX always includes a pull-up resistor), it is detected that... A falling edge on the T2EX pin generates a capture or reload signal.
2	**TR2**	Timer 2 Start/Stop Control Bit 0: Stop timer 2 1: Start Timer 2
1	**C/T2**	Timer 2 Timer/Counter Mode Selection: Pin 2 0: Timer mode, T2 pin used as I/O port. 1: Counter mode
0	**CP/RL2**	Capture/Reload Mode Selection Positioning 0: 16-bit timer/counter with reload capability 1: A 16-bit timer/counter with capture function, T2EX is the external capture signal for Timer 2. Input port

T2MOD (C9H) Timer **2** Operating Mode Register (Read/Write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol	-	-	-	-	-	-	T2OE	DCEN
read/	-	-	-	-	-	-	Reading/Writing	Reading/Writing
write power-on initial value	x	x	x	x	x	x	0	0

Position number	Bit symbol	illustrate
1	**T2OE**	Timer 2 output enable bit 0: Set T2 as a clock input or I/O port. 1: Set T2 as clock output
0	**DCEN**	Decrease count allowed bits 0: Timer 2 is disabled as an increment/decrement counter; Timer 2 is only used as an increment counter. 1: Enable Timer 2 as an increment/decrement counter
7~2	-	reserve

TMCON (8EH) Timer Frequency Control Register (Read/Write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Symbol	-	-	-	-	-	T2FD	T1FD	T0FD
read/write	-	-	-	-	-	Read/	Read/	Reading/Writing
power-on initial value	x	x	x	x	x	Write 0	Write 0	0

Position number	Bit symbol	illustrate
2	**T2FD**	T2 Input Frequency Selection Control 0: T2 frequency originates from fSYS/12 1: T2 frequency originates from fSYS

IE (A8H) Interrupt Enable Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	YES	EADC	ET2	-	ET1	EINT1	ET0	EINT0
read/write Read/write Power-on		Reading/Writing	Reading/Writing	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	x	0	0	0	0

Position number	Bit symbol	illustrate
5	**ET2**	Timer2 Interrupt Enable Control 0: Disable TIMER2 interrupt 1: Enable TIMER2 interruption

IP (B8H) Interrupt Priority Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	IPADC	IPT2	-	IPT1	POINT1	IPT0	IPINT0
read/write	-	Reading/Writing	Reading/Writing	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value x		0	0	x	0	0	0	0

Position number	Bit symbol	illustrate
5	**IPT2**	Timer2 Interrupt Priority 0: Sets the interrupt priority of Timer 2 to "low". 1. Set the interrupt priority of Timer 2 to "High".

11.2 T2 Working Mode

The operating modes and configuration methods of Timer 2 are shown in the table below:

C/T2	T2OE DCEN TR2 CP/RL2				RCLK TCLK 0		Way
X	0	X	1	1		0	0 16-bit capture
X	0	0	1	0	0	0	1. 16-bit Automatic Reload Timer
X	0	1	1	0	0	0	
X	0	X	1	X	1	X	2. Baud Rate Generator
					X	1	
0	1	X	1	X	0	0	3. For programmable clocks only
					1	X	3 Programmable time with baud rate generator
					X	1	Clock output
X	X	X	0	X	X	X	Timer 2 X stops, but the T2EX path remains open. Old Permission
1	1	X	1	X	X	X	Not recommended

Working mode **0: 16** -bit capture

In capture mode, the EXEN2 bit of T2CON has two options.

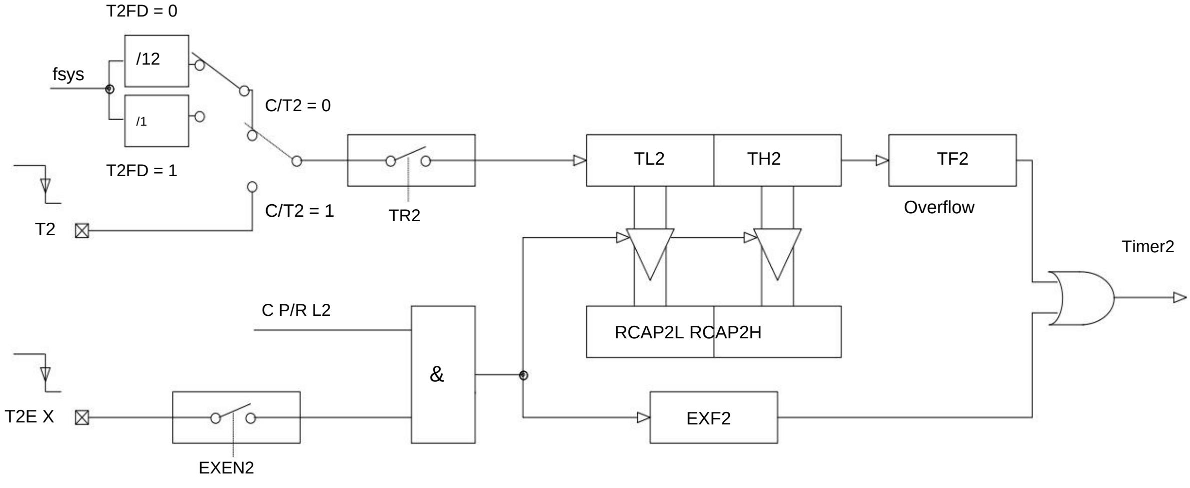
If EXEN2 = 0, Timer 2 acts as a 16-bit timer or counter. If ET2 is enabled, Timer 2 can set TF2 to overflow.

An interruption occurs.

If EXEN2 = 1, Timer 2 performs the same operation, but a falling edge on the external input T2EX can also cause a timer error in TH2 and TL2.

The previous values are captured into RCAP2H and RCAP2L respectively. Furthermore, a falling edge on T2EX can also cause EXF2 in T2CON to be set.

If ET2 is allowed, Like TF2, the EXF2 bit also generates an interrupt.



Mode 0: 16-bit capture

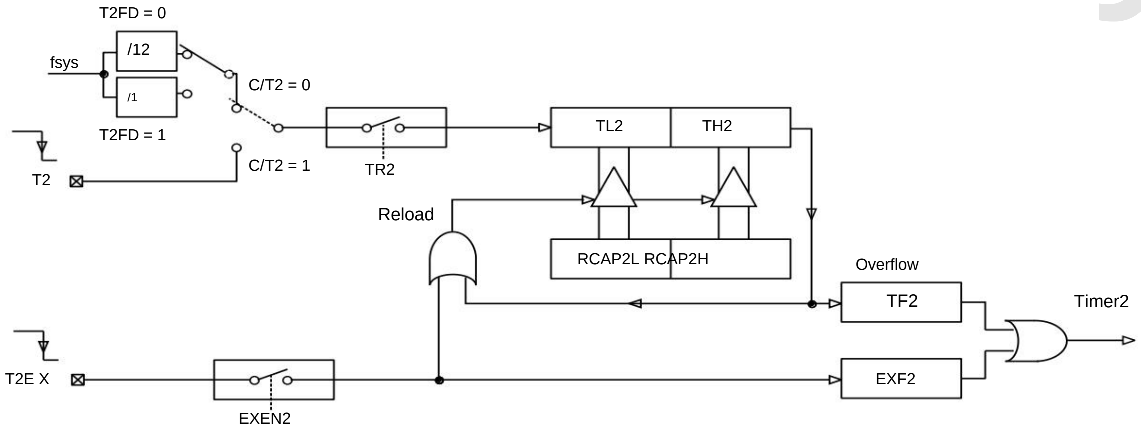
Operating Mode **1: 16** -bit Auto Reload Timer

In 16-bit auto-reload mode, Timer 2 can be selected to increment or decrement. This function is selected via the DCEN bit (decrement enabled) in T2MOD. After system reset, the DCEN bit is reset to 0, and Timer 2 increments by default. When DCEN is set to 1, whether Timer 2 increments or decrements depends on the level on the T2EX pin. When DCEN = 0, the two options are selected via the

EXEN2 bit in T2CON. If EXEN2 = 0, Timer 2 increments to 0xFFFFH, and after overflow,

the TF2 bit is set. Simultaneously, the timer automatically loads the 16-bit values of registers RCAP2H and RCAP2L written by the user software into registers TH2 and TL2. If EXEN2 = 1, an overflow or a falling edge on the external input T2EX triggers a 16-

bit reload, setting the EXF2 bit. If ET2 is enabled, both the TF2 and EXF2 bits can generate an interrupt.



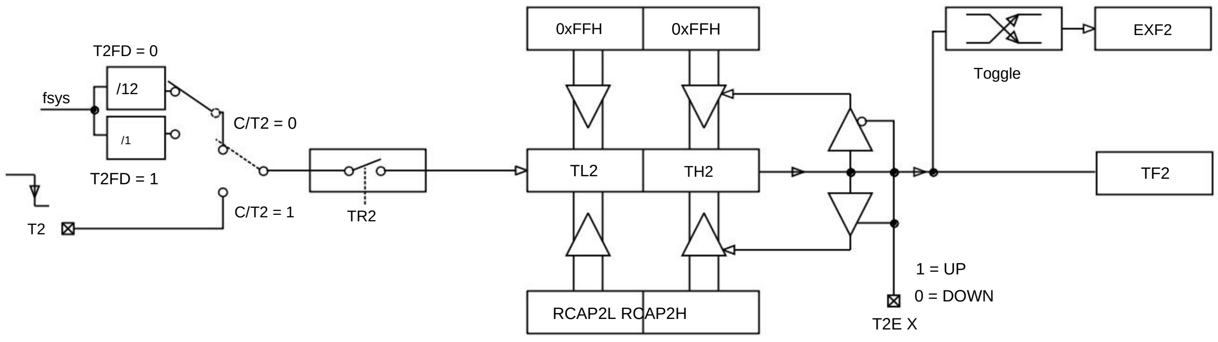
Mode 1: 16-bit Auto-Reload DCEN = 0

Setting the DCEN bit enables Timer 2 to increment or decrement. When DCEN = 1, the T2EX pin controls the counting direction, while EXEN2 has no effect.

Setting T2EX to 1 enables Timer 2 to increment. The timer overflows to 0xFFFFH, then the TF2 bit is set. Overflow also triggers RCAP2H. The 16-bit value on RCAP2L is reloaded into the timer register.

Setting T2EX to 0 causes Timer 2 to decrement. The timer overflows when the values of TH2 and TL2 equal the values of RCAP2H and RCAP2L. Start bit TF2, and simultaneously reload 0xFFFFH into the timer register.

Regardless of whether Timer 2 overflows, the EXF2 bit is used as the 17th bit of the result. In this operating mode, EXF2 is not used as an interrupt flag.



Mode 1: 16-bit Auto-Reload DCEN = 1

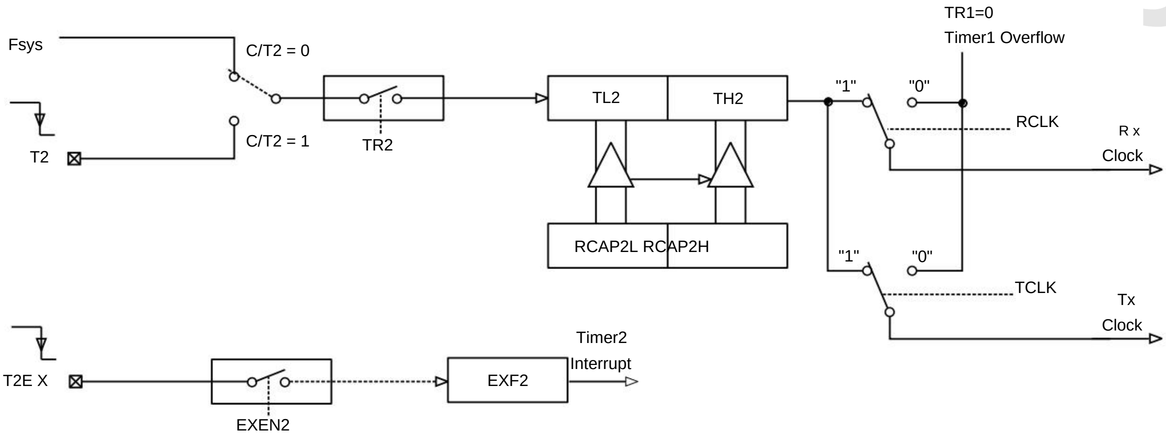
Operating Mode **2:** Baud Rate Generator

Select Timer 2 as the baud rate generator by setting TCLK and/or RCLK in the T2CON register. The baud rate of the receiver and transmitter can be adjusted. The difference lies in the timing of the timer. If Timer 2 is used as a receiver or transmitter, then Timer 1 will correspondingly function as another type of baud rate generator.

Setting TCLK and/or RCLK in the T2CON register puts Timer 2 into baud rate generator mode, which is similar to auto-reload mode. Timer 2 overflow will reload the values in the RCAP2H and RCAP2L registers into Timer 2's count, but will not generate an interrupt. If EXEN2 is set to 1, a falling edge on the T2EX pin will set EXF2, but will not cause a reload. Therefore, when Timer 2 is used as a baud rate transmitter, T2EX can act as an additional external interrupt. The baud rate in UART modes 1 and 3 is determined by the overflow rate of Timer 2 according to the following equation:

$$\text{BaudRate} = \frac{\text{fsys}}{[\text{RCAP2H,RCAP2L}]}$$ (Note: [RCAP2H, RCAP2L] must be greater than 0x0010)

The schematic diagram of Timer 2 as a baud rate generator is as follows:



Mode 2: Baud Rate Generator

Operating Mode **3:** Programmable Clock Output

In this method, T2 can be programmed to output a 50% duty cycle clock cycle: when C/T2 = 0; T2OE = 1, enabling Timer 2 as a clock generator. In this mode, T2 outputs a clock with a 50% duty cycle.

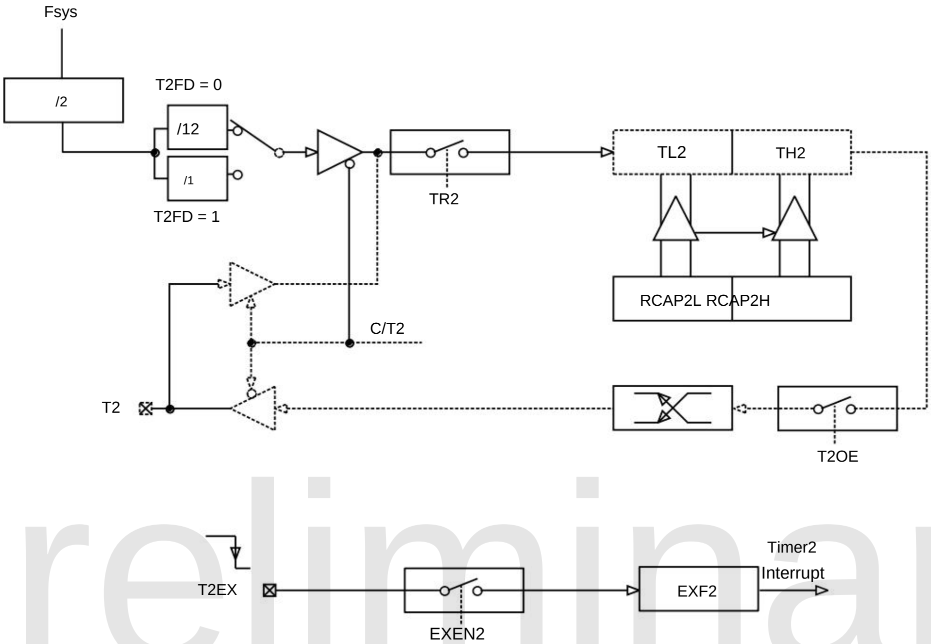
$$\text{Colck Out Frequency} = ; \frac{\text{fn2}}{(65536ÿ[\text{RCAP2H,RCAP2L}])\times 4}$$

Where fn2 is the clock frequency of Timer 2:

$$fn2 = \frac{fsys}{12}; T2FD = 0$$

$$fn2 = fsys; T2FD = 1$$

Timer 2 overflows without generating an interrupt; port T2 is used as the clock output.



Mode 3: Programmable clock output

Note:

1. Both TF2 and EXF2 can trigger an interrupt request from Timer 2, and they share the same vector address; 2. TF2 and EXF2 can be set to 1 by software when an event occurs or at any other time. Only software and hardware resets can clear them to 0; 3. When EA = 1 and ET2 = 1, setting TF2 or EXF2 to 1 will trigger an interrupt from Timer 2; 4. When Timer 2 is used as a baud rate generator, writing to TH2/TL2 or RCAP2H/RCAP2L will affect the accuracy of the baud rate and cause communication errors.

12 Multiplication and Division Machine

The SC92F836XB provides a 16-bit multiplier/divider, consisting of extended accumulators EXA0~EXA3, an extended B register EXB, and an arithmetic control register.

It consists of the OPERCON register. It can replace software to perform 16-bit × 16-bit multiplication and 32-bit/16-bit division operations.

Symbolic address	illustrate	7	6	5	4	3	2	1	0 Reset value
EXA0	E9H Extended Accumulator 0	EXA [7:0]							00000000b
EXA1	EAH Extended Accumulator 1	EXA [15:8]							00000000b
EXA2	EBH Extended Accumulator 2	EXA [23:16]							00000000b
EXA3	ECH Extended Accumulator 3	EXA [31:24]							00000000b
EXBL	EDH Extended B Register L	EXB [7:0]							00000000b
EXBH	EEH Extended B Register H	EXB [15:8]							00000000b

OPERCON (EFH) Operation Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol OPERS MD Read/Write Read/Write Read/			-	-	-	-	-	CHKSUMS
Write Power-on initial value 0 0			-	-	-	-	-	Reading/Writing
			x	x	x	x	x	0

Position number	Bit symbol	illustrate
7	**OPERS**	Operator Start Control for Multiplier/Divider Operations Writing "1" to this bit initiates a multiplication or division calculation; that is, this bit simply marks the start of the multiplication or division operation. The trigger signal indicates that the calculation is complete when this bit is zero. This bit can only be written with a 1 to be valid.
6	**MD**	Multiplication and division selection 0: Multiplication operation, writing the multiplicand and multiplier, and reading the product are as follows:

byte / Operand	Byte 3	Byte 2	Byte 1	Byte 0
multiplicand 16 bits,	-	-	EXA1	EXA0
multiplier 16 bits,	-	-	EXBH	EXBL
product 32 bits	EXA3	EXA2	EXA1	EXA0

1. Division operation: The writing of the dividend and divisor, and the reading of the quotient and remainder are as follows:

byte / Operands:	Byte 3	Byte 2	Byte 1	Byte 0
Dividend 32 bits,	EXA3	EXA2	EXA1	EXA0
Divisor 16 bits,	-	-	EXBH	EXBL
Quotient 32	EXA3	EXA2	EXA1	EXA0
bits, Remainder 16 bits	-	-	EXBH	EXBL

Note:

1. During the execution of arithmetic operations, reading or writing to the EXA and EXB data registers is prohibited.

2. The time required for multiplication and division operations to be converted is... 16/fSYSÿ

13 PWM

The SC92F836XB provides an independent counter that supports six PWM outputs: PWM0~5.

The PWM function of the SC92F836XB has the following features:

ÿ 10-bit PWM precision; ÿ Output

can be set to forward or reverse; ÿ

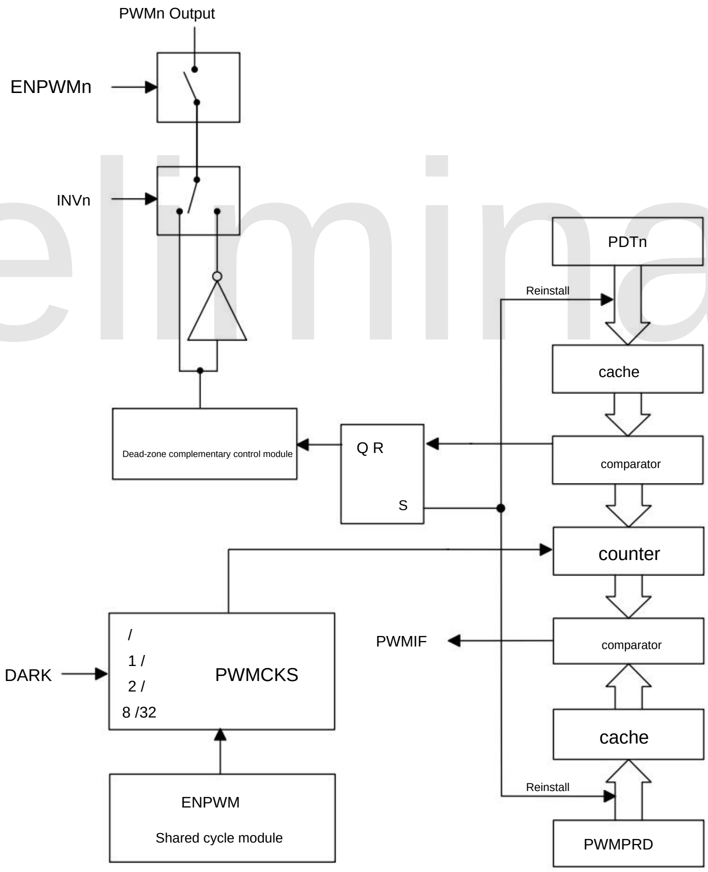
Available in independent and complementary modes:

1) In independent mode, PWM0~5 have the same cycle, but the duty cycle of each PWM output waveform can be set independently; 2) In complementary mode, three sets of complementary PWM waveforms with dead time can be output simultaneously.

ÿ Provides one PWM overflow interrupt.

The SC92F836XB's PWM supports adjustment of the period and duty cycle. The PWMCFG and PWMCON registers control the PWM state and period. During this period, the activation and duty cycle of each PWM channel can be adjusted individually.

13.1 PWM Structure Block Diagram



SC92F836XB PWM Block Diagram

13.2 PWM- related SFR registers

Symbol	address	description	7	6	5	4	3	2	1	0	Reset value
PWMCFG	D1H	PWM Setting Register	PWMCKS[1:0]		INV5	INV4	INV3	INV2	INV1	INV0	00000000b
PWMCON	D2H	PWM Control Register	ENPWM	PWMIF	ENPWM5	ENPWM4	ENPWM3	ENPWM2	ENPWM1	ENPWM0	00000000b
PWMPRD	D3H	PWM Period Setting Register	PWMPRD[9:2]								00000000b
PWMDTYA	D4H	PWM Duty Cycle Setting Register A	PWMPRD[1:0]		PDT2[1:0]		PDT1[1:0]		PDT0[1:0]		00000000b
PWMDTY0	D5H	PWM0 Duty Cycle Setting Register	PDT0[9:2]								00000000b
PWMDTY1	D6H	PWM1 Duty Cycle Setting Register	PDT1[9:2]								00000000b
PWMDTY2	D7H	PWM2 Duty Cycle Setting Register	PDT2[9:2]								00000000b
PWMDTYB	DCH	PWM duty cycle setting register B	PWMMOD	-	PDT5[1:0]		PDT4[1:0]		PDT3[1:0]		0x000000b
PWMDTY3	DDH	PWM3 Duty Cycle Setting Register / PWM dead time configuration register	PDT3[9:2]								00000000b
PWMDTY4	DEH	PWM4 Duty Cycle Setting Register	PDT4[9:2]								00000000b
PWMDTY5	DFH	PWM5 Duty Cycle Setting Register	PDT5[9:2]								00000000b
IE1	A9H	Interrupt Enable Register 1	-	-	-	ETK	EINT2	EBTM	EPWM	ESSI	xxx00000b
IP1	B9H	Interrupt Priority Control Register 1	-	-	-	IPTK	IPINT2	IPBTM	IPPWM	IPSSI	xxx00000b

13.3 PWM General Configuration Register

The PWM of the SC92F836XB is divided into independent mode and complementary mode. The registers shared by these two modes are as follows:

Users can select four PWM clock sources by configuring PWMCFG[7:6]. INV0~5 are used to select whether the outputs of PWM0~5 are inverted.

PWMPRD[9:0] is a shared period setting controller for six PWM channels. Whenever the PWM counter reaches the preset value of PWMPRD[9:0], the next... When a PWM CLK arrives, the counter will jump to 00h, meaning that the cycles of PWM0~5 are all (PWMPRD[9:0] + 1) * PWM clock.

Users can configure the shared cycle of PWM0~5 by configuring PWMPRD[7:0] and PWMDTYA[7:6].

Note: To ensure correct data writing, write operations to **the PWM** period register must follow the order of the lower **2** bits first, followed by the higher **8** bits.

IE1 (A9H) Interrupt Enable Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	ETC	ONE2	EBTM	EPWM	ESSI
reads/writes	-	-	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value	x	x	x	0	0	0	0	0

Position number	Bit symbol	illustrate
1	**EPWM**	PWM interrupt enable control 0: Disable PWM interrupt 1: Enable interrupt generation when PWM counter overflows.

IP1 (B9H) Interrupt Priority Register **1 (Read/Write)**

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	IPTK	POINT2	IPBTM	IPPWM	IPSSI
read/write	-	-	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value	x	x	x	0	0	0	0	0

Position number	Bit symbol	illustrate
1	**IPPWM**	PWM interrupt priority selection 0: Sets the PWM interrupt priority to "low". 1. Set the PWM interrupt priority to "high".

PWMCON (D2H) PWM control register **(read/write)**

Bit number	7	6	5	4	3	2	1	0
symbols	ENPWM	PWMIF	ENPWM5	ENPWM4	ENPWM3	ENPWM2	ENPWM1	ENPWM

								0
Read/	Read/Write Read/Write Read/Write			Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
Write power-on initial value 0		0	0	0	0	0	0	0

Position number	Bit symbol	illustrate
7	**ENPWM**	PWM module switching control (Enable PWM) 1: Allow the Clock to enter the PWM unit and start PWM operation. 0: The PWM unit stops working, and the PWM counter is cleared. PWMn remains connected to the output port. To use other functions multiplexed with the **PWMn** output, **ENPWMn** should be set to **0.**
6	**PWMIF**	PWM Interrupt Flag When the PWM counter overflows (that is, when it counts beyond PWMPRD), this bit will be... The hardware is automatically set to 1. If IE1[1] (EPWM) is also set to 1 at this time, a PWM interrupt is generated. Note: The six PWMs share a common cycle, and the PWM interrupt generated when overflow occurs is from the same interrupt vector.
5~0	**ENPWMx** **ÿx=0~5ÿ**	PWMx function switch 0: PWMx does not output to I/O. 1: PWMx output to IO

PWMCFG (D1H) PWM setting register (read/write)

Position	7	6	5	4	3	2	1	0
number symbol	PWMCKS[1:0] Read/		INV5	INV4	INV3	INV2	INV1	INV0
Write Read/Write Read/Write Read/Write Read/Write Read/Write Initial value 0 upon power-on						Read/	Read/	Reading/Writing
		0	0	0	0	Write 0	Write 0	0

Position	Bit symbol	illustrate
number 7~6	**PWMCKS[1:0]** PWM Clock Source Selector	00: fOSC 01: fOSC/2 10: fOSC/8 11: fOSC/32 For the definition of fOSC, see the block diagram in section 7.4, "High-Frequency System Clock Circuits".
5~0	**INVx** **ÿx=0~5ÿ**	PWMx output inverse control 0: The output of PWMx is not reversed. 1: PWMx output inverted

PWMPRD (D3H) PWM Period Setting Register (Read/Write)

Bit number **5** Symbol read/write Read/write Power-on initial value 0	7	6	5	4	3	2	1	0
	PWMPRD[9:2] Read/							
		Reading/Writing	Reading/Writing	Write Read/Write		Reading/Writing	Reading/Writing	Reading/Writing
		0	0	0	0	0	0	0

PWMDTYA (D4H) PWM Duty Cycle Setting Register **A (Read/Write)**

Position	7	6	5	4	3	2	1	0
number symbol	PWMPRD[1:0] Read/		PDT2[1:0]		PDT1[1:0] Read/		PDT0[1:0]	
Write Read/Write Read/Write Power-on initial			Write Read/Write Read/Write Read/Write 0 0				Reading/Writing	
value 0 0				0	0		0 0	

Position	Bit symbol	illustrate
number 7~6	**PWMPRD[9:0]** Period setting bits shared by PWM0 ~ PWM5;	This value represents the (period **–** 1) of the output waveforms of PWM0 ~ PWM5; that is, the PWM output... The output period value is (PWMPRD[9:0] + 1) * PWM clock;

PWMDTYB (DCH) PWM duty cycle setting register **B (read/write)**

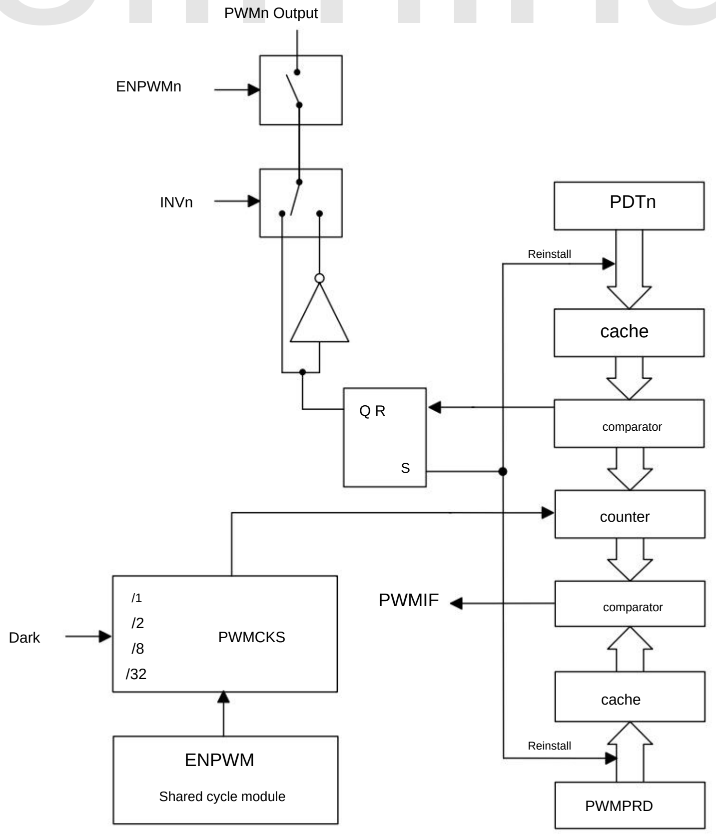
Bit-	7	6	5	4	3	2	1	0
numbered symbol PWMMOD - Read/Write Read/			PDT5[1:0]		PDT4[1:0] Read/		PDT3[1:0]	
Write Power-on initial value 0		-	Write Read/Write Read/Write Read/Write				Reading/Writing	
		x	0	0	0	0	0	0

Position number	Bit symbol	illustrate
7	**PWMMOD**	PWM mode settings: 0: Independent mode: Duty settings for PWM0~5 (6 PWM channels); 1. Complementary Mode: PWM0/3, PWM1/4, and PWM2/5 are divided into three groups. PWMs in the same group... The output pulse widths are the same, controlled by PDT0~2[9:0] respectively, and can be controlled by register PDT3. Set the dead time. Notice: If ENPWM is set to 1, the PWM module is enabled; however, if ENPWMn = 0, the PWM output is disabled. It is then turned off and used as a GPIO port. At this point, the PWM module can be used as a 10-bit timer. At this time, EPWM (IE1.1) is set to 1, and PWM will still generate interrupts.

13.4 Independent **PWM** Mode

In standalone mode (PWMMOD = 0), PWMDTY0~5, PWMDTYA, and PWMDTYB are used as duty cycle setting registers for PWM0~5 channels. Register. After configuring the PWM output state and period, the user can output at a fixed duty cycle by configuring the corresponding PWM channel's duty cycle register. PWM waveform.

13.4.1 Block Diagram of PWM Independent Mode



SC92F836XB PWM Independent Mode Block Diagram

13.4.2 PWM Independent Mode Duty Cycle Configuration

To ensure correct data writing, write operations to **the PWM** duty cycle register must follow the order of the lower **2** bits first, followed by the higher **8** bits.

PWMDTY0 (D5H) PWM0 Duty Cycle Setting Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	PDT0[9:2]							
read/write Read/write Power-on initial		Read/	Read/	Reading/Writing		Read/	Read/	Reading/Writing
value 0		Write 0	Write 0	0	0	Write 0	Write 0	0

PWMDTY1 (D6H) PWM1 Duty Cycle Setting Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	PDT1[9:2]							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

PWMDTY2 (D7H) PWM2 Duty Cycle Setting Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	PDT2[9:2]							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

PWMDTY3 (DDH) PWM3 Duty Cycle Setting Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	PDT3[9:2]							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

PWMDTY4 (DEH) PWM4 Duty Cycle Setting Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	PDT4[9:2]							
read/write Read/write Power-on initial		Read/	Read/	Reading/Writing		Read/	Read/	Reading/Writing
value 0		Write 0	Write 0	0	0	Write 0	Write 0	0

PWMDTY5 (DFH) PWM5 Duty Cycle Setting Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	PDT5[9:2]							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 7~0	**PDTx[9:2]** **ÿx=0~5ÿ**	Standalone mode: The PWMx duty cycle length is set in the high 8 bits. The high-level width of PWMx is (PDTx[9:0]) PWM clock cycles.

PWMDTYA (D4H) PWM Duty Cycle Setting Register **A (Read/Write)**

Position	7	6	5	4	3	2	1	0
number symbol	PWMPRD[1:0] Read/Write		PDT2[1:0] Read/		PDT1[1:0] Read/		PDT0[1:0]	
Read/Write Read/Write Power-on initial value 0			Write Read/Write		Write Read/Write		Reading/Writing	
		0	0	0	0	0	0	0

PWMDTYB (DCH) PWM duty cycle setting register **B (read/write)**

Bit number	7	6	5	4	3	2	1	0
symbol PWMMO	D	-	PDT5[1:0]		PDT4[1:0]		PDT3[1:0]	
Read/Write Read/Write Power-on		-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		x	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 5~0	**PDTx [1:0]** ÿx=0~5ÿ	The PWMx duty cycle length is set to the lower 2 bits; The high-level width of PWMx is (PDTx[9:0]) PWM clock cycles.

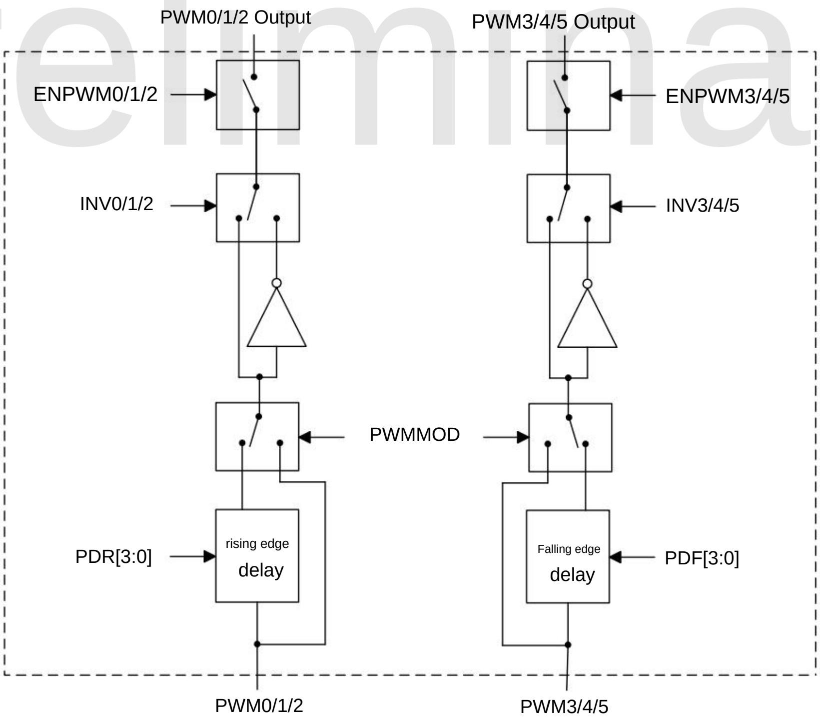
13.5 PWM Complementary Mode

When the SC92F836XB's PWM operates in complementary mode, the dead-time control module can prevent the effective time zones of the two complementary PWM signals from being interrupted. They overlap to ensure that a pair of complementary power switches driven by the PWM signal will not conduct simultaneously in practical applications.

In complementary mode (PWMMOD = 1), PWM0 and PWM3 are grouped together, with their duty cycles adjusted via PDT0[9:0]; PWM1 and PWM4 are grouped together. The duty cycle is adjusted via PDT1[9:0]; PWM2 and PWM5 are a group, and their duty cycles are adjusted via PDT2[9:0].

In complementary mode, registers PWMDTY4~5 are invalid, and the bits of register PWMDTY3 are redefined as the falling edge dead time of PWM3/4/5. Control bits PDF[3:0] and PWM0/1/2 rising edge dead time control bits PDR[3:0].

13.5.1 Block Diagram of PWM Complementary Mode



SC92F836XB PWM Complementary Mode Block Diagram

13.5.2 Duty Cycle Configuration for Complementary PWM Mode

To ensure correct data writing, write operations to **the PWM** duty cycle register must follow the order of the lower **2** bits first, followed by the higher **8** bits.

PWMDTY0 (D5H) PWM0 Duty Cycle Setting Register (Read/Write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol	**PDT0[9:2]**							
read/write Read/write Power-on initial		Read/	Read/	Reading/Writing		Read/	Read/	Reading/Writing
value 0		Write 0	Write 0	0	0	Write 0	Write 0	0

PWMDTY1 (D6H) PWM1 Duty Cycle Setting Register (Read/Write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol	**PDT1[9:2]**							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

PWMDTY2 (D7H) PWM2 Duty Cycle Setting Register (Read/Write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol	**PDT2[9:2]**							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

PWMDTYA (D4H) PWM Duty Cycle Setting Register **A (Read/Write)**

Position	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
number symbol	PWMPRD[1:0]	Read/Write	PDT2[1:0] Read/		PDT1[1:0] Read/		PDT0[1:0]	
Read/Write Read/Write	Power-on initial value	0	Write Read/Write		Write Read/Write		Reading/Writing	
		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 5~0	**PDTx [1:0]** **ÿx=0~3ÿ**	The PWMx duty cycle length is set to the lower 2 bits; The high-level width of PWMx is (PDTx[9:0]) PWM clock cycles.

13.5.3 Dead Time Configuration in PWM Complementary Mode

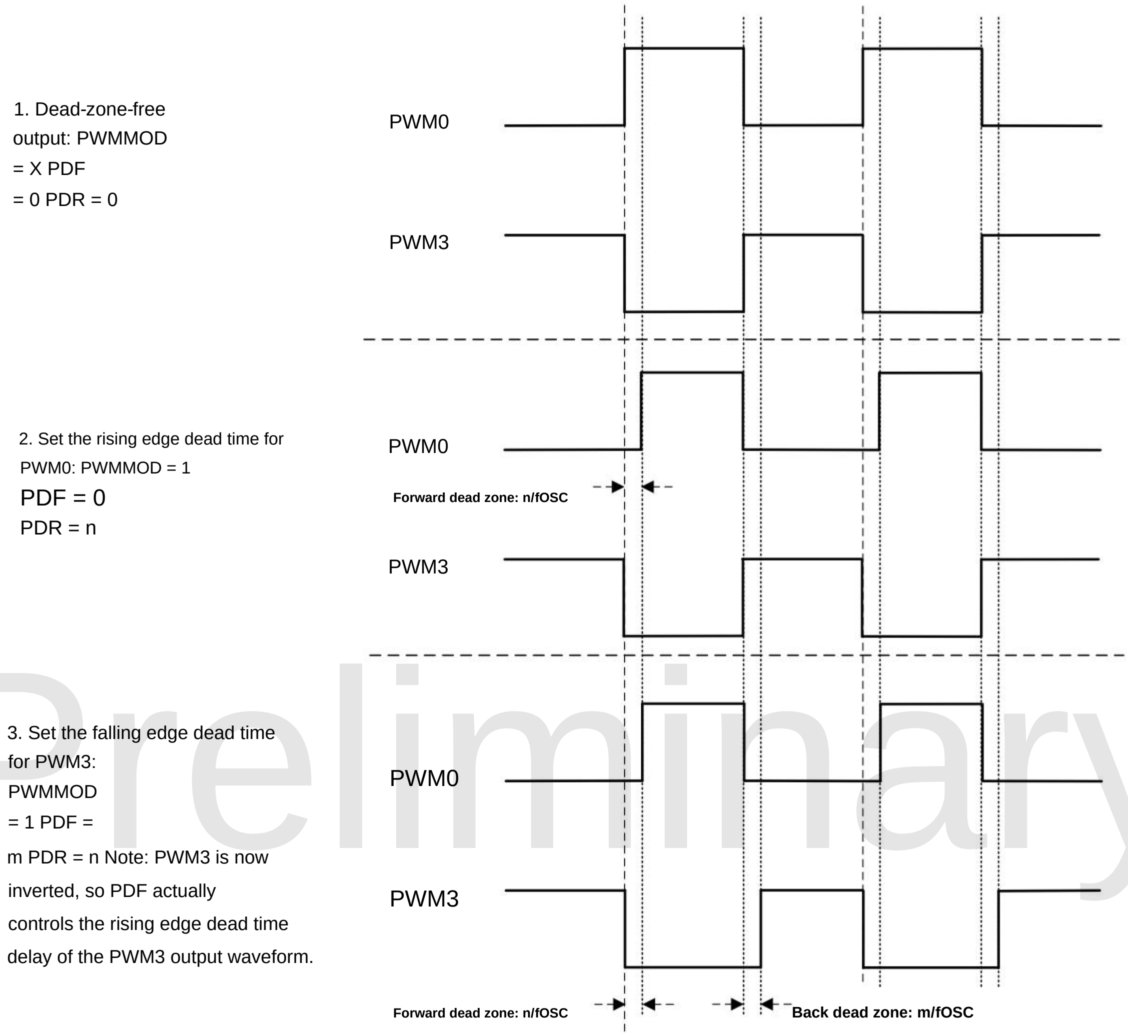
PWMDTY3 (DDH) PWM Dead Time Configuration Register (Read/Write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol	**PDF[3:0]** Read/				**PDR[3:0]**			
read/write Read/write Power-on initial		Write Read/Write		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing
value 0		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 7~4	**PDF[3:0]**	Complementary mode: PWM3/4/5 Falling Edge Dead Time = PDF[3:0] / fOSC
3~0	**PDR[3:0]**	Complementary mode: PWM0/1/2 Rising edge dead time = PDR[3:0] / fOSC

13.5.4 **PWM** Dead-Time Output Waveform

The following diagram shows the dead time adjustment waveforms of PWM0 and PWM3 in complementary mode. For easy differentiation, PWM3 has been reversed (INV3=1).

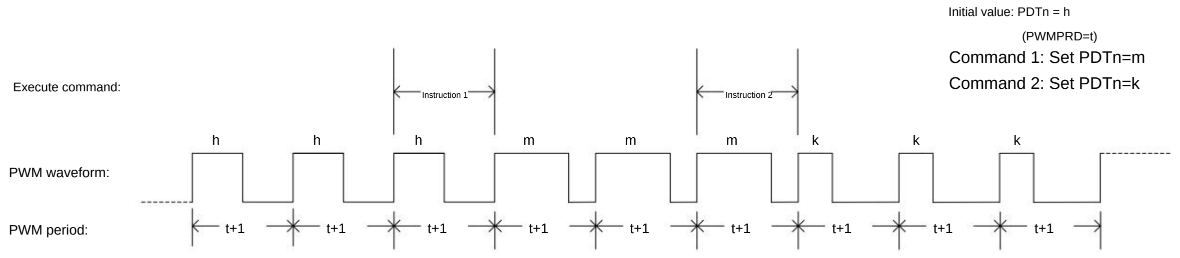


PWM dead-time output waveform

13.6 PWM Waveform and Usage

The effects of changes in each SFR parameter on the PWM waveform are as follows:

ÿ Duty cycle variation characteristics



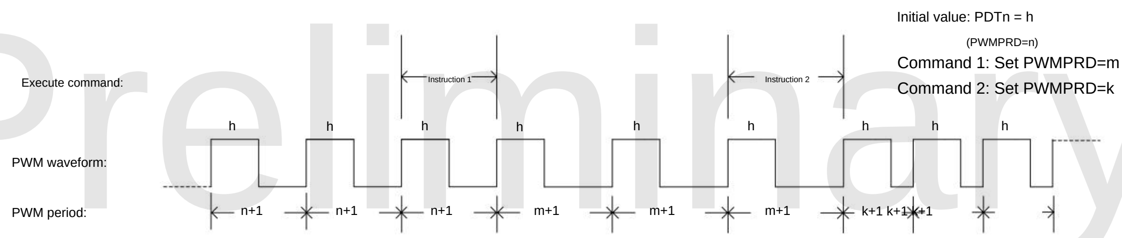
Duty cycle variation characteristic diagram

When the PWMn output waveform is active, if you need to change the duty cycle, you can do so by changing the value of the high-level setting register (PDTn). However, it should be noted that changing...

The duty cycle of PDTn does not change immediately; instead, it waits until the end of the current cycle before changing in the next. To ensure correct data writing, the PWM...

Write operations to the cycle and DUTY registers must follow the order of the lower 2 bits first, followed by the higher 8 bits. The relevant waveform output is shown in the figure above.

ÿ Periodic variation characteristics

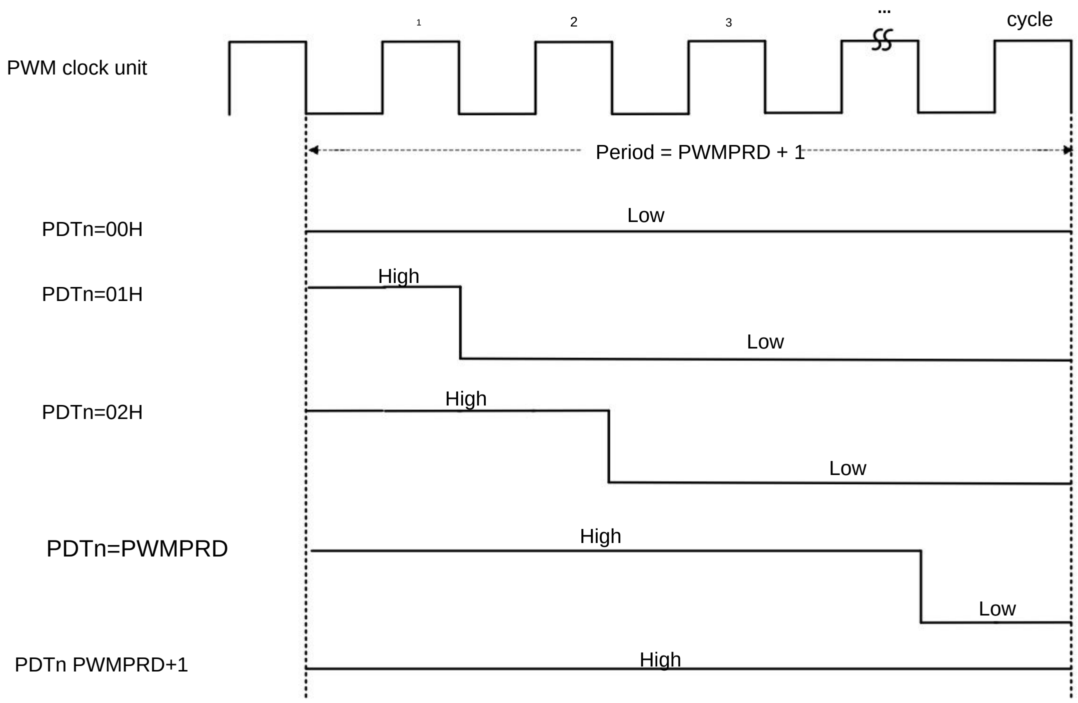


Periodic variation characteristic diagram

When the PWMn output waveform is in operation, if the period needs to be changed, this can be achieved by changing the value of the period setting register PWMPRD. This is similar to changing the duty cycle.

Changing the value of PWMPRD will not change the period immediately, but will wait until the end of the current period and then change it in the next period, as shown in the figure above.

ÿ Relationship between period and duty cycle



Relationship between period and duty cycle

The relationship between the period and duty cycle is shown in the figure above. This result assumes that the PWMn output reverse control (INVn) is initially 0. To obtain the opposite result, INVn can be set to 1.

14 GP I/O

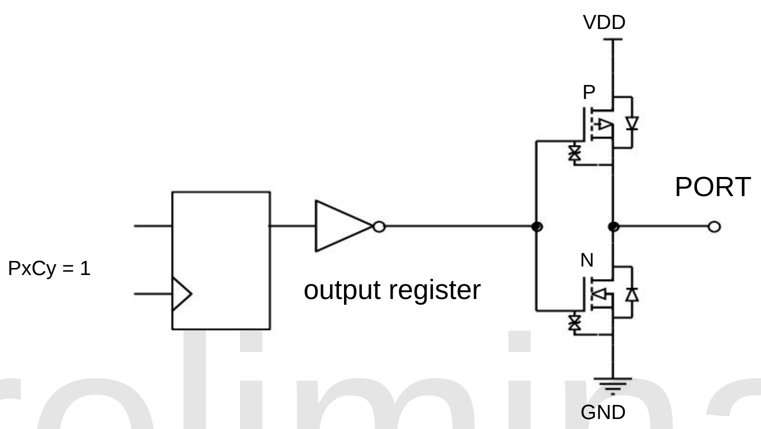
The SC92F836XB provides up to 26 controllable bidirectional GPIO ports. Input/output control registers are used to control the input/output status of each port. When a port is used as an input, each I/O port has an internal pull-up resistor controlled by PxPHy. These 26 I/O ports are multiplexed with other functions. P0.0~P0.4 can be used as COM drivers for LCD displays by setting half of VDD as the output voltage. Whether the I/O port is in input or output mode, the data read from the port data register reflects the actual status value of the port.

Note: Unused and unpackaged **I/O** ports must be set to push-pull output mode.

14.1 GPIO Structure Diagram

Strong push-pull output

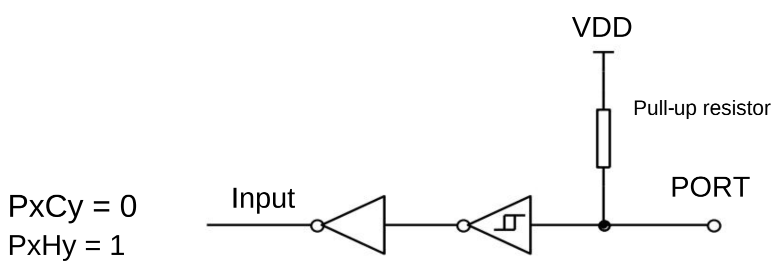
mode: In strong push-pull output mode, it can provide continuous high current drive: output high greater than 20mA, output low greater than 70mA. The port structure diagram of strong push-pull output mode is shown below:



Strong push-pull output mode

In pull-up input mode, a pull-

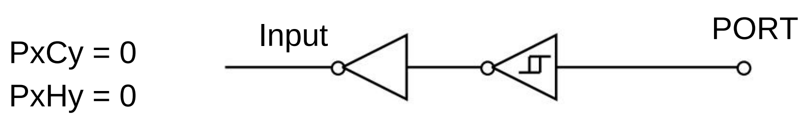
up resistor is always connected to the input port. A low-level signal is only detected when the input port voltage is pulled low. The port structure diagram for pull-up input mode is shown below:



Input mode with pull-up

The port structure diagram for high

-impedance input mode is shown below:



High-impedance input mode

14.2 I/O Port Related Registers

P0CON (9AH) P0 port input/output control register (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	P0C7	P0C6	P0C5	P0C4	P0C3	P0C2	P0C1	P0C0
read/write Read/write Power-on		Read/	Read/	Read/	Read/	Read/	Read/	Reading/Writing
initial value 0		Write 0	Write 0	Write 0	Write 0	Write 0	Write 0	0

P0PH (9BH) P0 port pull-up resistor control register (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	P0H7	P0H6	POH5	P0H4	P0H3	P0H2	P0H1	P0H0
read/write Read/write Power-on		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	0	0	0	0	0

P1CON (91H) P1 Port Input/Output Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	P1C7	P1C6	P1C5	P1C4	P1C3	P1C2	P1C1	P1C0
read/write Read/write Power-on		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	0	0	0	0	0

P1PH (92H) P1 port pull-up resistor control register (read/write)

Bit number **6 5** Symbol read/write Read/write Power-on initial value 0	7			4	3	2	1	0
	P1H7	P1H6	P1H5	P1H4	P1H3	P1H2	P1H1	P1H0
		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
		0	0	0	0	0	0	0

P2CON (A1H) P2 Port Input/Output Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	P2C7	P2C6	P2C5	P2C4	P2C3	P2C2	P2C1	P2C0
read/write Read/write Power-on		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	0	0	0	0	0

P2PH (A2H) P2 port pull-up resistor control register (read/write)

Bit number **6 5** Symbol read/write Read/write Power-on initial value 0	7			4	3	2	1	0
	P2H7	P2H6	P2H5	P2H4	P2H3	P2H2	P2H1	P2H0
		Read/	Read/	Read/	Read/	Read/	Read/	Reading/Writing
		Write 0	Write 0	Write 0	Write 0	Write 0	Write 0	0

P5CON (D9H) P5 Port Input/Output Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	-	-	-	P5C1	P5C0
read/	-	-	-	-	-	-	Reading/Writing	Reading/Writing
write power-on initial value x		x	x	x	x	x	0	0

P5PH (DAH) P5 port pull-up resistor control register (read/write)

Bit number **5** symbol reads/writes power-on initial value x	7	6		4	3	2	1	0
	-	-	-	-	-	-	P5H1	P5H0
	-	-	-	-	-	-	Reading/Writing	Reading/Writing
		x	x	x	x	x	0	0

Position	Bit symbol	illustrate
number 7~0	**PxCy** **(x=0~2.5, y=0~7)**	Px port input/output control: 0: Pxy is the input mode (initial power-on value) 1: Pxy is a push-pull output mode.
7~0	**PxHy** **(x=0~2.5, y=0~7)**	The pull-up resistor setting for port Px is only effective when PxCy=0: 0: Pxy is in high-impedance input mode (initial power-on value), pull-up resistor is off; 1: Pxy pull-up resistor turned on

P0 (80H) P0 port data register (read/write)

Bit number **6** Symbol read/write Read/write Power-	7		**5**	**4**	**3**	**2**	**1**	**0**
on initial	P0.7	P0.6	P0.5	P0.4	P0.3	P0.2	P0.1	P0.0
value 0		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
		0	0	0	0	0	0	0

P1 (90H) P1 port data register (read/write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol	P1.7	P1.6	P1.5	P1.4	P1.3	P1.2	P1.1	P1.0
read/write Read/write Power-on		Read/	Read/	Read/	Read/	Read/	Read/	Reading/Writing
initial value 0		Write 0	Write 0	Write 0	Write 0	Write 0	Write 0	0

P2 (A0H) P2 port data register (read/write)

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol	P2.7	P2.6	P2.5	P2.4	P2.3	P2.2	P2.1	P2.0
read/write Read/write Power-on		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	0	0	0	0	0

P5 (D8H) P5 port data register (read/write)

Bit number **6** symbol reads/writes power-on initial	7		**5**	**4**	**3**	**2**	**1**	**0**
value x	-	-	-	-	-	-	P5.1	P5.0
	-	-	-	-	-	-	Reading/Writing	Reading/Writing
		x	x	x	x	x	0	0

IOHCON (97H) IOH Set Register (Read/Write)

Position	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
number symbol	P2H[1:0] Read/		P2L[1:0] Read/		P0H[1:0] Read/		P0L[1:0]	
Write Read/Write	Read/Write Power-on	initial value 0	Write Read/Write		Write Read/Write		Reading/Writing	
		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 7~6	P2H[1:0]	P2 High 4 Bits IOH Settings 00: Set the high four bits of P2 IOH to level 0 (maximum); 01: Set the high four bits of P2's IOH to level 1; 10: Set the high four bits of P2 IOH to level 2; 11: Set the high four bits of P2 IOH to level 3 (minimum);
5~4	P2L[1:0]	P2 Low four bits IOH settings 00: Set the lower four bits of P2 IOH to level 0 (maximum); 01: Set the lower four bits of P2's IOH to level 1; 10: Set the lower four bits of P2 IOH to level 2; 11: Set the lower four bits of P2 IOH to level 3 (minimum);
3~2	P0H[1:0]	P0 high four bits IOH settings 00: Set the high four bits of P0 IOH to level 0 (maximum); 01: Set the high four bits of P0 IOH to level 1;

		10: Set the high four bits of P0 IOH to level 2; 11: Set the high four bits of P0 IOH to level 3 (minimum);
1~0	P0L[1:0]	P0 lower four bits IOH settings 00: Set the lower four bits of P0 IOH to level 0 (maximum); 01: Set the lower four bits of P0 IOH to level 1; 10: Set the lower four bits of P0 IOH to level 2; 11: Set the lower four bits of P0 IOH to level 3 (minimum);

15 Software LCD Driver

The P0.0~P0.4 pins of the SC92F836XB can be used as COM ports for a software LCD. In addition to their normal I/O functions, these pins can also output 1/2VDD voltage.

Pressure. Users can select the appropriate IO as the COM for the LCD driver according to their usage.

15.1 Software LCD Driver Related Registers

LCD driver-related SFR register description:

P0VO (9CH) P0 port **LCD** voltage output register (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	P04VO P03VO P02VO P01VO P00VO				
read/write	-	-	-	Read/	Read/	Read/	Read/	Reading/Writing
power-on initial value x		x	x	Write 0	Write 0	Write 0	Write 0	0

P0yVO (y=0~4) 0	**P0y** port output selection
	General I/O ports
1	The output voltage of port P0y is 1/2VDD;

OTCON (8FH) Output Control Register (Read/Write)

Position	7	6	5	4	3	2	1	0
number symbol	SSMOD[1:0] Read/Write		-	-	VOIRS[1:0] Read/Write		-	-
Read/Write Read/Write Power-on initial value 0			-	-	Read/Write		-	-
		0	x	x	0	0	x	x

Position	Bit symbol	illustrate
number 3~2	**VID[1:0]**	**LCD** voltage output port voltage divider resistor selection (select the appropriate driver based on the **LCD** screen size) 00: Turn off the internal voltage divider resistor (to save power) 01: Set the internal voltage divider resistor to 12.5K. 10: Set the internal voltage divider resistor to 37.5K. 11: Set the internal voltage divider resistor to 87.5K.

16 UART0

The SC92F836XB supports a full-duplex serial port, which can be easily used to connect to other devices or equipment, such as Wi-Fi modules or other components.

Driver chips for the UART communication interface, etc. The functions and characteristics of UART0 are as follows:

1. Three communication modes are available: Mode 0, Mode 1, and Mode 3;
2. Timer 1 or Timer 2 can be selected as the baud rate generator;
3. The completion of sending and receiving can generate an interrupt RI/TI, and the interrupt flag needs to be cleared by software.

SCON (98H) Serial Port Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	SM0	SM1	SM2	REN	TB8	RB8	OF	RI
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

Position number	Bit symbol	illustrate
7~6	**SM0~1**	Serial communication mode control bits 00: Mode 0, 8-bit half-duplex synchronous communication mode, transmitting and receiving serial data on the RX pin. The TX pin is used as the transmit shift clock. Each frame transmits and receives 8 bits, with the least significant bit received or transmitted first. 01: Mode 1, 10-bit full-duplex asynchronous communication, consisting of 1 start bit, 8 data bits, and 1... Composed of stop bits, the communication baud rate is variable; 10: Retained; 11: Mode 3, 11-bit full-duplex asynchronous communication, consisting of 1 start bit, 8 data bits, and one... It consists of a programmable 9th bit and 1 stop bit, allowing for variable communication baud rate.
5	**SM2**	Serial communication mode control bit 2. This control bit is only valid for modes 2 and 3. 0: Set RI to generate an interrupt request whenever a complete data frame is received; 1. When a complete data frame is received, RI will only be set and an interrupt generated if RB8=1. beg.
4	**REN**	Receive enable control bit 0: Data reception is not allowed; 1: Allow receiving data.
3	**TB8**	Only valid for modes 2 and 3, and is the 9th bit of the transmitted data.
2	**RB8**	Only valid for modes 2 and 3, and is the 9th bit of the received data.
1	**OF**	Send interrupt flag
0	**RI**	Receive interrupt flag

SBUF (99H) Serial Port Data Buffer Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	SBUF[7:0]							
read/write Read/write Power-on		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	0	0	0	0	0

Position	The sign bit	illustrate
number 7~0	**SBUF[7:0]** serial port data buffer register	SBUF contains two registers: a transmit shift register and a receive latch. The data in SBUF will be sent to the transmit shift register, and the transmit process will begin. Reading SBUF will return... Return the contents of the latch.

PCON (87H) Power Management Control Register (Write-only, *Unreadable *****)

Bit number	7	6	5	4	3	2	1	0
symbol SMOD Read/Write Write-only		-	-	-	-	-	STOP	IDL
Power-on initial value 0		-	-	-	-	-	Write only	Write only
		x	x	x	x	x	0	0

Position number	Bit symbol	illustrate
7	**SMOD**	The baud rate multiplier setting bit is only valid in mode **0 (SM0~1 =** 00): 0: The serial port operates at 1/12 of the system clock. 1: The serial port operates at 1/4 of the system clock.

16.1 Baud Rate of Serial Communication

In Mode 0, the baud rate can be programmed to be 1/12 or 1/4 of the system clock, determined by the SMOD (PCON.7) bit. When SMOD is 0, the serial port... It operates at 1/12 of the system clock. When SMOD is 1, the serial port operates at 1/4 of the system clock.

In Mode 1 and Mode 3, the baud rate can be selected from the overflow rate of Timer 1 or Timer 2.

Set TCLK (T2CON.4) and RCLK (T2CON.5) to 1 respectively to select Timer 2 as the baud clock source for TX and RX (see Timer for details). (Chapter). Regardless of whether TCLK or RCLK is logic 1, Timer 2 will operate in baud rate generator mode. If TCLK and RCLK are logic 0, Timer 2 will operate in baud rate generator mode.

Device 1 serves as the baud clock source for Tx and Rx.

The baud rate formulas for Mode 1 and Mode 3 are shown below, where [TH1, TL1] are the 16-bit counter register of Timer 1, [RCAP2H, ...

RCAP2L] is the 16-bit overload register for Timer 2.

1. When using Timer 1 as a baud rate generator, Timer 1 must stop counting, i.e., TR1=0:

$$\text{BaudRate} = \frac{\text{fsys}}{[\text{TH1,TL1}]}$$ (Note: [TH1,TL1] must be greater than 0x0010)

2. Use Timer 2 as a baud rate generator:

$$\text{BaudRate} = \frac{\text{fsys}}{[\text{RCAP2H,RCAP2L}]}$$ (Note: [RCAP2H, RCAP2L] must be greater than 0x0010)

17. SPI/TWI/UART 3-in-1 serial interface SSI

The SC92F836XB integrates a 3-to-1 serial interface circuit (SSI), which facilitates the connection of the MCU with devices or equipment with different interfaces.

The user can configure the SSI interface as any of the SPI, TWI, and UART communication modes by configuring the SSMOD[1:0] bits of the OTCON register.

The formula has the following characteristics:

1. The SPI mode can be configured to either master or slave mode.
2. In TWI mode communication, it can only act as a slave device.
3. The UART mode can operate in mode 1 (10-bit full-duplex asynchronous communication) and mode 3 (11-bit full-duplex asynchronous communication).

The specific configuration method is as follows:

OTCON (8FH) Output Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol SSMOD[1:0] Read/Write Read/Write Read/			-	-	VOIRS[1:0] Read/		-	-
Write Power-on initial value 0			-	-	Write Read/Write		-	-
		0	x	x	0	0	x	x

Position	Bit symbol	illustrate
number 7~6	**SSMOD[1:0]**	**SSI** communication mode control bit 00: SSI Off 01: SSI is set to SPI communication mode; 10: SSI is set to TWI communication mode; 11: SSI is set to UART communication mode;

17.1 SPI

SSMOD[1:0] = 01, the three-to-one serial interface SSI is configured as an SPI interface. Serial External Device Interface (SPI) is a high-speed serial communication...

The interface allows the MCU to perform full-duplex, synchronous serial communication with peripheral devices (including other MCUs).

17.1.1 SPI Operation Related Registers

SSCON0 (9DH) SPI Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol SPEN Read/Write Read/		-	MSTR CPOL CPHA SPR2				SPR1	SPR0
Write Power-on initial value 0		-	Read/	Read/	Read/	Read/	Read/	Reading/Writing
		x	Write 0	Write 0	Write 0	Write 0	Write 0	0

Position number	Bit symbol	illustrate
7	**SPAIN**	**SPI** Enable Control 0: Disable SPI 1: Enable SPI
5	**MSTR**	**SPI** Master-Slave Selection 0: SPI is for slave device

		1: SPI is the master device
4	**CPOL**	Clock polarity control bit 0: SCK is low in the idle state. 1: SCK is high in the idle state.
3	**CPHA**	Clock phase control bit 0: Data is acquired at the first edge of the SCK cycle. 1: Data acquired at the second edge of the SCK cycle
2~0	**SPR[2:0]**	**SPI** clock rate selection bit 000ÿfSYS /4 001ÿfSYS /8 010ÿfSYS /16 011ÿfSYS /32 100ÿfSYS /64 101ÿfSYS /128 110ÿfSYS /256 111ÿfSYS /512
6	-	reserve

SSCON1 (9EH) SPI Status Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	SPIF WCOL		-	-	TXE	FOUR	-	TBIE
read/write Read/write Power-on		Reading/Writing	-	-	Reading/Writing	Reading/Writing	-	Reading/Writing
initial value 0		0	x	x	0	0	x	0

Position number	Bit symbol	illustrate
7	**SPIF**	**SPI** data transfer flag 0: Cleared by software 1: Indicates that data transmission is complete, set by hardware.
6	**WCOL**	Write the conflict flag 0: Cleared to 0 by the software, indicating that write conflicts have been resolved. 1: The hardware sets it to 1, indicating that a collision has been detected.
3	**TXE**	Send buffer null flag 0: Send buffer is not empty 1: If the send buffer is empty, it must be cleared by the software.
2	**FOUR**	Transmission direction selection bit 0: MSB sent first 1: LSB sent first
0	**TBIE**	Send buffer interrupt enable control bit 0: Do not allow transmission interruption 1: Enable transmit interrupts. When SPIF=1, TBIE=1 will generate an SPI interrupt.
5~4,1	-	reserve

SSDAT (9FH) SPI Data Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	SPD[7:0]							
read/write Read/write Power-on		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
initial value 0		0	0	0	0	0	0	0

Position number	Bit symbol	illustrate
7~0	**SPD[7:0]**	**SPI** Data Buffer Register

		Data written to SSDAT is placed into the transmit shift register. Data from the receive shift register is obtained when SSDAT is read.

17.1.2 Signal Description

Master Output Slave Input (MOSI):

This signal connects the master device and a slave device. Data is serially transmitted from the master device to the slave device via MOSI; it is the master device's output and the slave device's input.

Master Input Slave Output (MISO):

This signal connects the slave device and the master device. Data is serially transmitted from the slave device to the master device via MISO; it is the slave device's output and the master device's input. When... The SPI configuration for slave devices is not selected, and the slave device's MISO pin is in a high-impedance state.

SPI Serial Clock (SCK): The SCK

signal is used to control the synchronous movement of input and output data on the MOSI and MISO lines. One byte is transmitted on the line every 8 clock cycles. If the slave device is not selected, the SCK signal is ignored by the slave device.

17.1.3 Working Mode

SPI can be configured into either master or slave mode. Configuration and initialization of the SPI module are accomplished by setting the SSCON0 register (SPI control register) and SSCON1 (SPI status register). After configuration, data transfer is performed by setting SSCON0, SSCON1, and SSDAT (SPI data register). During SPI communication, data is synchronously shifted in and out serially. The serial clock line (SCK) powers the data on

the two serial data lines (MOSI and MISO).

Data movement and sampling must be synchronized. If a slave device is not selected, it cannot participate in activities on the SPI bus.

When the SPI master device transmits data to the slave device via the MOSI line, the slave device responds by sending data back to the master device via the MISO line. This achieves synchronous full-duplex transmission of data transmission and reception within the same clock cycle. The transmit shift register and receive shift register use the same special function address. Writing to the SPI data register SSDAT writes to the transmit shift register, and reading from the SSDAT register retrieves the data from the receive shift register. Some devices have an SPI interface with an SS pin (slave select pin, active low). When communicating with the

SC92F836XB's SPI, the connection method of the SS pins of other devices on the SPI bus needs to be determined according to different communication modes. The following table lists the connection methods of the SS pins of other devices on the SPI bus under different communication modes of the SC92F836XB's SPI:

SC92F836XB SPI: Master/Slave Mode	for Other Devices on the SPI Bus	In slave mode, the SS (slave select) pin has one master and one slave pin. A master-multiple-	
		slave configuration (SC92F836XB) has	
		multiple I/O pins connected to	the slave's SS pin. Before data transmission, the slave device's SS pin must be set low.
From the pattern	Main mode	One main and one subordinate raise the price.	

Main Mode

ÿ Mode Activation:

The SPI master device controls the initiation of all data transfers on the SPI bus. When the MSTR bit in the SSCON0 register is set to 1, SPI operates in master mode, and only one master device can initiate a transfer. ÿ Transmit: In SPI master mode, writing one byte of data

to the SPI data

register SSDAT will write the data to the transmit shift buffer. If the transmit shift register already contains data, the master SPI generates a WCOL signal to indicate that the write is too fast. However, the data in the transmit shift register is unaffected, and the transmission is not interrupted. Alternatively, if the transmit shift register is empty, the master device immediately and serially shifts the data from the transmit shift register onto the MOSI lines according to the SPI clock frequency on SCK. When the transfer is complete, the SPIF bit in the SSCON1 register is set to 1. If SPI interrupts are enabled, an interrupt will also be generated when the SPIF bit is set to 1. ÿ Reception: When the master device transmits data to the slave device via the MOSI line, the corresponding slave device simultaneously transmits the contents of its transmit shift register to the master device's receive shift register via the MISO

line, achieving full-

duplex operation. Therefore, setting the SPIF flag to 1 indicates both transmission completion and data reception completion. Data received by the slave device is stored in the master device's receive shift register according to the MSB or LSB priority transmission direction. When a byte of data has been completely shifted into the receive register, the processor can obtain the data by reading the SSDAT register.

From the pattern

ÿ Mode Activation:

When the MSTR bit in the SSCON0 register is cleared to 0, SPI operates in slave mode.

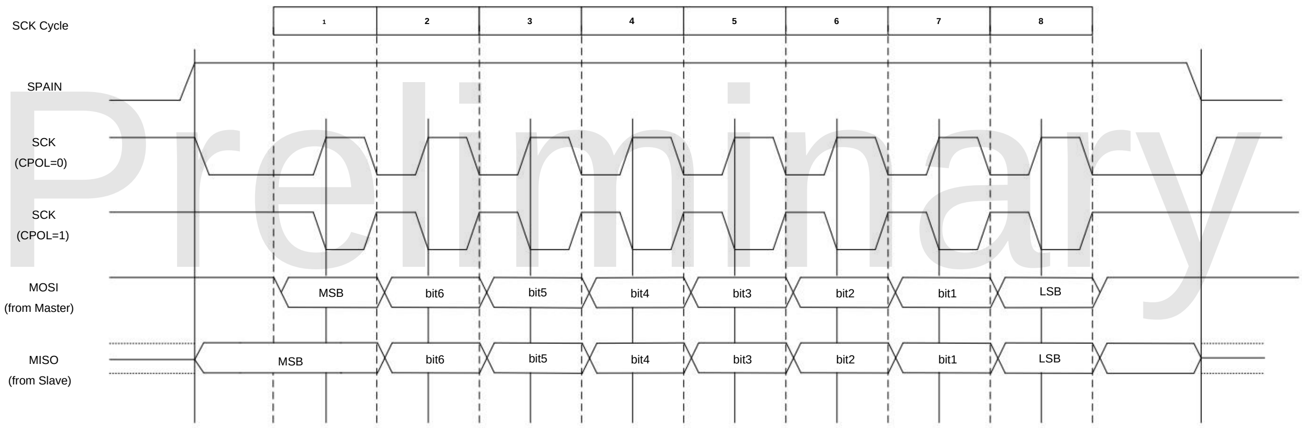
Sending and Receiving:

In slave mode, data is shifted in via the MOSI pin and shifted out via the MISO pin according to the SCK signal controlled by the master device. A bit counter records... The number of edges for SCK is such that when the receive shift register shifts in 8 bits of data (one byte) while the transmit shift register shifts out 8 bits of data (one byte), The SPIF flag is set to 1. Data can be obtained by reading the SSDAT register. If SPI interrupts are enabled, this will also occur when SPIF is set to 1. An interrupt is generated. At this point, the receive shift register retains its existing data and the SPIF bit is set to 1, so the SPI slave device will not receive any data. Until SPIF is cleared to 0. The SPI slave device must write the data to be transmitted into the transmit shift register before the master device begins a new data transmission. If no data is written before transmission begins, the slave device will send a "0x00" byte to the master device. If the SSDAT write operation occurs during transmission... In this case, if the WCOL flag of the SPI slave device is set to 1, that is, if the transfer shift register already contains data, the WCOL bit of the SPI slave device will be set to 1. Setting it to 1 indicates a write SSDAT conflict. However, the data in the shift register is unaffected, and the transfer is not interrupted.

17.1.4 Transmission Method

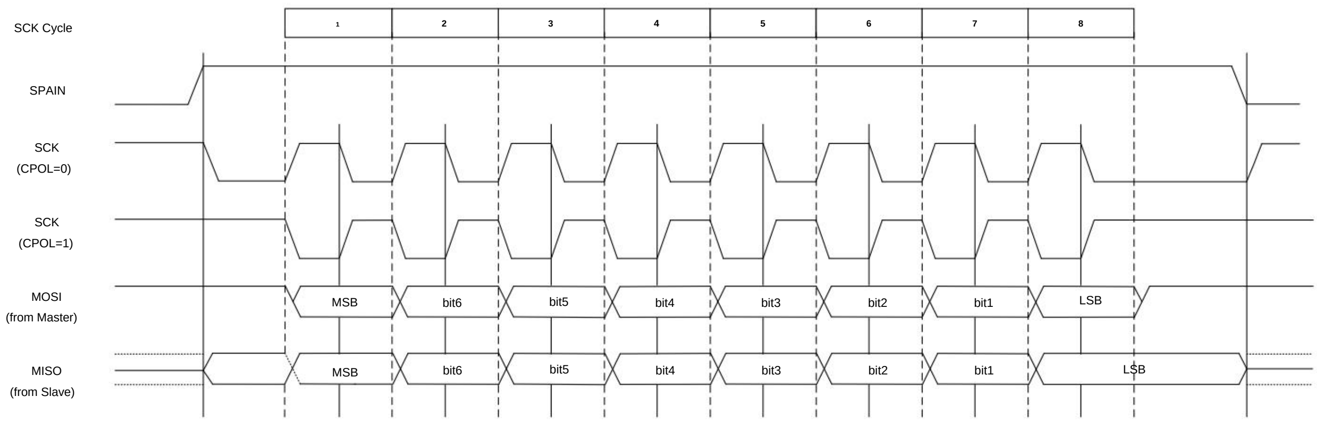
By setting the CPOL and CPHA bits of the SSCON0 register via software, users can select four combinations of SPI clock polarity and phase. The CPOL bit defines the clock polarity, i.e., the idle level state, and has little impact on the SPI transmission format. The CPHA bit defines the clock phase, i.e., the clock speed. The clock edge that allows data sampling shift. In master-slave communication, the clock polarity and phase settings should be consistent between the two devices.

When CPHA = 0, data is captured on the first edge of SCK, and the slave device must have the data ready before the first edge of SCK.



CPHA = 0 Data Transmission Diagram

When CPHA = 1, the master device outputs data to the MOSI line on the first edge of SCK, and the slave device takes the first edge of SCK as the start of transmission. The signal, SCK, begins capturing data on its second edge; therefore, the user must complete the write operation to SSDAT within the first two edges of SCK. This data transmission... The output mode is the preferred mode of communication between a master device and a slave device.



CPHA = 1 Data Transmission Diagram

17.1.5 Error Detection

Writing to the SSDAT register during the transmission of a data sequence will cause a write conflict, setting the WCOL bit in the SSCON1 register to 1.

It will not cause an interruption, and the transmission will not be stopped. The WCOL bit needs to be cleared to 0 by software.

17.2 TWI

SSMOD[1:0] = 10, the three-in-one serial interface SSI is configured as a TWI interface. The SC92F836XB can only act as a slave device in TWI communication.

SSCON0 (9DH) TWI Control Register (Read/Write)

Bit	7	6	5	4	3	2	1	0
symbol TWEN TWIF Read/Write Read/			-	GCA	AA	STATE[2:0]		
Write Read/Write Power-on initial value			-	Read/Write 0		Read/	Read/	Reading/Writing
0 0			x		0	Write 0	Write 0	0

Position number	Bit symbol	illustrate
7	**TWEN**	TWI Enable Control 0: Turn off TWI 1: Open TWI
6	**TWIF**	TWI interrupt flag 0: Reset by software 1. The interrupt flag is set to 1 by hardware under the following conditions. ÿ The address of the first frame was successfully matched. ÿ Successfully received or sent 8 bits of data ÿ Restart ÿ The slave device receives a stop signal
4	**GCA**	General address response flags 0: Non-response generic address 1: When GC is set to 1 and a general address match occurs, this bit is set to 1 by hardware and automatically cleared to zero.
3	**AA**	Receive enable bit 0: Do not allow receiving information sent by the host. 1: Allow receiving information sent by the host.
2~0	**STATE[2:0]** State machine status flags	000: The slave device is in idle state, waiting for TWEN to be set to 1 and detecting the TWI start signal. When the slave device... The machine will jump to this state after receiving the stop condition. 001: The slave device is receiving the first frame address and read/write bits (bit 8 is the read/write bit, 1 for read, 0 for write). (For writing). The slave device will jump to this state after receiving the start condition. 010: Slave data reception status 011: Slave device data transmission status 100: In the slave data transmission state, when the master sends a UACK (acknowledgment bit is high), Jump to this state and wait for a restart or stop signal. 101: When the slave device is in transmit mode, writing 0 to AA will enter this state, waiting for a restart. No. or stop signal.
5	-	reserve

SSCON1 (9EH) TWI Address Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	TWA[6:0]							GC
read/write Read/write Power-on		Read/	Read/	Read/	Read/	Reading/Writing		Reading/Writing
initial value 0		Write 0	Write 0	Write 0	Write 0	0	0	0

Position	Bit symbol	illustrate
number 7~1	**TW[6:0]**	TWI Address Register

0	**GC**	TWI Universal Address Enable 0: Do not respond to general addresses 1: Allow responses to generic addresses

SSDAT (9FH) TWI Data Buffer Register (Read/Write)

Bit number 7 Symbol read/		6	5	4	3	2	1	0
write	TWDAT[7:0]							
Read/write Power-on initial		Read/	Read/	Read/Write Read/Write 0 0		Read/	Read/	Reading/Writing
value 0		Write 0	Write 0			Write 0	Write 0	0

Position	Bit symbol	illustrate
number 7~0	**TWDAT[7:0]**	TWI Data Buffer Register

17.2.1 Signal Description

TWI clock signal line (SCL)

This clock signal is generated by the master unit and connected to all slave units. One byte of data is transmitted every nine clock cycles. The first eight cycles are used for data transmission. The last clock is used as the receiver's acknowledgment clock.

TWI data signal line (SDA)

SDA is a bidirectional signal line. When idle, it should be at a high level, pulled high by the pull-up resistor on the SDA line.

17.2.2 Working Mode

The SC92F836XB's TWI communication is only available in slave mode:

ÿ Mode Activation:

The mode starts when the TWI enable flag is turned on (TWEN = 1) and a start signal is received from the host.

The slave device enters the state of receiving the first frame address (STATE[2:0] = 001) from idle mode (STATE[2:0] = 000) and waits for the master's first frame. Frame data. The first frame of data is sent by the master and includes 7 address bits and 1 read/write bit. All slave devices on the TWI bus will receive the first frame of data from the master. One data frame. After the master sends the first data frame, it releases the SDA signal line. If the address sent by the master matches the value in a slave's own address register... The same indicates that the slave device has been selected. The selected slave device will determine the 8th bit on the bus, which is the data read/write bit (=1, read command; =0, write command). (The command is executed), then the SDA signal line is occupied, and a low-level acknowledge signal is given to the host in the 9th clock cycle of SCL, after which the bus is released.

Once a slave device is selected, it will enter different states depending on the read/write bits:

ÿ Non-generic address response, slave receive mode:

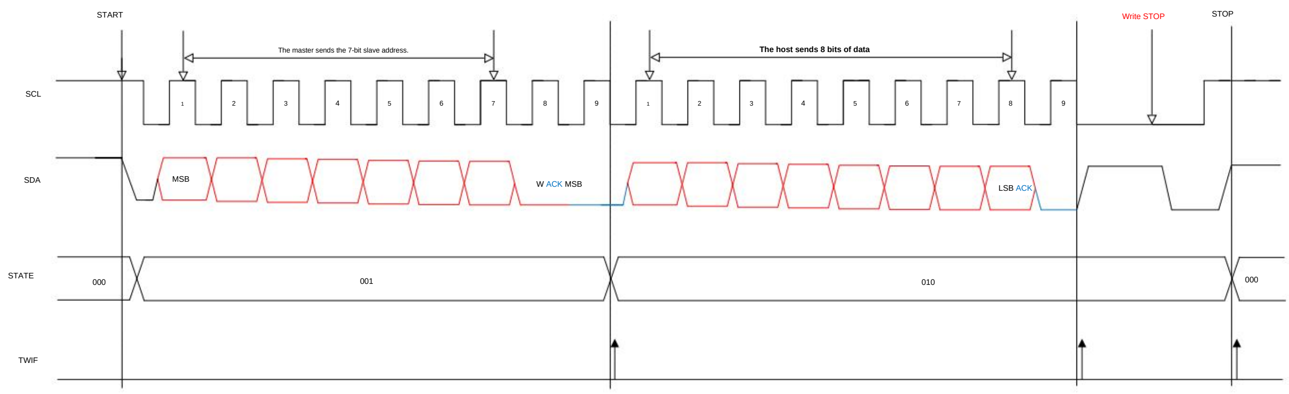
If the read/write bit received in the first frame is write (0), the slave device enters the slave receive state (STATE[2:0] = 010) and waits to receive from the master. The data is sent. Every 8 bits sent by the master, the bus is released, and the slave waits for the acknowledgment signal in the 9th cycle.

1. If the slave device's response signal is low, the master device can communicate in the following three ways:

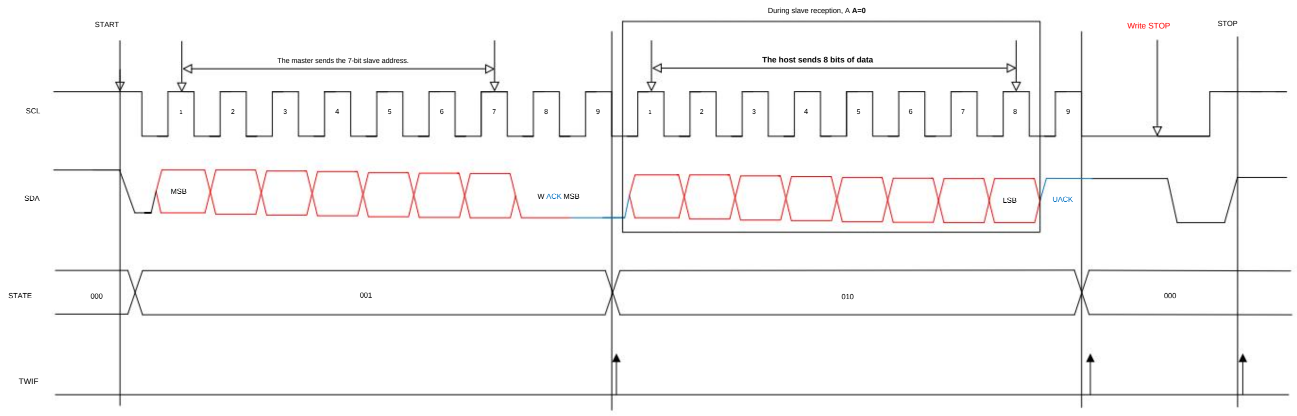
1) Continue sending data;

2) Resend the start signal, at which point the slave device re-enters the state of receiving the first frame address (STATE[2:0] = 001);

3) Send a stop signal to indicate that the current transmission is over, the slave device returns to the idle state, and waits for the master device to start the next time.



2. If the slave device responds with a high level (during reception, the AA value in the slave device register is rewritten to 0), it indicates that the current byte has been transmitted. After completion, the system will proactively end the transmission and return to the idle state (STATE[2:0] = 000), no longer receiving data from the host. according to.



ÿ Non-generic address response, slave sending mode:

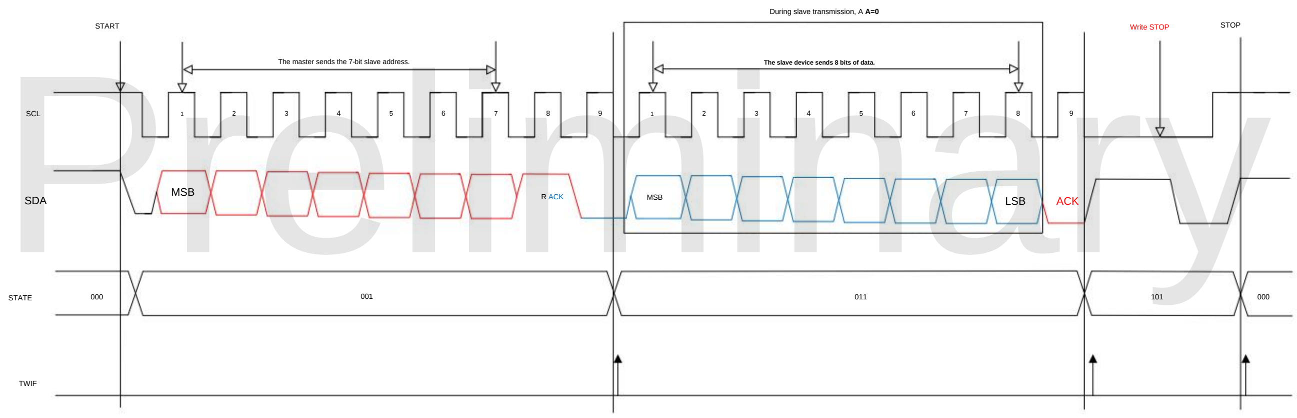
If the read/write bit received in the first frame is read (1), the slave will occupy the bus and send data to the master. The slave releases the bus after every 8 bits of data are sent.

Release the bus and wait for the host's response:

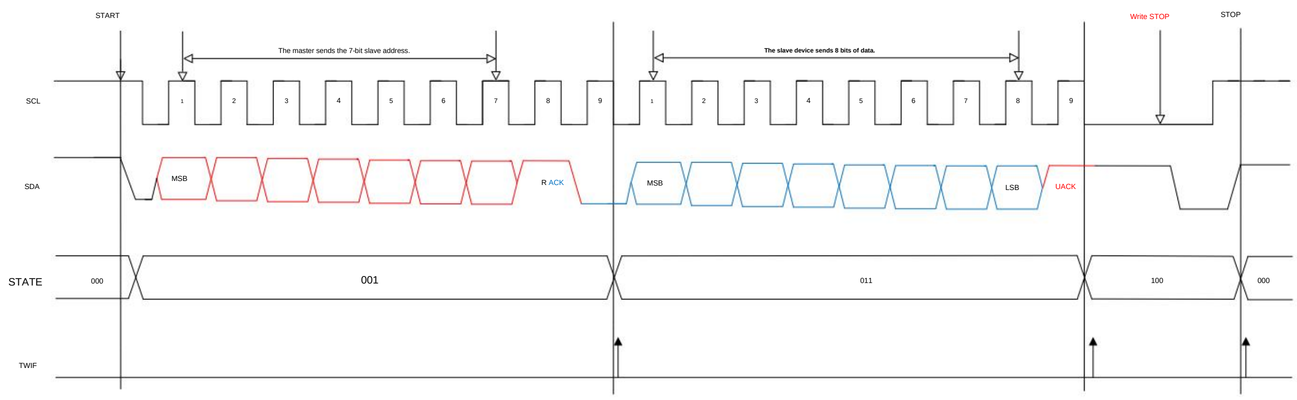
1. If the master responds with a low level, the slave continues to send data. During transmission, if the AA value in the slave register is changed...

If set to 0, the transmitter will actively terminate the transmission and release the bus after transmitting the current byte, waiting for a stop or restart signal from the host.

ÿSTATE[2:0] = 101ÿÿ



2. If the master responds with a high level, the slave will set STATE[2:0] = 100 and wait for the master to stop or restart signal.



ÿ Response from a generic address:

When GC=1, general addresses are allowed. The slave device enters the state of receiving the first frame address (STATE[2:0] = 001), and receives the first frame.

The address bits in the frame data are 0x00, at which point all slave devices respond to the master. The read/write bit sent by the master must be write (0), and all slave devices will respond.

After receiving, it enters the data reception state (STATE[2:0] = 010). The host releases the SDA line every 8 data packets sent and reads the SDA line.

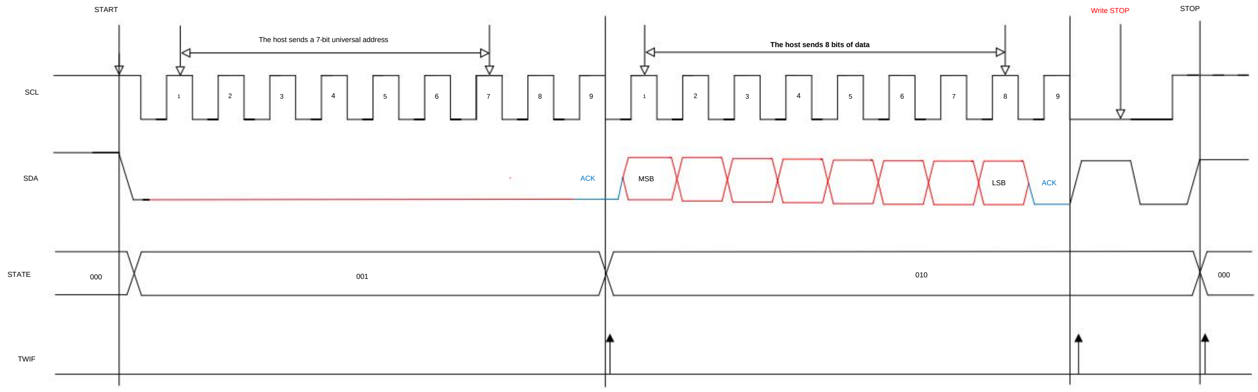
state:

1. If there is a response from the slave device, the master device can communicate in the following three ways:

1) Continue sending data;

2) Restart;

3) Send a stop signal to end the current communication.



2. If there is no response from the slave device, SDA is in an idle state.

Note: When using a universal address in a master-slave mode, the read/write bit sent by the master cannot be in the read (1) state; otherwise, except for the device sending data, All other devices on the bus will respond.

17.2.3 Operating Procedures

The operating steps for TWI in a three-in-one serial port are as follows:

ÿ Configure SSMOD[1:0] and select TWI mode;

ÿ Configure the SSCON0 TWI control register;

ÿ Configure the SSCON1 TWI address register;

ÿ If the slave device receives data, it waits for the interrupt flag TWIF in SSCON0 to be set to 1. The interrupt flag is reset every 8 bits of data received by the slave device. Set to 1. The interrupt flag needs to be manually cleared.

ÿ If the slave device is sending data, the data to be sent must be written into TWDAT, and TWI will automatically send the data. An interrupt will occur every 8 bits sent. The TWIF flag will then be set to 1.

17.3 UART1

SSMOD[1:0] = 11, the three-to-one serial interface SSI is configured as a UART interface.

SSCON0 (9DH) Serial Port **1 Control Register (Read/Write)**

Bit number 7 Symbol read/		6	5	4	3	2	1	0
write	SM0	-	SM2	REN	TB8	RB8	OF	RI
Read/write Power-on initial		-	Read/	Read/	Read/	Read/	Read/	Reading/Writing
value 0		x	Write 0	Write 0	Write 0	Write 0	Write 0	0

Position number	Bit symbol	illustrate
7	**SM0**	Serial communication mode control bits 0: Mode 1, 10-bit full-duplex asynchronous communication, consisting of 1 start bit, 8 data bits, and 1 stop bit. The stop bit configuration allows for variable communication baud rate. 1: Mode 3, 11-bit full-duplex asynchronous communication, consisting of 1 start bit, 8 data bits, and one selectable bit. The programming consists of the 9th bit and 1 stop bit, allowing for variable communication baud rate;
5	**SM2**	Serial communication mode control bit 2, this control bit is only valid for mode 3. 0: Set RI to generate an interrupt request whenever a complete data frame is received; 1. When a complete data frame is received, RI will only be set and an interrupt generated if RB8=1. beg.
4	**REN**	Receive enable control bit 0: Data reception is not allowed;

		1: Allow receiving data.
3	**TB8**	Only valid for mode 3, and is the 9th bit of the transmitted data.
2	**RB8**	Only valid for mode 3, and is the 9th bit of the received data.
1	**OF**	Send interrupt flag
0	**RI**	Receive interrupt flag
6	-	reserve

SSCON1 (9EH) Serial Port **1** Baud Rate Control Register Low Byte (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	BAUDL [7:0]							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

SSCON2 (95H) Serial Port **1** Baud Rate Control Register High Bit (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	BAUDH [7:0]							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 7~0	**BAUD [15:0]** Serial port baud rate control	$$\text{BaudRate} = \frac{\text{fsys}}{\text{BAUD1H, BAUD1L}}$$ **Note: [BAUD1H,BAUD1L]** must be greater than **0x0010**

SSDAT (9FH) Serial Port Data Buffer Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	SBUF[7:0]							
read/write Read/write Power-on initial		Read/	Read/	Read/Write Read/Write 0 0		Read/	Read/	Reading/Writing
value 0		Write 0	Write 0			Write 0	Write 0	0

Position	The sign bit	illustrate
number 7~0	**SBUF[7:0]** serial port data buffer register	SBUF contains two registers: a transmit shift register and a receive latch. The data in SBUF will be sent to the transmit shift register, and the transmit process will begin. Reading SBUF will return... Return the contents of the latch.

18 analog-to-digital converter **(ADC)**

The SC92F836XB has a built-in 12-bit, 11-channel high-precision successive approximation ADC, and external 10-channel ADC and other I/O functions are also supported. The ADC also has an internal channel selectable to 1/4 VDD, which, in conjunction with the internal 2.4V reference voltage, is used to measure the VDD voltage.

The ADC reference voltage can be selected in two ways:

ÿ VDD pin (i.e., directly internal VDD);

ÿ The reference voltage output by the internal regulator is precisely 2.4V (at this time, the MCU supply voltage VDD must not be lower than 2.9V).

Note: The clock source of the **ADC** circuit is fixed at **fHRC =** 24MHz and will not change with the switching of internal and external system clocks.

18.1 ADC- related registers

ADCCON (ADH) ADC Control Register (Read/Write)

Bit number **7** Symbol		6	5	4	3	2	1	0
ADCEN ADCS EOC/ADCIF Read/Write Read/Write Power-on				ADCIS[4:0]				
initial value 0		Reading/Writing	Reading/Writing	Reading/Writing	Read/Write Read/Write Read/Write Read/Write			
		0	0	0	0	0	0	n

Position number	Bit symbol	illustrate
7	**ADCEN**	Power on the ADC 0: Power off the ADC module 1: Turn on the ADC module power.
6	**ADCS**	ADC Start Writing "1" to this bit initiates an ADC conversion; that is, this bit is only used to trigger the ADC conversion. Signal. This bit can only be written with a 1 for it to be valid. Note: After writing "1" to **ADCS ,** do not change the interrupt flag before setting the interrupt flag **EOC/ADCIF.** Write operation to **ADCCON** register
5	**EOC/ADCIF**	Conversion complete / ADC interrupt request flag (End Of Conversion / ADC Interrupt Flag) 0: Conversion not yet complete 1: ADC conversion complete. User software needs to clear the process. ADC Conversion Complete Flag (EOC): This bit is hard-coded after the user sets the ADCS to start conversion. The value is automatically cleared to 0; after the conversion is complete, this bit will be automatically set to 1 by the hardware. ADC interrupt request flag ADCIF: This bit also serves as an interrupt request flag for the ADC. If the user enables the ADC... If an interrupt occurs, the user must clear this bit using software.
4~0	**ADCIS[4:0]**	ADC Input Selector 00000: Select AIN0 as the ADC input. 00001: Select AIN1 as the input of the ADC 00010: Select AIN2 as the input of the ADC 00011: Select AIN3 as the input of the ADC 00100: Select AIN4 as the input of the ADC 00101: Select AIN5 as the input of the ADC 00110: Select AIN6 as the input of the ADC 00111: Select AIN7 as the input of the ADC 01000: Select AIN8 as the input of the ADC 01001: Select AIN9 as the input of the ADC 01010~11110: Reserved 11111: The ADC input is 1/4 VDD, which can be used to measure the power supply voltage.

ADCCFG2 (AAH) ADC Setting Register **2 (Read/Write)**

Bit number **7 6** Symbol read/write power-on			5	4	3	2	1	0
initial	-	-	-	-	LOWSP	ADCCK[2:0]		
value x	-	-	-	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
		x	x	x	0	0	0	0

Position number	Bit symbol	illustrate
3	**LOWSP**	ADC Sampling Clocks Selector 0: Set the ADC sampling time to 6 ADC sampling clock cycles. 1: Set the ADC sampling time to 36 ADC sampling clock cycles. LOWSP controls the sampling clock frequency of the ADC, while the conversion clock frequency of the ADC is determined by... ADCCK[2:0] control is unaffected by the LOWSP bit. The ADC needs to go through 6 or 36 ADC sampling clocks plus 14 ADC conversion clocks to complete the process. It can complete the entire process from sampling to conversion, therefore, in practical applications, the ADC can complete the entire process from sampling to conversion. The total conversion time is calculated as follows: LOWSP=0ÿTADC1=(6+14)/fADCÿ LOWSP=1ÿTADC2=(36+14)/fADC
2~0	**ADCCK[2:0]**	ADC Sampling Clocks Selector 000: Set the ADC clock frequency fADC to fHRC/32; 001: Set the ADC clock frequency fADC to fHRC/24; 010: Set the ADC clock frequency fADC to fHRC/16; 011: Set the ADC clock frequency fADC to fHRC/12; 100: Set the ADC clock frequency fADC to fHRC/8; 101: Set the ADC clock frequency fADC to fHRC/6; 110: Set the ADC clock frequency fADC to fHRC/4; 111: Set the ADC clock frequency fADC to fHRC/3 Note: The clock source of the **ADC** circuit is fixed at **fHRC =** 24MHz and will not change with the internal or external system clock. It changes with the switching of the clock.
7~4	-	reserve

ADCCFG0 (ABH) ADC Setting Register **0 (Read/Write)**

Bit number **7 6** symbol EAIN7 Read/Write Read/Write			**5**	**4**	**3**	**2**	**1**	**0**
Power-on initial value 0		EAIN6	EAIN5	EAIN4	EAIN3	EAIN2	EAIN1	EAIN0
		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
		0	0	0	0	0	0	0

ADCCFG1 (ACH) ADC Setting Register **1 (Read/Write)**

Bit number **7 6** Symbol read/write power-on initial			**5**	**4**	**3**	**2**	**1**	**0**
value x	-	-	-	-	-	-	EAIN9	EAIN8
	-	-	-	-	-	-	Reading/Writing	Reading/Writing
		x	x	x	x	x	0	0

Position number	Bit symbol	illustrate
0	**EAINx** **(x=0~9)**	**ADC** Port Setting Register 0: Set AINx as an I/O port 1: Set AINx as the ADC input and automatically remove the pull-up resistor.

OP_CTM1 (C2H@FFH) Customer Option Register **1 (Read/Write)**

Bit number	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
symbol VREFS XTLHF Read/Write Read/Write Read/			-	-	IAPS[1:0] Read/		-	-
Write Power-on initial value n			-	-	Write Read/Write		-	-
		n	x	x	n	n	x	x

Position number	Bit symbol	illustrate

7	**VREFS**	Reference voltage selection (initial value is loaded from **Code Option ,** user can modify the setting). 0: Set the ADC's VREF to VDD 1: Set the ADC's VREF to the internally accurate 2.4V.

ADCVL (AEH) ADC Conversion Value Register (Low Bit) (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	ADCV[3:0] Read/				-	-	-	-
read/write Read/write Power-on		Write Read/Write		Reading/Writing	-	-	-	-
initial value 0		0	0	0	x	x	x	x

ADCVH (AFH) ADC Conversion Value Register (High Bit) (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	ADCV[11:4] Read/							
read/write Read/write Power-on		Read/	Read/	Write Read/Write		Read/	Read/	Reading/Writing
initial value 0		Write 0	Write 0	0	0	Write 0	Write 0	0

Position	Bit symbol	illustrate
number 11~4	**ADCV[11:4]**	The high 8 bits of the ADC conversion value
3~0	**ADCV[3:0]**	The lower 4 bits of the ADC conversion value

IE (A8H) Interrupt Enable Register (Read/Write)

The symbol for position number 6	7		5	4	3	2	1	0
	YES	EADC Read/	ET2	-	ET1	EINT1	ET0	EINT0
Write Read/Write Power-on initial		Reading/Writing	Reading/Writing	-	Reading/Writing	Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	x	0	0	0	0

Position number	Bit symbol	illustrate
6	**EADC**	ADC interrupt enable control 0: Do not allow EOC/ADCIF to generate interrupts. 1: Enable EOC/ADCIF to generate interrupts.

IP (B8H) Interrupt Priority Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	IPADC	IPT2	-	IPT1	POINT1	IPT0	IPINT0
read/write	-	Read/	Read/	-	Read/	Read/	Read/	Reading/Writing
power-on initial value x		Write 0	Write 0	x	Write 0	Write 0	Write 0	0

Position number	Bit symbol	illustrate
6	**IPADC**	ADC interrupt priority selection 0: Sets the interrupt priority of the ADC to "low". 1. Set the interrupt priority of the ADC to "high".

18.2 ADC Conversion Steps

The actual steps required for users to perform ADC conversion are as follows:

ÿ Configure the ADC input pins; (Set the bit corresponding to AINx as the ADC input. Usually, the ADC pins are pre-fixed.)

ÿ Set the ADC reference voltage Vref and the frequency used for ADC conversion;

ÿ Turn on the ADC module power;

ÿ Select the ADC input channel; (Set the ADCIS bit to select the ADC input channel)

ÿ Start ADCS, conversion begins;

ÿ Wait for EOC/ADCIF=1. If ADC interrupt is enabled, an ADC interrupt will occur, and the user needs to clear the EOC/ADCIF flag to 0 via software.

ÿ Obtain 12 bits of data from ADCVH and ADCVL, converting the most significant bit first and then the least significant bit, completing the conversion in one step;

ÿ If the input channel is not changed, repeat steps 5-7 for the next conversion.

Note: Before setting **IE[6](EADC),** it is best for the user to clear **EOC/ADCIF** using software , and to do so after the **ADC** interrupt service routine has finished executing. Also clear the **EOC/ADCIF** to avoid continuously generating **ADC** interrupts.

19 Dual-mode touch circuit

The SC92F836XB has a built-in 23-channel dual-mode capacitive touch circuit, configurable for high-sensitivity or high-reliability modes, and features the following:

1. The high-sensitivity mode is suitable for touch applications that require high sensitivity, such as air-button touch and proximity sensing.
2. The high-reliability mode has strong anti-interference capabilities and can pass the 10V dynamic CS test.
3. Enables 23 touch buttons and related functions.
4. Highly flexible software library support, low development difficulty
5. Supports automated debugging software and intelligent development.
6. The touch module can enter a low-power mode when the MCU is in STOP mode, and the overall power consumption of the chip can be as low as [redacted] when a single touch button wakes it up.
11uA

Note: The clock source of the dual-mode touch circuit is fixed at **fHRC =** 24MHz and will not change with the switching of internal and external system clocks.

19.1 Power Consumption Modes of Touch Circuits

The SC92F836XB allows touch scanning to be enabled in STOP Mode: this reduces the overall power consumption of the MCU, thus meeting the requirements for low power consumption.

Touch applications with low power consumption requirements.

Users can understand that the touch circuit of the SC92F836XB has two power consumption modes:

1. Normal operating mode
2. Low-power operation mode

The two power consumption modes are defined as follows:

CPU	Normal operating mode	Low power operation mode
	RUNÿNormal modeÿ	StopÿSTOP Modeÿ
Touch circuit	RUN	RUN

19.2 Touch Mode

The SC92F836XB's dual-mode touch circuitry provides two touch modes:

1. High Sensitivity Mode
2. High Reliability Mode

Users can select the touch button library file provided by Saiyuan (which can be downloaded from the Saiyuan official website) and quickly and easily implement the required functions by using the touch button library file.

Touch functionality.

Users can select the most suitable touch mode for the current application using the information in the table below:

illustrate	High sensitivity mode ÿ	High Reliability Mode
Features	High anti-interference capability, can be controlled by 3V dynamic CS ÿ Ultra-high sensitivity	ÿ Strong anti-interference capability, can withstand 10V dynamic CS ÿ Lower power consumption
Applicable Applications	ÿ General Touch Button Applications ÿ Application of air-touch buttons ÿ Proximity sensing applications ÿ Touch applications requiring high sensitivity	ÿ Applications requiring strong anti-interference capabilities ÿ Applications requiring 10V dynamic CS
To select high-sensitivity mode, load the high-sensitivity touch library into the project; to select high-reliability mode, load the high-reliability touch library into the project.		
The documentation titled "Saiyuan SC92F_93F Series TouchKey MCU Application Guide"	Related chapters: ÿ 2 ÿ 4 SC92F8XXX_HIGHRELIABILITY_LIB_T1 Library Description SC92F8XXX_SC93F8XXX_HIGHSENSITIVE_LIB_ T1 Library Description ÿ 3 SC92F8XXX_SC93F8XXX_HIGHSENSITIVE_LIB_ T2 Library Description	Application Guide for SC92F/93F Series TouchKey MCUs Related chapters:
The corresponding library file	SC92F8X6XB_HighSensitive_Lib_Tn_Vx.xxLIB requires attention to the following:	SC92F8X6XB_HighReliability_Lib_Tn_Vx.x.x.LIB
	ÿ The T1 library is used for spring-type applications. ÿ The T2 library is used for air gesture applications, and the number of buttons must be at least 3. More than one	It can only be used in spring-type applications.
Selection Instructions	Under normal circumstances, it is recommended to use this high sensitivity mode for better performance. Use it for experience.	High reliability mode is recommended in only two situations: ÿ Requires 10V dynamic CS ÿ Lower power consumption current is required, and the current cannot be reduced in high-sensitivity mode. full

20 EEPROM and IAP Operations

The SC92F836XB's IAP operating space range offers two selectable modes:

The operating modes of EEPROM and IAP are as follows:

1. The 128 bytes of EEPROM at the highest internal address can be used for data storage;

2. IAP operations can be performed within the entire 8 Kbyte range of the IC's ROM space and the 128 bytes of EEPROM, primarily for remote program updates.

New use.

The IAP operation space selection is used as the Code Option when writing to the IC using the programmer:

OP_CTM1 (C2H@FFH) Customer Option Register **1 (Read/Write)**

Bit number	7	6	5 4		3	2	1	0
symbol VREFS XTLHF	read/write read/write	power-on initial	-	-	IAPS[1:0] Read/		-	-
value n		Reading/Writing	-	-	Read/Write	Write	-	-
		n	x	x	n	n	x	x

Position	Bit symbol	illustrate
number 3~2	**IAPS[1:0]**	EEPROM and IAP space range selection 0: IAP operations are prohibited in the Code area; only the EEPROM area can be used for data storage. 01: The last 0.5K Code area allows IAP operations (1E00H~1FFFH) 10: The last 1K Code area allows IAP operations (1C00H~1FFFH) 11: All Code areas are allowed for IAP operations (0000H~1FFFH)

20.1 EEPROM / IAP Operation Related Registers

EEPROM / IAP operation related register description:

Symbol	address description	7	6	5	4	3	2	1	0	Reset value
IAPKEY	F1H IAP Protection Register	IAPKEY[7:0]								00000000b
IAPADL	F2H IAP Write to Low-order Address Register	IAPDR[7:0]								00000000b
IPADH	F3H IAP Write to High-Number Register	-	-	-	IAPADR[12:8]					xxx00000b
IAPADE	F4H IAP Write to Extended Address Register	IAPADER[7:0]								00000000b
IAPDAT	F5H IAP Data Register	IAPDAT[7:0]								00000000b
IAPCT	F6H IAP Control Register	-	-	-	-	PAYTIMES [1:0]		CMD[1:0]		xxxx0000b

IAPKEY (F1H) IAP Protection Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	IAPKEY[7:0]							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 7~0	**IAPKEY[7:0]** Enables EEPROM/IAP Function and Operation Time Limit Settings	Write a non-zero value n, representing: ÿ Enable EEPROM/IAP function; ÿ If no write command is received after n system clock cycles, the EEPROM/IAP function will be disabled. Close again.

IAPADL (F2H) IAP write to the low-order address register (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	IAPDR[7:0]							
read/write Read/write Power-on initial		Read/	Read/	Reading/Writing		Read/	Read/	Reading/Writing
value 0		Write 0	Write 0	0	0	Write 0	Write 0	0

Position	Bit symbol	illustrate
number 7~0	**IAPDR[7:0]**	The lower 8 bits of the EEPROM/IAP write address

IAPADH (F3H) IAP write to the high-order address register (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	IAPADR[12:8]				
read/write	-	-	-	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
power-on initial value x		x	x	0	0	0	0	0

Position	Bit symbol	illustrate
number 4~0	**IAPADR[12:8]**	The high 5 bits of the EEPROM/IAP write address
7~5	-	reserve

IAPADE (F4H) IAP write to extended address register (read/write)

Bit number 5 Symbol read/write Read/write Power-on initial value 0	7	6		4	3	2	1	0
	IAPADER[7:0]							
		Read/	Read/	Reading/Writing		Read/	Read/	Reading/Writing
		Write 0	Write 0	0	0	Write 0	Write 0	0

Position	Bit symbol	illustrate
number 7~0	**IAPADER[7:0]**	IAP Extended Address: 0x00: Both MOVC and IAP programming are performed on the code. 0x01: Read operations are allowed only for the User ID area; write operations are not permitted. 0x02: Both MOVC and IAP programming are performed on EEPROM. Other: Reserved

IAPDAT (F5H) IAP Data Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	IAPDAT[7:0]							
read/write Read/write Power-on initial		Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 7~0	**IAPDAT**	Data written via IAP

IAPCTL (F6H) IAP Control Register (Read/Write)

Bit number	7	6	5	4	3	2	1	0
symbol	-	-	-	-	PAYTIMES[1:0] Read/Write		CMD[1:0]	
read/write	-	-	-	-	Read/Write		Reading/Writing	
power-on initial value x		x	x	x	0	0	0	0

Position	Bit symbol	illustrate
number 3~2	**PAYTIMES[1:0]** CPU Hold Time setting	during EEPROM/IAP write operations. 00: Set CPU HOLD TIME to 4ms @ 12/6/2MHz 01: Set CPU HOLD TIME to 2ms @ 12/6/2MHz 10: Set CPU HOLD TIME to 1ms @ 12/6/2MHz 11: Retain Note: The CPU holds the PC pointer, while other functional modules continue to operate; the interrupt flag will... It is saved and enters an interrupt after the Hold period ends, but only the last one can be saved from multiple interrupts. Second-rate. Selection recommendation: If VDD is between 2.7V and 5.5V, a 10V ohmmeter can be selected. VDD is between 2.4V and 5.5V; you can choose 01 or 00.

1~0	**CMD[1:0]**	EEPROM / IAP write operation commands 10: Write Other: Reserved Note: At least **8 NOPs** must be added after the EEPROM/IAP write operation statement . This ensures that subsequent instructions can be executed correctly after the **IAP** operation is completed!

20.2 EEPROM / IAP Operation Procedure

The EEPROM/IAP writing process of SC92F836XB is as follows:

ÿ Write to IAPADE[7:0], 0x00: Select the Code area to perform IAP operation; 0x02: Select the EEPROM area to perform EEPROM operation. Read and write operations;

ÿ Write to IAPDAT[7:0] (Prepare the data to be written to EEPROM/IAP);

ÿ Write {IAPADR[12:8], IAPADR[7:0]} (Prepare the target address for EEPROM/IAP operations);

ÿ Write IAPKEY[7:0] Write a non-zero value n (enable EEPROM/IAP protection, and no write command is received within n system clock cycles). (This will disable EEPROM/IAP).

ÿ Write to IAPCTL[3:0] (set CPU Hold time, write CMD[1:0] to 1, 0, CPU Hold and start EEPROM/IAP writing) enter);

ÿ Once the EEPROM/IAP write operation is complete, the CPU will continue with subsequent operations;

Notice:

1. When programming an IC, if "Disable IAP operation in Code area" is selected via Code Option, then when IAPADE[7:0]=0x00 (selected) The Code area is inoperable via IAP, meaning data cannot be written; data can only be read using the MOVC instruction.
2. When IPADE=0x01 or 0x02, MOVC and writing are performed on the EEPROM or IFB area. If an interrupt occurs at this time, Furthermore, if the interrupt contains a MOVC operation, the MOVC result may be incorrect, leading to program malfunction. To avoid this, please... Before performing any operation with IPADE=0x01 or 0x02, the user must disable the global interrupt (EA=0). After the operation is complete, set IPADE to 0x00. Enable global interrupt (EA=1).

20.2.1 128 BYTES Independent EEPROM Operation Routine

```
#include "intrins.h"
unsigned char EE_Add; unsigned
char EE_Data;
unsigned char code * POINT =0x0000;
```

EEPROM write operation C **demo program :**

```
EA = 0;                     // Guan Zong interrupted
IAPADE = 0x02;              //Select EEPROM region
IAPDAT = YES_Data;          //Send data to EEPROM data register
IAPADH = 0x00;              //Higher addresses are written as 0x00 by default
IAPADL = EE_Add;            //Write the low-order bits of the target address to EEPROM
IAPKEY = 0xF0;              //This value can be adjusted according to the actual situation; it is necessary to ensure that after this instruction is executed and before the value of IACTL is assigned,
                            //The time interval must be less than 240 (0xf0) system clock cycles; otherwise, the IAP function will be disabled.
                            // Special attention should be paid when enabling interrupts.
IAPCTL = 0x0A; _nop_();     // Perform EEPROM write operation, 1ms @ 12M/6M/2M;
_nop_();                    //Wait (at least 8 _nop_() are needed)

_nop_();
_nop_();
_nop_();
_nop_();
_nop_();
_nop_();
_nop_();
IAPADE = 0x00;              //Return to ROM region
```

EA = 1; // Enable global interrupt

EEPROM read operation C **demo program :**

EA = 0; // // Guan Zong interrupted

IAPADE = 0x02; // Read Select EEPROM region

EE_Data = *(POINT + EE_Add); // Return the value of IAP_Add into IAP_Data

IAPADE = 0x00; // Enable ROM region to prevent MOVC operations from accessing the EEPROM.

EA = 1; global interrupt

20.2.2 8 KBYTES CODE Area IAP Operation Routine

```
#include "intrins.h"
unsigned int IAP_Add; unsigned
char IAP_Data;
unsigned char code * POINT =0x0000;
```

IAP write operation C **demo program :**

```
IAPADE = 0x00;                    //Select Code area
IAPDAT = IAP_Data;                // Send data to IAP data register
IAPADH = (unsigned char)((IAP_Add >> 8)); // Write the high-order bits of the IAP target address
IAPADL = (unsigned char)IAP_Add; // This value can be adjusted     // Write the low-order bits of the IAP target address
IAPKEY = 0xF0; // The             according to the actual situation; it is necessary to ensure that after this instruction is executed and before the value of IAPCTL is assigned,
            time interval must be less than 240 (0xf0) system clock cycles, otherwise the IAP function will be disabled;
                                  // Special attention should be paid when enabling interrupts.
IAPCTL = 0x0A; _nop_();           // Perform IAP write operation, 1ms @ 12M/6M/2M;
_nop_();                          //Wait (at least 8 _nop_() are needed)

_nop_();
_nop_();
_nop_();
_nop_();
_nop_();
_nop_();
_nop_();
```

A **C demo** program for **IAP** read operations :

```
IAPADE = 0x00;                    //Select Code area
IAP_Data = *( POINT+IAP_Add);     // Read the value of IAP_Add into IAP_Data
```

Note: IAP operations within the 8 Kbytes Code area carry certain risks and require users to implement appropriate security measures in their software.
Improper use may result in the user's program being rewritten! Unless the user absolutely needs this function (such as for remote program updates), it is not recommended that the user use it.

21. CHECK SUM module

The SC92F836XB has a built-in checksum module that can be used to generate 16-digit checksum values for program code in real time. Users can utilize this...

The checksum is compared with the theoretical value to check whether the contents of the program area are correct.

Note: The checksum value is the cumulative sum of all data in the entire program area, that is, all data in the address units **0000H~1FFDH .** If there are...

Residual values from the user's previous operation can cause **the checksum** value to differ from the theoretical value. Therefore, it is recommended that users erase or write to the entire **code** area.

After performing operation **0,** the code is burned to ensure that **the check sum** value is consistent with the theoretical value.

21.1 Registers related to CHECK SUM verification operation

CHKSUML (FCH) Check Sum result register low byte (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	CHKSUML[7:0]							
read/write Read/write	Power-on initial	Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 7~0	**CHKSUML [7:0]**	Check Sum result register low byte

CHKSUMH (FDH) Check Sum result register high-order bits (read/write)

Bit number	7	6	5	4	3	2	1	0
symbol	CHKSUMH[7:0]							
read/write Read/write	Power-on initial	Reading/Writing	Reading/Writing	Reading/Writing		Reading/Writing	Reading/Writing	Reading/Writing
value 0		0	0	0	0	0	0	0

Position	Bit symbol	illustrate
number 7~0	**CHKSUMH [7:0]**	Check Sum result register high byte

OPERCON (EFH) Operation Control Register (Read/Write)

Bit number 7 Symbol read/write		6	5	4	3	2	1	0
power-	-	-	-	-	-	-	-	CHKSUMS
on initial	-	-	-	-	-	-	-	Reading/Writing
value x		x	x	x	x	x	x	0

Position number	Bit symbol	illustrate
0	**CHKSUMS**	Check sum operation starts trigger control (Start) Write "1" to this bit to begin a check sum calculation. Only writing "1" to this bit is valid.

22 Electrical Characteristics

22.1 Limiting Parameters

symbol	parameter	Minimum value	Maximum value	UNIT
VDD/VSS DC supply voltage		-0.3	5.5	In
Voltage ON any pin input/output voltage Pin		-0.3	VDD+0.3	In
FACING	Operating ambient temperature	-40	85	ÿ
TSTG	Storage temperature	-55	125	ÿ

22.2 Recommended working conditions

symbol		Minimum value	Maximum value	UNIT	System clock frequency
VDD	Parameters:	2.9	5.5	In	>12MHz
VDD	Operating	2.4	5.5	In	ÿ12MHz
FACING	voltage, Operating ambient temperature	-40	85	ÿ	·

22.3 DC Electrical Characteristics

(VDD = 5V, TA **=** +25ÿ, unless otherwise specified)

symbol	parameter	Minimum value	typical value	maximum value	unit	Test conditions
Current						
Iop1	Operating current	·	8.3	·	mA	fSYS =12MHz
Iop2	Operating current	·	6.7	·	mA	fSYS =6MHz
Iop3	Operating current	·	5.7	·	mA	fSYS =2MHz
Ipd1	Standby current (Power Down Mode)	·	0.7	1.0	ÿA	
IIDL1	standby current (IDLE mode)	·	6.0	·	m.a.	
IBTM	Base Timer Operating Current	·	5.4	7.0	ÿA	BTMFS[3:0]= 1000 are generated every 4.0 seconds. Interruption
IWDT	WDT current	·	6.1	7.3	ÿA	WDTCKS[2:0]= 000 WDT Overflow Time 500ms
ITK1	Touch key operating current (high sensitivity) Sensitivity mode)	·	1.3	1.7	m.a.	
ITK2	Touch key operating current (high) (Based on model)	·	0.4	0.6	m.a.	
IO port characteristics						
HIV1	Input high voltage	0.7VDD	·	VDD+0.3	V	
VIL1	Input low voltage	-0.3	·	0.3VDD	V	
HIV2	Input high voltage	0.8VDD	·	VDD	V	Schmidt trigger input: RST/tCK/SCK
In IL2	Input low voltage	-0.2	·	0.2VDD	V	
IOL1	Output low current	·	40	·	mA	VPin =0.4V
IOL2	Output low current	·	70	·	mA	VPin =0.8V
IOH1	Output high current P1/P5	·	20	·	mA	VPin=4.3V

IOH2	High current output P1/P5	-	10	-	mA	VPin=4.7V
IOH3	High current output P0/P2	-	20	-	mA	VPin=4.3V Pxyz=0, IOH level 0
	High output current P0/P2	-	9.8	-	mA	VPin=4.3V Pxyz=1, IOH Level 1
	High output current P0/P2	-	5.0	-	mA	VPin=4.3V Pxyz=2, IOH Level 2
	High output current P0/P20~P23	-	1.8	-	mA	VPin=4.3V Pxyz=3, IOH Level 3
	High output current P24~P27	-	2.6	-	mA	VPin=4.3V Pxyz=3, IOH Level 3
IOH4	High output current P0/P2	-	9.6	-	mA	VPin=4.7V Pxyz=0, IOH level 0
	High output current P0/P2	-	4.4	-	mA	VPin=4.7V Pxyz=1, IOH Level 1
	High output current P0/P2	-	2.3	-	mA	VPin=4.7V Pxyz=2, IOH Level 2
	High output current P0/P20~P23	-	0.8	-	mA	VPin=4.7V Pxyz=3, IOH Level 3
	High output current P24~P27	-	1.2	-	mA	VPin=4.7V Pxyz=3, IOH Level 3
RPH1	The pull-up	-	33	-	kÿ	
resistor serves as the internal reference voltage of the ADC, at 2.4V.						
VDD24	Internal reference voltage 2.4V, output voltage 2.38V.		2.40	2.42	V	TA=-40~85ÿ

(VDD = 3.3V, TA = +25ÿ, unless otherwise specified)

symbol	parameter	Minimum value	typical value	maximum value	unit	test conditions
Current						
Iop4	Operating current	-	5.8	-	mA	fSYS =12MHz
Iop5	Operating current	-	4.8	-	mA	fSYS =6MHz
Iop6	Operating current	-	4.2	-	mA	fSYS =2MHz
Ipd2	Standby current (Power Down Mode)	-	0.7	1.0	a	
IIDL2	standby current (IDLE mode)	-	5.9	-	m.a.	
ITK3	Touch key operating current (high) Sensitivity mode)	-	1.2	1.5	m.a.	
ITK4	Touch key operating current (high) (Reliable mode)	-	0.3	0.4	m.a.	
IO port characteristics						
HIV3	Input high voltage,	0.7VDD	-	VDD+0.3 V		
VIL3	input low voltage,	-0.3	-	0.3VDD V		
HIV4	input high voltage,	0.8VDD	-	VDD	V	Schmidt trigger input: 0.2VDD V
VIL4	input low voltage;	-0.2	-	RST/tCK/SCK		
IOL3	output low current,	-	31	-	mA	VPin =0.4V
IOL4	output low current,	-	55	-	mA	VPin=0.8V
IOH5	output high current.	-	6	-	mA	VPin=3.0V
RPH2	The pull-up	-	56	-	kÿ	
resistor serves as the internal reference voltage for the ADC, 2.4V.						
VDD24	Internal reference voltage 2.4V, output voltage 2.38V.		2.40	2.42	V	TA=-40~85ÿ

22.4 AC Electrical Characteristics

(VDD = 2.4V ~ 5.5V, TA = 25ÿ, unless otherwise specified)

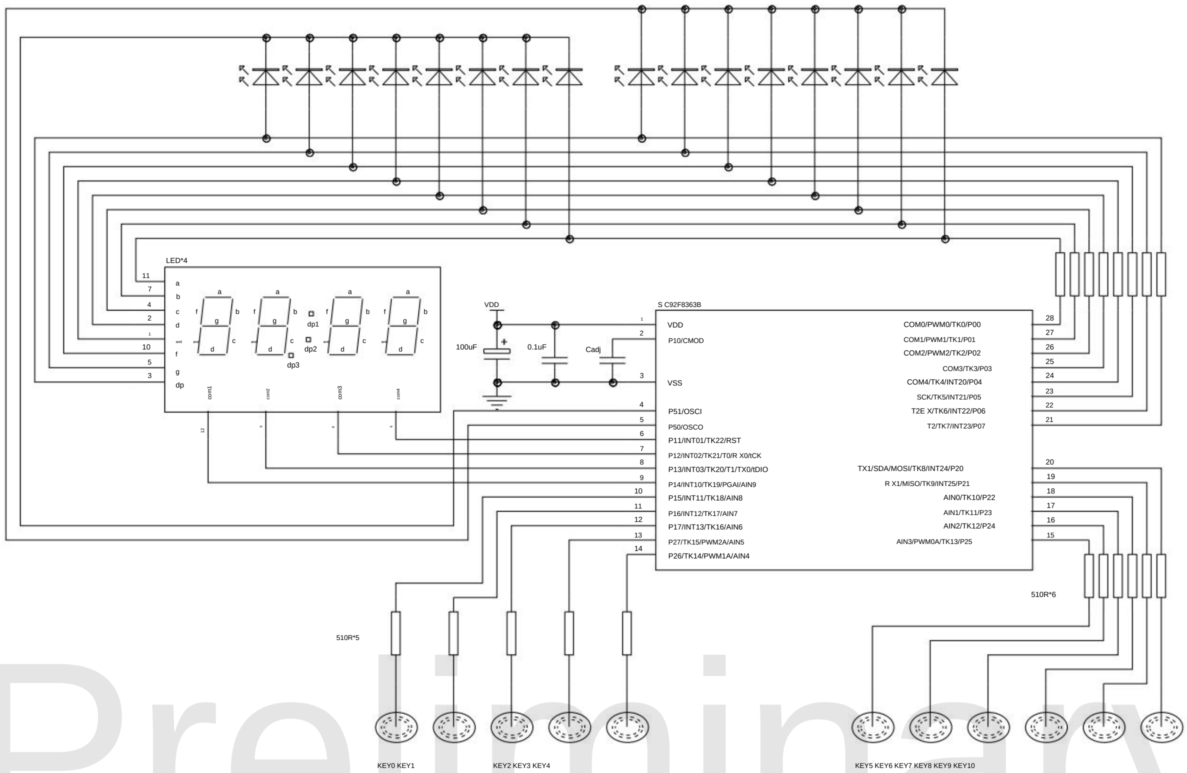
Symbolic parameter Tosc1		Minimum value,	typical value, maxim	um value, unit, test	conditions	
	External high-frequency oscillator start-up time	-	9	-	ms	external 16MHz crystal Vibration
Tosc2	External high-frequency oscillator start-up time	-	18	-	ms	external 8MHz crystal Vibration
Tosc3	External high-frequency oscillator start-up time	-	35	-	ms	external 4MHz crystal Vibration
TPOR	Power On Reset Time	-	1	1.5	ms	
TPDW	When waking up from Power Down mode between	-	1	1.5	ms	
Peat	Reset pulse width	18	-	-	ÿs	low level active
HRC	RC oscillation stability	23.76	24	24.24	MHz	VDD=3.0~5.5V TA=-20~85 ÿ

22.5 ADC Electrical Characteristics

(TA = 25ÿ, unless otherwise specified)

symbol	parameter	Minimum value	typical value, maxim	um value, unit, test	conditions	
WHAT	Power supply voltage	2.4	5.0	5.5	In	
No.	accuracy	-	12	-	bit	GNDÿVAINÿVDD
ONLY	ADC input voltage	GND	-	VDD	In	
RAIN	ADC input resistance	1	-	-	Mÿ	VIN=5V
IADC1	ADC conversion current 1	-	2.7	3.2	mA	ADC module turned on VDD=5V
IADC2	ADC conversion current 2	-	2.1	2.5	mA	ADC module turned on VDD=3.3V
DNL	Differential Nonlinear Error	-	±1	-	LSB	VDD=5V VREF=5V
INL	Integral Nonlinearity Error Offset Error Full-	-	±2	-	LSB	
NO	Scale Error Total	-	±10	-	LSB	
IF	Absolute Error	-	0	-	LSB	
EAD		-	±10	-	LSB	
TADC1	ADC conversion time 1	-	10	-	ÿs	ADC Clock = 2MHz ADC sampling period = 6
TADC2	ADC conversion time 2	-	20	-	ÿs	ADC Clock = 1 MHz ADC sampling period = 6

23 Application Circuits



SC92F8363B Touch Key Application Circuit Diagram

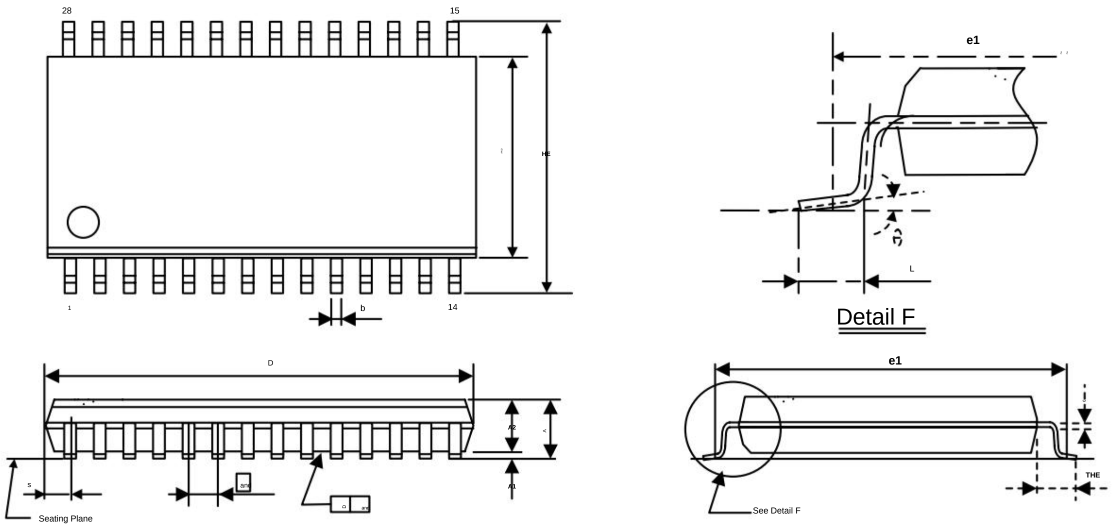
24 Ordering Information

Product Number	Packaging	Package
SC92F8363BM28U	SOP28L	Tube
SC92F8363BX28U	TSSOP28L	Tube
SC92F8362BM20U	SOP20L	Tube
SC92F8362BX20U	TSSOP20L	Tube
SC92F8361BM16U	SOP16L	Tube

25 Package Information

SC92F8363BM28U

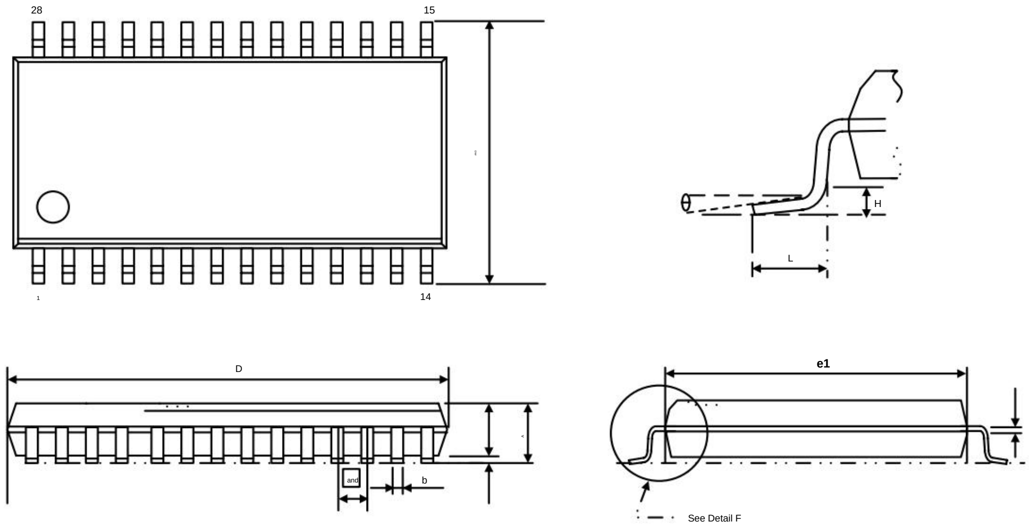
SOP28L (300mil) dimensions (unit: millimeters)



symbol	mm (millimeters)		
	Minimum		maximum
A	2.47	Standard 2.56	2.65
A1	0.100	0.200	0.300
A2	2.240	2.340	2.440
b	0.39	--	0.48
C	0.254(BSC)		
D	17.80	18.00	18.20
AND	7.374	7.450	7.574
HE	10.100	10.300	10.500
and	1.270(BSC)		
L	0.7	0.85	1.0
THE	1.3	1.4	1.5
ÿ	0ÿ	-	8ÿ
S	0.745(BSC)		

SC92F8363BX28U

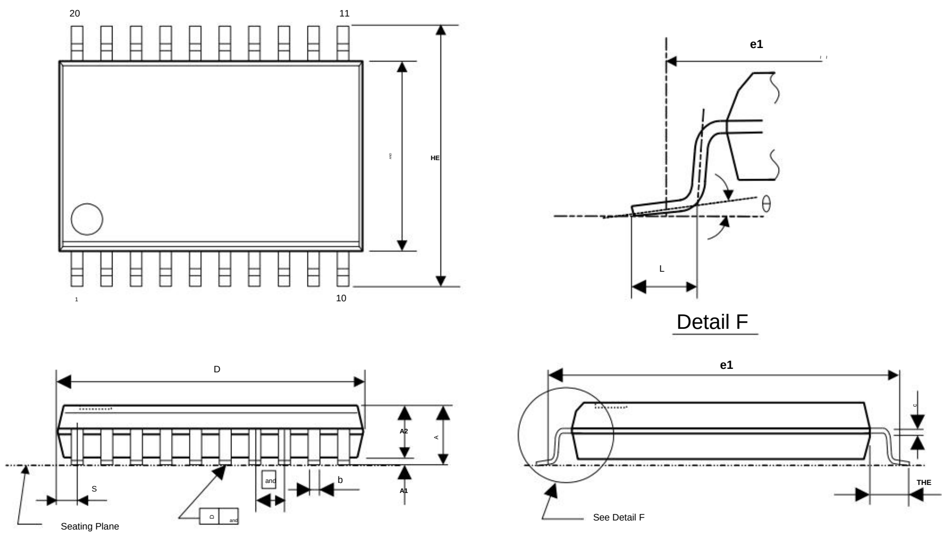
TSSOP28 External Dimensions Unit: millimeters



symbol	mm (millimeters)		
	Minimum	normal	maximum
A	-	-	1.200
A1	0.050	-	0.150
A2	0.800	-	1.050
b	0.190	-	0.300
c	0.090	-	0.200
D	9.600	-	9.800
AND	6.250	-	6.550
e1	4.300	-	4.500
and	0.65(BSC)		
L	-	-	1.0
ÿ	0ÿ	-	8ÿ
H	0.05	-	0.25

SC92F8362BM20U

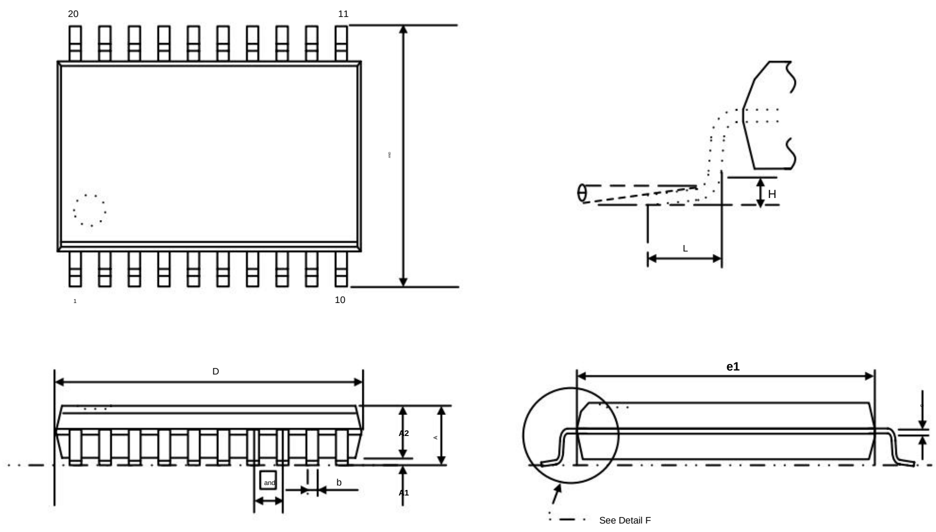
SOP20L (300mil) dimensions (unit: millimeters)



symbol	mm (millimeters)		
	Minimum	normal	maximum
A	2.47	2.56	2.65
A1	0.100	0.200	0.300
A2	2.240	2.340	2.440
b	0.39	--	0.47
C	0.254(BSC)		
D	12.60	12.80	13.00
AND	7.30	7.50	7.70
HE	10.100	10.300	10.500
and	1.27(BSC)		
L	0.700	0.850	1.000
THE	1.30	1.40	1.50
ÿ	0ÿ	-	8ÿ
S	0.660(BSC)		

SC92F8362BX20U

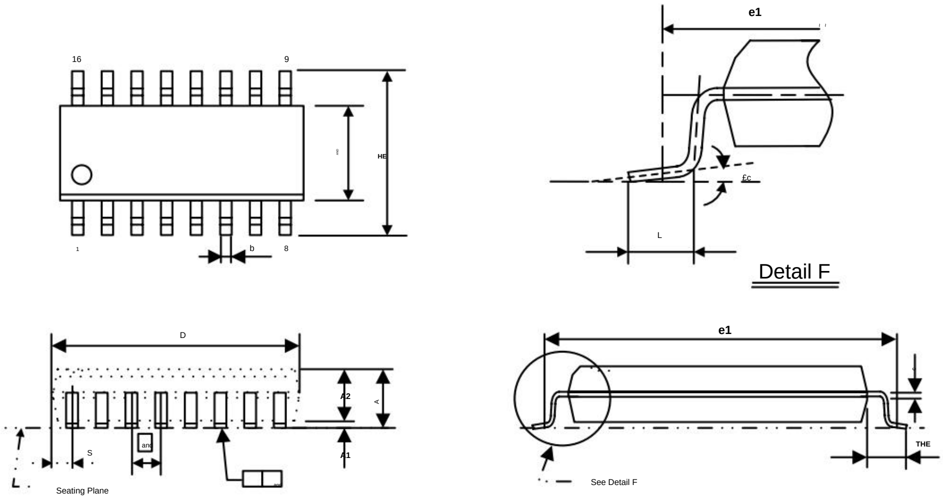
TSSOP20L external dimensions (unit: mm)



symbol	mm (millimeters)		
	Minimum	normal	maximum
A	-	-	1.200
A1	0.050	-	0.150
A2	0.800	-	1.050
b	0.190	-	0.300
c	0.090	-	0.200
D	6.400	-	6.600
AND	6.20	-	6.60
e1	4.300	-	4.500
and	0.65(BSC)		
L	-	-	1.00
ÿ	0ÿ	-	8ÿ
H	0.05		0.15

SC92F8361BM16U

SOP16L (150mil) dimensions **(unit: millimeters)**



symbol	mm (millimeters)		
	Minimum	normal	maximum
A	1.500	1.625	1.750
A1	0.050	0.1375	0.225
A2	1.35	1.45	1.55
b	0.39	0.435	0.48
C	0.254(BSC)		
D	9.700	9.900	10.100
AND	3.700	3.900	4.100
HE	5.800	5.950	6.100
and	1.27(BSC)		
L	0.500	0.650	0.800
THE	0.950	1.050	1.150
ÿ	0ÿ	·	8ÿ
S	0.505(BSC)		

26 Specification Change Log

Version	Record	date
V0.7 modifies Chapter 25 Encapsulation Information.		December 2020
In V0.6, T2CON Timer 2 information table is changed from T2 to T2EX. In the C		November 2020
V0.5	language example, 83H is changed to 0x83. 2. Update the TBIE bit description 3. Corrected the description regarding whether TK can wake up STOP. 4. Update the descriptions related to the GPIO section. 5. Increase the maximum value of IBTM. 6. Update the text and image descriptions for the TWI chapter. 7. Updated the touch control section description 8. Change IIC to TWI in the internal block diagram. 9. Update the description of OP_HRCR	October 2018
V0.4 Modify ADC electrical parameters. V0.3		July 2018
Delete water level detection function description.		July 2018
V0.2	1. Correct the baud rate calculation formula for T2 working mode 2. 2. Description of the newly added water level detection function 3. Update the BIT description in BTMFS.	June 2018
V0.1 First Edition		November 2017